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CHAPTER 10

GASEOUS EXCHANGE

MULTIPLE CHOICE QUESTIONS

Each question has four possible answers. Circle the correct answer.

(1) Organisms need energy in the form of:
(a) GMP (b) GTP (c) AMP (d) ATP

(2) In plants, gaseous exchange occurs through:
(a) Stomata (b) Cuticle (c) Lenticels (d) All of these

(3) The aquatic plants get oxygen from:
(a) Soil (b) Water (c) Air (d) None of these

(4) The nasal cavity opens into:
(a) Glottis (b) Epiglottis (c) Pharynx (d) Larynx

(5) Muscular passage common to both food and air:
(a) Pharynx (b) Larynx (c) Trachea (d) Oesophagus

(6) Which organ is also called as voice box?
(a) Nostrils (b) Pharynx (c) Larynx (d) Glottis

(7) Vibrations of which produce sound?
(a) Trachea (b) Oesophagus (c) Vocal cards (d) Nostrils

(8) Movements of which organs result in speech?
(a) Lips (b) Cheeks (c) Tongue (d) All of these

(9) Which organs have ciliated and glandular cells?
(a) Trachea (b) Bronchi (c) Both a & b (d) None of these

(10) How many lungs are present in thoracic cavity?
(a) 4 (b) 3 (c) 2 (d) 1

(11) Which thick muscular structure in present below the lungs?
(a) Diaphragm (b) Trachea (c) Intercoastals (d) Ribs

(12) How many lobes are present in left lung?
(a) 1 (b) 2 (c) 3 (d) 4

(13) The physical movements associated with gaseous exchange are called:
(a) Breathing (b) Peristalsis (c) Swallowing (d) Chewing

(14) During inspiration, the volume of the thoracic cavity:
(a) Increases (b) Decreases (c) Remains constant (d) Lessens

(15) During expiration, the rib muscles:
(a) Contract (b) Relax (c) Remain as such (d) None of these

(16) The respiration centre that controls breathing is located in:
(a) Brain (b) Kidney (c) Heart (d) Lungs

(17) To which organs the respiratory centre sends messages?
(a) Intercoastal muscles (b) Diaphragm (c) Both a and b (d) None of these

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(18) **Amount of oxygen in inspired air:**
(a) 21% (b) 16% (c) 18% (d) 14%

(19) **Amount of carbon dioxide in inspired air:**
(a) 4% (b) 0.04% (c) 0.5% (d) 2%

(20) **Amount of nitrogen in inspired air:**
(a) 70% (b) 72% (c) 76% (d) 79%

(21) **Water vapours in inspired air:**
(a) Variable (b) Saturated (c) Almost none (d) All of these

(22) **Dust particles in inspired air:**
(a) Variable (b) Saturated (c) Almost none (d) All of these

(23) **The inflammation of the bronchi or bronchioles:**
(a) Asthma (b) Pneumonia (c) Emphysema (d) Bronchitis

(24) **An infection of lungs:**
(a) Asthma (b) Pneumonia (c) Emphysema (d) Bronchitis

(25) **The symptoms of which disease vary from person to person?**
(a) Asthma (b) Pneumonia (c) Emphysema (d) Bronchitis

(26) **Virus of which disease can cause pneumonia?**
(a) Hepatitis (b) Polio (c) AIDS (d) Influenza

(27) **Inhalers are used to treat:**
(a) Pneumonia (b) Lung cancer (c) Asthma (d) Emphysema

(28) **How many different chemicals are present in cigarette smoke?**
(a) Less than 400 (b) Less than 4000 (c) More than 4000 (d) More than 2000

(29) **How many persons die each year by lung cancer world wide?**
(a) 1000000 (b) 1,100,000 (c) 1,200,000 (d) 1,30,0000

(30) **At what percentage, the rate of smoking is increasing in developing countries?**
(a) 1.4% (b) 1.6% (c) 2.4% (d) 3.4%

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ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	d	2	d	3	b	4	c	5	a
6	c	7	c	8	d	9	c	10	c
11	a	12	b	13	a	14	a	15	b
16	a	17	c	18	a	19	b	20	d
21	a	22	a	23	d	24	b	25	d
26	d	27	c	28	c	29	d	30	d

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SHORT QUESTIONS

Q. No. 1 Define cellular respiration.

CELLULAR RESPIRATION

The process in which the carbon-hydrogen bonds in the food are broken by oxidation reduction reactions and the energy is transformed into ATP is called cellular respiration.

Q. No. 2 What is aerobic respiration?

AEROBIC RESPIRATION

The process of respiration in which oxygen is used and there is complete oxidation of the food material producing carbon dioxide and water is called aerobic respiration.

Q. No. 3 In what form energy is available to organisms?

ENERGY AVAILABILITY TO ORGANISMS

The energy is available to organisms in the form of ATP (Adenosine Triphosphate) for their activities.

Q. No. 4 Define gaseous exchange.

GASEOUS EXCHANGE

The process of taking in of oxygen and giving out of carbon dioxide is termed as gaseous exchange.

Q. No. 5 Define breathing.

BREATHING

The process through which animals take air in their bodies to get oxygen from it and then give out the air for getting rid of carbon dioxide is called breathing.

Q. No. 6 What is the difference between respiration and breathing?

DIFFERENCE BETWEEN RESPIRATION AND BREATHING

The respiration involves the mechanical and the biochemical processes whereas breathing is only the mechanical or physical process of exchange of gases.

Q. No. 7 How speech is produced?

PRODUCTION OF SPEECH

The vibrations in vocal cards and the movements of lips, cheeks, tongue and jaws produce specific sounds which result in speech.

- Speech is an ability that only humans are gifted with and this is one of the characteristics which has put human beings superior to all.

Q. No. 8 What is the function of the mucus secreted by the epithelial cells of trachea and bronchi?

FUNCTION OF MUCUS

The glandular cells of trachea and bronchi secrete mucus which moistens the air and trap any fine particles of dust or bacteria that have escaped from the nasal cavity.

Q. No. 9 What is the function of cilia present in trachea and bronchi?

FUNCTION OF CILIA

The cilia present in trachea and bronchi beat with an upward motion so that the foreign particles along the mucus are sent to the oral cavity from where it may be either swallowed or coughed out.

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Q. No. 10 Explain whether breathing movements are involuntary or voluntary.

BREATHING MOVEMENTS

The breathing movements are involuntary to a large extent. However, we can control the rate of breathing but not for a long time.

Q. No. 11 Why the percentage of respiratory disorders is higher in Pakistan?

RESPIRATORY DISORDERS IN PAKISTAN

The percentage of respiratory disorders is higher in Pakistan because of the more concentration of air pollutants not only in the urban but also in rural atmosphere.

Q. No. 12 What was the death rate of pneumonia before the discovery of antibiotics?

DEATH RATE OF PNEUMONIA

Prior to the discovery of antibiotics, one-third of pneumonia patients died from the infection.

Q. No. 13 What is the death rate of lung cancer?

DEATH RATE OF LUNG CANCER

Lung cancer is the most common cause of cancer-related deaths. It is responsible for more than 1.3 million deaths world wide annually.

Q. No. 14 What will happen to lungs, if a person stops smoking?

DISCONTINUATION OF SMOKING

If a person stops smoking, the chance to develop cancer decreases as damage to the lungs is repaired and contaminant particles are gradually removed.

Q. No. 15 What is nicotine? What are its bad effects?

NICOTINE

Nicotine is a powerful poison and was widely used as an insecticide in the past.

Bad Effects:

When inhaled through tobacco smoking, it reaches our circulatory system and not only hardens the walls of the arteries but also damages the brain tissues.

Q. No. 16 What are the observations about rates of smoking by WHO?

RATES OF SMOKING BY WHO

According to the WHO, the rates of smoking have declined in the developed world. In the developing world, it is rising by 3.4% per year as of 2002.

Q. No. 17 When World No Tobacco Day is celebrated?

WORLD NO TOBACCO DAY

The World No Tobacco Day is celebrated on 31st of May every year.

Q. No. 18 What is the percentage of passive smokers to develop heart diseases and lung cancer?

PASSIVE SMOKERS

Non-smokers who are exposed to second hand smoke (passive smoke) at home or work increase their heart disease risk by 25-30% and their lung cancer risk by 20-30%.

Q. No. 19 How does smoking affect the social life of a person?

EFFECT OF SMOKING ON SOCIAL LIFE

Smoking affects the social life of a person. Smokers may face social unacceptance because other people may not want to be exposed to other's smoke.

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LONG QUESTIONS

Q. No. 1 Explain gaseous exchange in plants.

GASEOUS EXCHANGE IN PLANTS

Lack of Organ System Level:

Plants have no organs or systems for the exchange of gases with the environment. Every cell of the plant body exchanges gases with the environment by its own.

Gaseous Exchange through Cuticle:

In young stems and leaves, some gaseous exchange occurs through the cuticle which is present over their epidermis.

Gaseous Exchange through Stomata:

The leaves and young stems have stomata in their epidermis. The gaseous exchange occurs through these stomata.

Gaseous Exchange through Air Spaces:

The inner cells of the leaves (mesophyll) and stems have air spaces among them, which help in the exchange of gases.

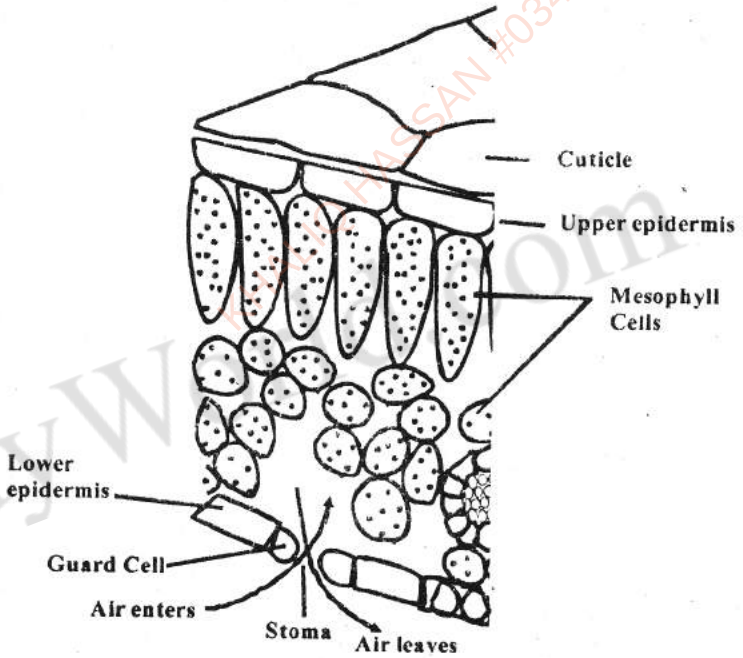

Figure: Gaseous Exchange in a Leaf

Leaf Cells during Day Time:

During the day time, when the mesophyll cells of leaves are carrying out photosynthesis and respiration side by side, the oxygen produced in photosynthesis is utilized in cellular respiration. Similarly, the carbon dioxide produced during cellular respiration is utilized in photosynthesis.

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Leaf Cells during Night Time:

During night, when there is no photosynthesis occurring, the leaf cells get oxygen from the environment and release carbon dioxide through stomata.

Gaseous Exchange through Lenticels:

In woody stems and mature roots, the entire surface is covered by bark which is impervious to gases or water. The lenticels allow air to pass through them.

Lenticels: The pores in the layer of bark are called as lenticels.

Presence: The lenticels are slightly more raised than the general surface of the stem.

Gaseous Exchange through Roots:

Gases diffuse in and out of the general surface of the young roots. The gases are found in the soil surrounding the roots.

Gaseous Exchange in Aquatic Plants:

The aquatic plants get the oxygen dissolved in water and release carbon dioxide in the water.

Q. No. 2 Write a note on human respiratory system.

HUMAN RESPIRATORY SYSTEM

The human respiratory system consists of two parts:

1. The Air Passageway
2. The Lungs

1. THE AIR PASSAGEWAY

The air passageway consists of the parts through which the outside air comes in the lungs and after the exchange of gases it goes out. The passage of air consists of the following parts:

1. Nostrils
2. Nasal cavity
3. Internal Nostrils
4. Pharynx
5. Larynx
6. Trachea
7. Bronchi
8. Bronchioles
9. Alveolar Ducts
10. Alveoli

1. Nostrils:

The nasal cavity opens to the outside through the openings called nostrils.

2. Nasal cavity:

The nose encloses the nasal cavity. The nasal cavity is divided into two portions by a wall. Each portion is lined by fine hairs and mucous which filter the dust particles from the air. The mucous also moistens and warms the incoming air and keeps its temperature nearly equal to that of the body.

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3. Internal Nostrils:

The nasal cavity opens into pharynx by means of two small openings called internal nostrils.

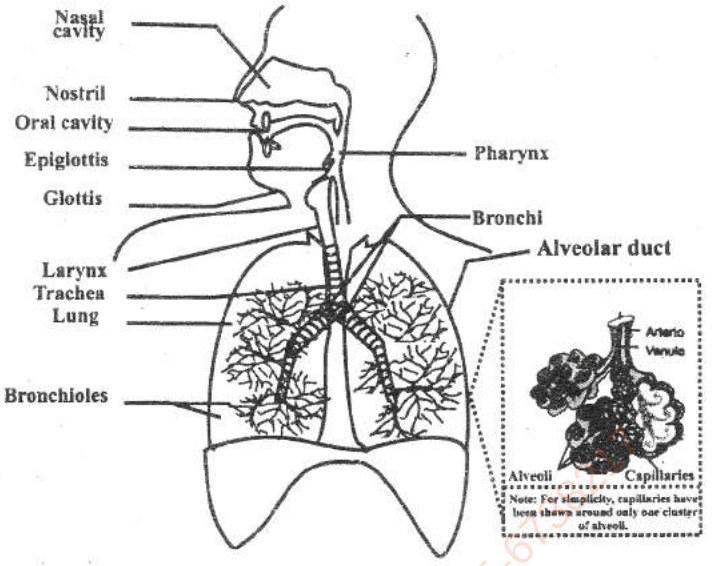

Figure: The air passageway and the lungs

4. Pharynx:

Pharynx is a muscúlar passage and is common to both food and air. It extends to the opening of oesophagus and the larynx.

5. Larynx:

The air goes from the pharynx into the larynx. The larynx is the box made of cartilage. It is present between pharynx and trachea. It is also called as voice box.

Glottis:

Glottis is a narrow opening at the floor of the pharynx which leads to larynx.

Epiglottis:

The glottis is guarded by a flap of tissue called the epiglottis.

Vocal Cards:

Two pairs of fibrous bands called vocal cards are stretched along the larynx. The vocal cards vibrate when the air passes through them. This vibration produces sounds.

6. Trachea:

Larynx continues to the trachea, which is also called as wind pipe. It is about 12 cm long tube which lies in front of the oesophagus. There are C-shaped cartilaginous rings in the wall of trachea. The cartilages keep the trachea from collapsing even when there is no air in it.

7. Bronchi:

On entering the chest cavity, the trachea divides into two smaller tubes called bronchi. The singular of bronchi is bronchus. The bronchi also have cartilaginous plates in their walls. Each bronchus enters into the lung of its side and then divides into smaller branches.

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8. Bronchioles:

The bronchi continue dividing in the lungs until they make several fine tubes called bronchioles.

9. Alveolar Ducts:

The bronchioles progressively lose the cartilage and they become narrower. The bronchioles end as fine tubules called the alveolar ducts.

10. Alveoli:

Each alveolar duct opens into a cluster of pouches called alveoli. The alveoli form the respiratory surface in human body. Each alveolus is a sac like structure lined by a single layer of epithelial cells. It is bound on the outside by a network of capillaries.

Blood Circulation:

The pulmonary artery from the heart containing deoxygenated blood enters the lungs and branches into arterioles and then into capillaries which surround the alveoli. These then join together to form the venules which form pulmonary vein. The pulmonary vein carries the oxygenated blood back to heart.

2. THE LUNGS

All the alveoli on one side constitute a lung.

Number:

There is a pair of lung in the thoracic cavity.

Chest Wall:

The chest wall is made up of 12 pairs of ribs and the rib muscles called the intercoastal muscles.

Diaphragm:

A thick muscular structure called diaphragm is present below the lungs.

Size:

The left lung is slightly smaller and has two lobes and the right lung is bigger and has three lobes.

Elasticity:

The lungs are spongy and elastic organs.

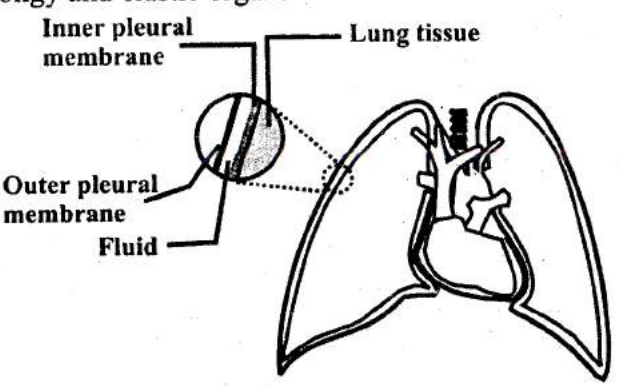

Figure: Lungs and Pleural Membranes

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Blood Circulation:

The lungs have blood vessels that are the branches of pulmonary arteries and veins.

Protection:

Each lung is enclosed by two membranes called the outer pleural membrane and the inner pleural membrane. The membranes enclose a fluid which provides lubrication for the free expanding and contracting of the lungs.

Q. No. 3 Explain the mechanism of breathing.

THE MECHANISM OF BREATHING

Breathing:

The physical movements associated with the gaseous exchange are called breathing.

Phases of Breathing:

There are two phases of breathing:

1. Inhalation
2. Exhalation

1. Inspiration or Inhalation:

During inspiration, the rib muscles contract and ribs are raised. At the same time the dome-shaped diaphragm contracts and is lowered. These movements increase the area of the thoracic cavity, which reduces the pressure on lungs. As a result, the lungs expand and the air pressure within them also decreases. The air from outside rushes into the lungs to equalize the pressure on both sides.

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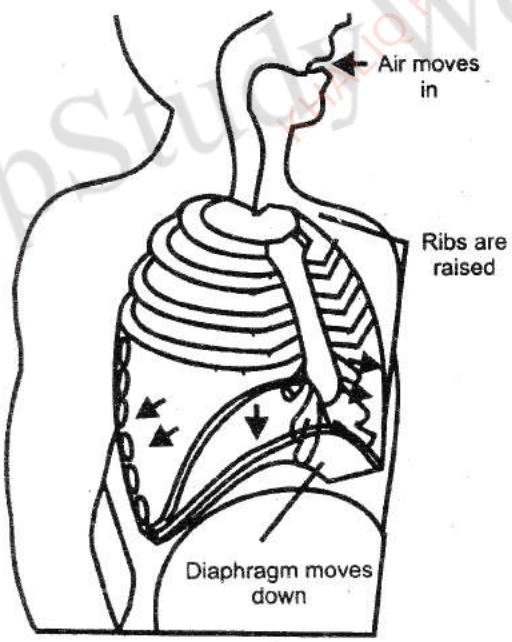

Figure: Steps of Inhalation

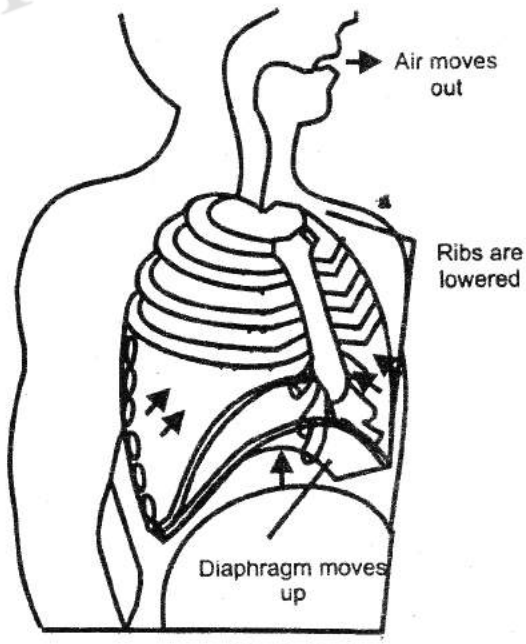

Figure: Steps of Exhalation

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2. Expiration or Exhalation:

After the gaseous exchange in the lungs, the impure air is expelled out in exhalation. The rib muscles relax bringing the ribs back to the original position. The diaphragm muscles also relax and it gets its raised dome shape. This reduces the space in the chest cavity and increases the pressure on lungs. The lungs contract and the air is expelled out of them.

Normal Breathing Rate:

Humans breathe 16-20 times per minute in normal circumstances i.e. at rest.

Control of Breathing Rate:

The rate of breathing is controlled by the respiratory centre in the brain. The respiratory centre is sensitive to the connect ration of carbon dioxide in the blood.

Stimulation of Respiratory Centre:

When we do exercise or some hard job our muscle cells carry out cellular respiration at greater rate. It results in the production of more carbon dioxide which is released in the blood. This greater than normal concentration of carbon dioxide stimulates the respiratory centre of brain.

Response by Respiratory Centre:

The respiratory centre sends messages to the rib muscles and diaphragm to increase the rate of breathing so that the excess carbon dioxide present in blood can be removed out of body.

Maximum Breathing Rate:

During exercise or other hard physical works the breathing rate may increase up to 30-40 times per minute.

Q. No. 4 Give a comparison between inspired and expired air.

COMPARISON BETWEEN THE INSPIRED AND THE EXPIRED AIR

Feature	Inspired Air	Expired Air
Amount of oxygen	21%	16%
Amount of carbon dioxide	0.04%	4%
Amount of nitrogen	79%	79%
Amount of water vapours	Variable	Saturated
Amount of dust particles	Variable	Almost none
Temperature	Variable	Almost equal to body temperature

Q. No. 5. Write a note on bronchitis.

BRONCHITIS

Definition:

The inflammation of the bronchi or bronchioles is called bronchitis.

Outcomes:

It results in excessive secretions of mucus into the tubes, leading to the swelling of tubular walls and narrowing of tubes.

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Causes:

It is caused by:

- Viruses
- Bacteria
- Exposure to chemical irritants (e.g. tobacco smoke)

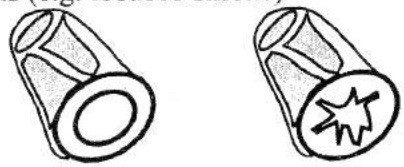


Figure: Normal Bronchus **Inflamed Bronchus**

Types of Bronchitis:

There are two major types of bronchitis i.e. acute and chronic.

Acute Bronchitis:

The acute bronchitis usually lasts about two weeks and patients recover with no permanent damage to the bronchi or bronchioles.

Chronic Bronchitis:

In chronic bronchitis, the bronchi develop chronic inflammation. It usually lasts for three months to two years. The majority of the people diagnosed with chronic bronchitis are 45 years of age or older.

Symptoms:

Symptoms of bronchitis include:

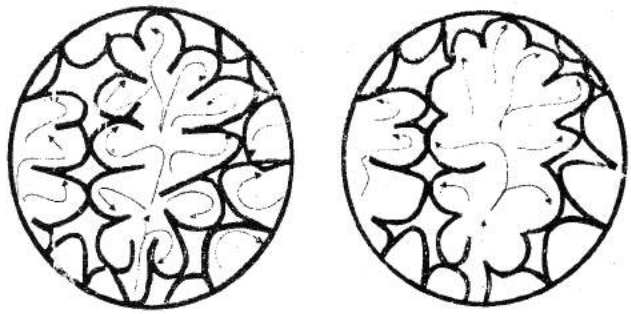
- Cough
- Mild wheezing
- Fever
- Chills
- Shortness of breath especially when doing hard job

Q. No. 6 Write a note on emphysema.

EMPHYSEMA

Introduction:

Emphysema is the destruction of the walls of the alveoli.



Normal **Emphysema**

Figure: The Alveoli

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Outcomes:

It results in larger sacs but with less surface area for gaseous exchange. As lung tissue breaks down, the lungs do not come back to their original shape after exhalation. So air cannot be pushed out and is trapped in the lungs.

Symptoms:

The symptoms of emphysema include:

- Shortness of breadth
- Fatigue
- Recurrent respiratory infections
- Weight loss

Appearance of Symptoms:

By the time the symptoms of emphysema appear, the patient has usually lost 50% to 70% of lung tissue.

Serious Complications:

The level of oxygen in blood may get so low that it causes serious complications.

Q. No. 7 Write a note on pneumonia.

PNEUMONIA

Introduction:

Pneumonia is an infection of lungs.

Double Pneumonia:

If this infection affects both lungs, it is called double pneumonia.

Common Cause:

The most common cause of pneumonia is a bacterium, *Streptococcus pneumoniae*.

Other Causes:

- Viral infections (influenza virus)
- Fungal infections

Mode of Infection:

When the causative organisms enter the alveoli, they settle there and grow in number. They break the lung tissues and the area becomes filled with fluid and pus.

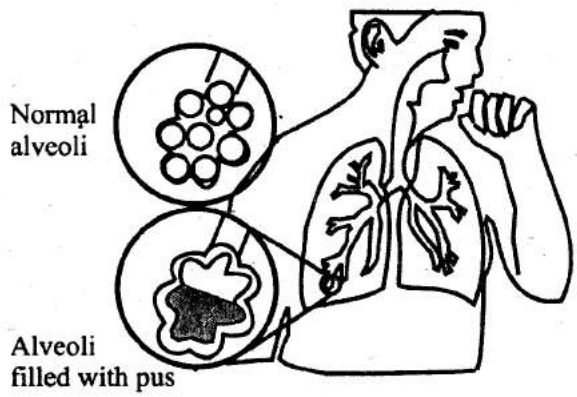

Figure: Pneumonia

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Symptoms:

The symptoms of pneumonia include:

- Cold
- High fever
- Shivering
- Cough
- Sputum production
- Short of breath

Skin Colour:

The patient's skin colour may change and become dusky or purplish. It is due to poor oxygenation of blood.

Prevention:

Vaccines are available to prevent pneumonia caused by *S. pneumoniae*.

Treatment:

Antibiotics are used in the treatment of this type of pneumonia.

Fatal Disease:

Prior to the discovery of antibiotics, one-third of pneumonia patients died from the infection.

Q. No. 8 Write a note on asthma.

ASTHMA

Introduction:

Asthma is a form of allergy.

Outcomes:

There is

- Inflammation of the bronchi
- More mucous production
- Narrowing of the airways

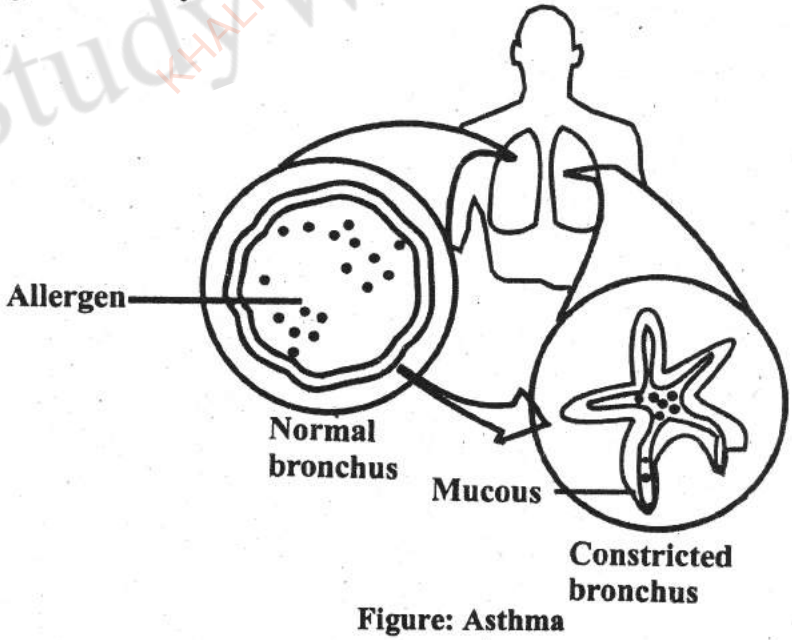

Figure: Asthma

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Sensitivity:

In asthma patients, the bronchi and bronchioles become sensitive to different allergens. When exposed to any of such allergens, the sensitive airways show immediate and excessive response of constriction. In this condition, the patient feels difficulty in breathing.

Allergens:

The agents that cause allergy are called allergens.

Examples:

- Dust
- Smoke
- Perfumes
- Pollens

Symptoms:

The symptoms of asthma vary from person to person. The major symptoms include:

- Shortness of breath (especially with exertion or at night)
- Wheezing (whistling sound when breathing out)
- Cough
- Chest tightness

Treatment:

The chemicals with ability to dilate the bronchi and bronchioles are used in the treatment of asthma. Such medicine is given in the form of inhalers.

Q. No. 9 Write a note on lung cancer.

LUNG CANCER

Introduction:

Lung cancer is a disease of uncontrolled cell divisions in the tissues of the lung. The cells continue to divide without any control and form tumours.

Malignant Tumours:

The cellular growth may also invade adjacent tissues beyond the lungs.

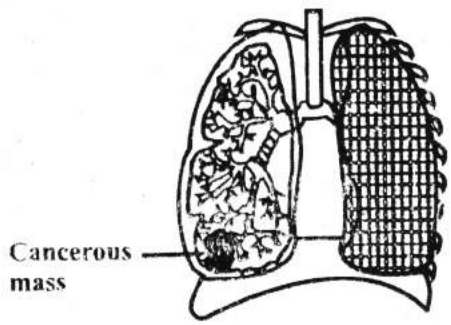

Figure: Lung Cancer

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Symptoms:

The most common symptoms are:

- Shortness of breath
- Coughing (including coughing up blood)
- Weight loss

Major Cause:

Smoking is the main cause of lung cancer. This risk of lung cancer is significantly lower in nonsmokers. Cigarette smoke contains over 50 known carcinogens.

Other Causes:

The main causes include:

- Carcinogens
- Ionizing radiation
- Viral infection

Passive smoking:

The inhalation of smoke from another's smoking is called passive smoking. It is also a cause of lung cancer. The smoke from the burning end of a cigarette is more dangerous than the smoke from the filter end.

Prevention:

Eliminating tobacco smoking is a primary goal in the prevention of lung cancer.

Role of WHO:

The World Health Organization has called for governments to stop tobacco advertising to prevent young people from taking up smoking.

Q. No. 10 Describe bad effects of smoking.

BAD EFFECTS OF SMOKING

Smoking is harmful due to the chemicals in cigarettes and smoke. Tobacco smoke contains over 4,000 different chemicals, out of which at least 50 are carcinogens and many are poisonous.

Misperception:

Many people think that lung cancer is the only smoking-related disease and it is the number one cause of death among smokers. But it is not right. Cigarette smoke affects the body from head to toe. Smokers have a much higher risk of developing a number of life threatening diseases.

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Carcinogenic:

Smoking may also lead to the cancers in:

- Kidneys
- Oral cavity
- Larynx
- Breast
- Bladder
- Pancreas

Damage to Respiratory System:

Many chemicals in tobacco smoke damage the air passageway, which lead to emphysema and other respiratory disorders.

Effect on Circulatory System:

Smoking also has effects on the circulatory system. The carbon monoxide present in tobacco smoke lessens the oxygen-carrying capacity of haemoglobin.

Cause of Arteiosclerosis:

Many other chemicals in smoke increase the production of blood platelets. When platelets are more than the normal numbers, the make the blood viscous and it can lead to arteriosclerosis.

Lung Infections:

Smokers are at greater risk of developing infections, particularly in the lungs. For example, smoking increases the risk of tuberculosis by two to four times, and of pneumonia by four times.

Effect on Teeth:

Smoking is also responsible for weakening and staining the teeth. Tooth loss is 2 to 3 times higher in smokers than in non-smokers.

Effect of Nicotine:

Nicotine is a powerful poison and was widely used as an insecticide in the past. When inhaled through tobacco smoking, it reaches our circulatory system and not only hardens the walls of the arteries but also damages the brain tissues.

Passive Smokers:

Non-smokers who are exposed to second hand smoke (passive smoke) at home or work increase their heart disease risk by 25-30% and their lung cancer risk by 20-30%.

Effect of Smoking on Social Life:

Smoking affects the social life of a person. Smokers may face social unacceptance because other people may not want to be exposed to other's smoke.

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REVIEW QUESTION:

MULTIPLE CHOICE QUESTIONS

1. **The process of gaseous exchange involves:**
   (a) Breakdown of C-H bonds to yield energy.
   (b) Physical movements that take air in and out of body.
   (c) Getting oxygen from the air and removing carbon dioxide.
   (d) Transport of oxygen by the blood to different parts of the body.
2. **Most of the gaseous exchange in a leaf occurs through:**
   (a) Stomata (b) General surface (c) Cuticle (d) Lenticels
3. **How many bronchi are there in the air passageway?**
   (a) One (b) Two (c) Many (d) None
4. **Where does the gaseous exchange occur in humans?**
   (a) Pharynx (b) Trachea (c) Bronchi (d) Alveoli
5. **Which structure actively helps in taking the air out of lungs?**
   (a) Nasal cavity (b) Bronchus (c) Bronchiole (d) Diaphragm
6. **The primary chemical stimulus for breathing is the concentration of:**
   (a) Carbon dioxide in blood (b) Oxygen in blood
   (c) Carbon dioxide in muscles (d) Oxygen in muscles
7. **Point out the FALSE statement about respiration:**
   (a) Gases can easily pass through the walls of the alveoli
   (b) Gas exchange in lungs is very efficient because lungs provide large surface area
   (c) In emphysema the walls of alveoli break and there is more surface area
   (d) Dust particles can damage the lung by irritating the inner alveoli surface.
8. **A disease involving the breakdown of air sacs of the lungs is:**
   (a) Pneumonia (b) Bronchitis (c) Asthma (d) Emphysema
9. **Which process does not occur in the nasal cavity?**
   b) Trapping of large dust particles (b) Humidification of the inhaled air
   (c) Warming of the inhaled air (d) Exchange of gases
10. **What type of blood vessels surrounds the alveoli?**
    (a) Artery (b) Arteriole (c) Capillary (d) Vein

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	c	2	a	3	b	4	d	5	d
6	a	7	c	8	d	9	d	10	c

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SHORT QUESTIONS

Q.2 Write short answer to the following questions.

1. Differentiate between breathing and cellular respiration.

DIFFERENCE BETWEEN BREATHING AND CELLULAR RESPIRATION

Breathing	Cellular Respiration
Definition: The process through which animals take air in their bodies to get oxygen from it and then give out the air for getting rid of carbon dioxide is called breathing.	***Definition:*** The process in which the carbon-hydrogen bonds in the food are broken by oxidation reduction reactions and the energy is transformed into ATP is called cellular respiration.
Energy Production: No energy is produced.	***Energy Production:*** Energy is produced in the form of ATP.
Level: It occurs at organ system level and respiratory system is involved	***Level:*** It occurs at cell level.

2. Trace the path of air from the nasal cavity to the alveoli.

PATH OF AIR FROM THE NASAL CAVITY TO THE ALVEOLI

1. Nasal cavity
2. Internal nostrils
3. Pharynx
4. Larynx
5. Trachea
6. Bronchi
7. Bronchioles
8. Alveolar ducts
9. Alveoli

3. How will you differentiate between a stoma and a lenticel?

DIFFERENCE BETWEEN STOMA AND LENTICEL

Stoma	Lenticel
Formation A pore opens when guard cells become turgid.	***Formation:*** Develops from cambium.
Presence: Present on lower epidermis of leaf.	***Presence:*** Present on stem of dicot stem.
Function: Helps in gaseous exchange during night	***Function:*** Helps gaseous exchange during night

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UNDERSTANDING THE CONCEPT

1. **How do the different parts of the plant body exchange gases with the environment?**

Consult Long Question No. 1

2. **Write down the steps of inhalation and exhalation.**

Consult Long Question No. 3

3. **State the signs and symptoms, causes and treatments of bronchitis, emphysema and pneumonia.**

Consult Long Question No. 5, 6 and 7

4. **How does the tobacco smoke damage the respiratory system?**

DAMAGE OF TOBACCO SMOKE TO RESPIRATORY SYSTEM

Smoking is harmful due tc the chemicals in cigarettes and smoke. Tobacco smoke contains over 4,000 different chemicals, out of which at least 50 are carcinogens and many are poisonous.

Misperception:

Many people think that lung cancer is the only smoking-related disease and it is the number one cause of death among smokers. But it is not right. Cigarette smoke affects the body from head to toe. Smokers have a much higher risk of developing a number of life threatening diseases.

Damage to Respiratory System:

Many chemicals in tobacco smoke damage the air passageway, which lead to emphysema and other respiratory disorders.

Lung Infections:

Smokers are at greater risk of developing infections, particularly in the lungs. For example, smoking increases the risk of tuberculosis by two to four times, and of pneumonia by four times.

Passive Smokers:

Non-smokers who are exposed to second hand smoke (passive smoke) at home or work increase their heart disease risk by 25-30% and their lung cancer risk by 20-30%.

Effect of Smoking on Social Life:

Smoking affects the social life of a person. Smokers may face social unacceptance because other people may not want to be exposed to other's smoke.

THE TERMS TO KNOW

Alveolar duct:
Fine tubules at the end of bronchioles, open into alveoli

Alveolus:
A sac like structure present next to the alveolar duct

Asthma:
An inflammation of the bronchi that causes narrowing and swelling of the airways

Breathing:
The process through which animals take air in their bodies to get oxygen and then give out the air for getting rid of carbon dioxide

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Bronchioles:
Fine tubules formed by the division of bronchi

Bronchitis:
Inflammation of the bronchi or bronchioles

Bronchus:
The part of the air passage way formed by the division of trachea

Diaphragm:
The muscular structure that forms the floor of the chest cavity

Emphysema:
A disease in which the walls of the alveoli are destroyed

Exhalation:
The phase of breathing in which air is expelled from the lungs

Gaseous exchange:
Taking in of oxygen and giving out of carbon dioxide by organisms

Inhalation:
The phase of breathing in which air is drawn into the lungs

Larynx:
The part of the air passage way between pharynx and trachea

Lenticels:
Pores in the bark of woody stems and mature roots

Nasal cavity:
Hollow space in the nose; opens to the outside through nostrils; divided into two portions by a wall

Nostril:
The opening of the nasal cavity

Pneumonia:
The infection of one or both lungs; caused by specific bacteria, viruses or fungi; the infected part of the lung becomes filled with fluid and pus

Trachea:
Wind pipe; part of the air passage way between larynx and bronchi

Vocal cords:
Two pairs of fibrous bands called vocal cards are stretched along the larynx. The vocal cards vibrate when the air passes through them. This vibration produces sounds.

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CHAPTER 11

HOMEOSTASIS

MULTIPLE CHOICE QUESTIONS

Each question has four possible answers. Circle the correct answer.

(1) The maintenance of the internal conditions of the body at equilibrium, despite changes in the external environment is called:
(a) Excretion (b) Thermoregulation (c) Osmoregulation (d) Homeostasis

(2) Which one is produced as a by product of photosynthesis?
(a) Glucose (b) Oxygen (c) Carbon dioxide (d) Water

(3) The appearance of drops of water on the tips or edges of leaves is called:
(a) Excretion (b) Transpiration (c) Guttation (d) Respiration

(4) The waste product of coniferous trees:
(a) Resins (b) Gums (c) Latex (d) Mucilage

(5) The waste product of rubber plant:
(a) Resins (b) Gums (c) Latex (d) Mucilage

(6) Resin is the waste product of:
(a) Coniferous trees (b) Keekar (c) Rubber plant (d) Lady finger

(7) Latex is the waste product of:
(a) Coniferous trees (b) Keekar (c) Rubber plant (d) Lady finger

(8) Example of hydrophytes:
(a) Cactus (b) Sea grass (c) Water lily (d) All of these

(9) Example of halophyte:
(a) Cactus (b) Sea grass (c) Water lily (d) Both a and b

(10) Which organ performs role in the maintenance of body temperature?
(a) Heart (b) Pancreas (c) Spleen (d) Skin

(11) Shape of human kidney:
(a) Oval (b) Circular (c) Triangular (d) Bean shaped

(12) Dimensions of human kidney:
(a) 10cm long, 8cm wide, 3 cm thick (b) 10cm long, 5cm wide, 4cm thick
(c) 10cm long, 5cm wide, 5 cm thick (d) 10cm long, 4cm wide, 5cm thick

(13) Which side of kidney faces vertebral column?
(a) Concave (b) Convex (c) Both a and b (d) None of these

(14) The base of ureter:
(a) Renal cortex (b) Renal medulla (c) Pyramids (d) Renal pelvis

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(15) Number of nephrons is each kidney:
(a) One million (b) Two million (c) Three million (d) Four million

(16) Capillaries of glomerulus arise from:
(a) Afferent arteriole (b) Efferent arteriole (c) Both a and b (d) None of these

(17) Main function of kidney:
(a) Respiration (b) Photosynthesis (c) Urine formation (d) Coordination

(18) The ascending limb of loop of Henle allows the reabsorption of:
(a) Salts (b) Water (c) Both a and b (d) None of these

(19) In urine chemical composition, the amount of sodium ions:
(a) 0.34 g/l (b) .17 g / R (c) 2.29 g / L (d) 1.83 g/L

(20) In urine chemical composition, the amount of potassium ions:
(a) 1.250 g/l (b) .750 g / l (c) 1.750 g / l (d) 2.250 g/L

(21) The regulation of the concentration of water and salts is blood and other body fluids:
(a) Excretion (b) Respiration (c) Osmoregulation (d) Thermoregulation

(22) When there is excess water in body fluids, kidneys form:
(a) Hypotonic urine (b) Hypertonic urine (c) Isotonic urtne (d) Both a and b

(23) The kidney stones may be formed by the deposition of crystals of:
(a) Calcium oxalate (b) Calcium phosphate (c) Ammonium phosphate (d) All of these

(24) Symptoms of kidney stones include:
(a) Pain in kidney (b) Vomiting (c) Frequent urination (d) All of these

(25) Date of birth of Abu Nasr al-Farabi:
(a) 871 (b) 872 (c) 873 (d) 874

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ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	b	3	c	4	a	5	l
6	a	7	c	8	c	9	b	10	d
11	d	12	b	13	a	14	d	15	a
16	a	17	c	18	a	19	b	20	b
21	c	22	a	23	d	24	d	25	b

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SHORT QUESTIONS

Q. No. 1 What is the metabolic waste?

METABOLIC WASTE

Any material that is produced during body metabolism and that may harm the body is called metabolic waste.

Q. No. 2 Define transpiration.

Transpiration

The loss of water from plant surface in the form of vapours is called transpiration.

Q. No. 3 What is the difference between transpiration and guttation?

DIFFERENCE BETWEEN TRANSPIRATION AND GUTTATION

Transpiration	Guttation
Definition: The loss of water from plant surface in the form of vapours is called transpiration.	***Definition:*** The appearance of drops of water on the tips or edges of leaves is called guttation.
Time: It takes place in day time.	***Time:*** It takes place at night time.
Types: It has three types: • Stomatal transpiration • Cuticular transpiration • Lenticular transpiration	***Types:*** It has no other type.
Examples: All plants	***Examples:*** Some grasses

Q. No. 4. Why guttation should not be confused with due?

CONFUSION BETWEEN GUTTATION AND DUE

Guttation is not to be confused with due, which condenses from the atmosphere onto the plant surface.

Q. No. 5 Write names of excretory products in plants.

EXCRETORY PRODUCTS IN PLANTS

The following are the excretory products in plants:

- Oxygen
- Carbon dioxide
- Water
- Calcium oxalate

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- Latex
- Resins
- Gums
- Mucilage

Q. No. 6 Which organs work for homeostasis in humans?

ORGANS FOR HOMEOSTASIS IN HUMANS

The following organs work for homeostasis in humans:

Lungs:

Lungs remove excess carbon dioxide and keep it in balance.

Skin:

Skin performs role in the:

- Maintenance of body temperature
- Removal of excess water and salts

Kidneys:

The kidneys filter the following from the blood and form urine:

- Excess water
- Salts
- Urea
- Uric acid

Q. No. 7 What are 'Goosebumps'?

GOOSEBUMPS

Formation:

Contraction of small muscles attached to hairs forms 'Goosebumps'.

Function:

It creates an insulation blanket of warm air.

Q. No. 8 Explain role of lungs in excretion of carbon dioxide.

ROLE OF LUNGS IN EXCRETION OF CARBON DIOXIDE

Lungs maintain the concentration of carbon dioxide in the blood. Our cells produce carbon dioxide when they perform cellular respiration. From cells, carbon dioxide diffuses into tissue fluid and from here into blood. Blood carries carbon dioxide to lungs from where it is removed in air.

Q. No. 9 Which materials do not filter through the glomerular capillaries and why?

The following materials are not filtered through the glomerular capillaries:

- Blood cells
- Proteins

Reason:

They are relatively larger in size.

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Q. No. 10 What is the percentage of originally filtered volume of glomerular filtrate that forms urine?

At the final stage, urine is only 1% of the originally filtered volume.

Q. No. 11 What volume of urine is produced by an average human adult per day?

The typical volume of urine produced by an average adult is around 1.4 liters per day.

Q. No. 12 What is the composition of the kidney stones?

COMPOSITION OF KIDNEY STONES

The kidney stones are composed of the crystals of the following salts:

- Calcium oxalate
- Calcium phosphate
- Ammonium phosphate
- Uric acid

Q. No. 13 What causes the materials to move from glomerular capillaries to Bowman's capsule?

Blood pressure causes the materials to move from glomerular capillaries to Bowman's capsule

Q. No. 14 What is the contribution of Abu Nasr al-Farabi?

CONTRIBUTION OF ABU NASR AL-FARABI

Period: 872-951

Contribution:

He was a prominent scientist who wrote many books that contained information about kidney diseases.

Q. No. 15 What is the contribution of Abu al-Qasim al-Zahrawi?

CONTRIBUTION OF ABU AL-QASIM AL-ZAHRAWI

He is also known as Albucasis.

Period: 936-1013

Contribution:

He is considered to be Islam's greatest surgeon who invented many surgical procedures including the surgical removal of stones from the urinary bladder.

Publication:

His encyclopedia, Al-Tasrif ("The Method"), contained over 200 surgical medical instruments he personally designed.

Q. No. 16 Define kidney failure.

KIDNEY FAILURE

The complete or partial failure of kidney to function is called kidney failure.

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Q. No. 17 What are the leading causes of kidney failure?

LEADING CAUSES OF KIDNEY FAILURE

The following are the leading causes of kidney failure:

- Diabetes mellitus
- Hypertension

Q. No. 18 Enlist the causes of kidney failure.

CAUSES OF KIDNEY FAILURE

The following are the causes of kidney failure:

- Diabetes mellitus
- Hypertension
- Sudden interruption in the blood supply to kidney
- Drug overdosage

Q. No. 19 What are the symptoms of kidney failure?

SYMPTOMS OF KIDNEY FAILURE

The following are the symptoms of kidney failure:

- High level of urea in blood
- High level of waste materials in blood
- Vomiting
- Nausea
- Weight loss
- Frequent urination
- Blood in urine
- Swelling of legs, feet and face
- Shortness of breath

Q. No. 20 How is kidney failure treated?

TREATMENT OF KIDNEY FAILURE

The kidney failure is treated with:

- Dialysis
- Kidney transplant

Q. No. 21 What problems may arise after kidney transplant.

PROBLEMS AFTER TRANSPLANT

Problems after a transplant may include:

- Transplant rejection
- Infections
- Imbalances in body salts
- Bone problems
- Ulcers

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LONG QUESTIONS

Q. No. 1 What is homeostasis? Explain with examples.

HOMEOSTASIS

Definition:

The maintenance of the internal conditions of the body at equilibrium, despite changes in the external environment is called homeostasis.

Examples:

Body Temperature:

The core temperature of human body remains at about 37°C despite fluctuations in the surrounding air temperature.

Blood Glucose Level:

The blood glucose level remains about 1g per litre despite eating a meal rich in carbohydrates.

Importance of Homeostasis:

Body cells need the internal environment in which conditions do not change much. Stable internal conditions are important for the efficient functioning of enzymes.

Types of homeostasis

The following are some types of homeostasis:

1. Osmoregulation
2. Thermoregulation
3. Excretion

1. Osmoregulation:

Definition:

The maintenance of the amounts of water and salts in body fluids (i.e. blood and tissue fluids) is called homeostasis.

Importance:

The relative amounts of water and salts in body fluids and inside cells control the processes of diffusion and osmosis, which are essential for the functioning of cells.

2. Thermoregulation:

Definition:

The maintenance of internal body temperature is called thermoregulation.

Example:

The enzymes of body work best at particular temperatures (optimum temperature). Any change in body temperature may affect the functioning of enzymes.

3. Excretion:

The elimination of metabolic wastes from body to maintain the internal conditions at equilibrium is called excretion.

Q. No. 2 How extra carbon dioxide and oxygen is removed by plants?

REMOVAL OF EXTRA CARBON DIOXIDE

During Day Time:

In daytime, the carbon dioxide produced during cellular respiration is utilized in photosynthesis and hence it is not a waste product.

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During Night Time:

At night, it is surplus because there is no utilization of carbon dioxide. It is removed from the tissue cells by diffusion.

Leaves and Stems:

In leaves and young stems, carbon dioxide escapes out through stomata.

Roots:

In young roots, carbon dioxide diffuses through the general root surface, especially through root hairs.

Removal of Oxygen

Oxygen is produced in mesophyll cells only during daytime, as a by-product of photosynthesis. After its utilization in cellular respiration, the mesophyll cells remove the extra amount of oxygen through stomata.

Q. No. 3 How do plants get rid of extra water?

REMOVAL OF EXTRA WATER

Plants obtain water from soil and it is also produced in the body during cellular respiration. Plants store large amount of water in their cells for turgidity.

Extra water is removed from plant body by transpiration.

Transpiration:

The loss of water from plant surface in the form of vapours is called transpiration.

During Night Time:

At night, transpiration usually does not occur because most plants have their stomata closed.

Guttation:

The appearance of drops of water on the tips or edges of leaves is called guttation.

Explanation:

If there is high water content is soil, water enters the roots and is accumulated in xylem vessels. Some plants force this water through special pores, present at leaf tips or edges, and form drops.

Example:

- Some grasses

Guttation versus Dew:

Guttation is not to be confused with dew, which condenses from the atmosphere onto the plant surface.

Q. No. 4 Explain the removal of metabolic waste products in plants.

REMOVAL OF METABOLIC WASTES

Metabolic waste:

Any material that is produced during body metabolism and that may harm the body is called metabolic waste.

Plants deposit many metabolic wastes in their bodies as harmless insoluble materials.

Calcium Oxalate:

Calcium oxalate is deposited in the form of crystals in the leaves and steams of many plants.

- Tomato

Resins:

- Coniferous trees

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Gums:

- Keekar

Latex:

- Rubber plant

Mucilage:

- Carnivorous plants
- Ladyfinger

Leaf Fall:

In trees which shed their leaves yearly, the excretory products are removed from body during leaf fall.

Q. No. 5 Explain osmotic adjustments in plants.

OSMOTIC ADJUSTMENTS IN PLANTS

On the basis of the available amount of water and salts, plants are divided into three groups.

1. Hydrophytes
2. Xerophytes
3. Halophytes

1. Hydrophytes:

Occurrence:

Hydrophytes are the plants which live completely or partially submerged in freshwater. Such plants do not face the problem of water shortage.

Mechanism:

They have developed mechanisms for the removal of extra water from their cells. Hydrophytes have broad leaves with a large number of stomata on their upper surface. This characteristic helps them to remove the extra amount of water.

Example:

- Water lily

2. Xerophytes:

Occurrence:

Xerophytes live in dry environments.

Cuticle:

They possess thick and waxy cuticle over their epidermis, to reduce water loss from internal tissues.

Number of Stomata:

They have less number of stomata to reduce the rate of transpiration.

Deep Roots:

Such plants have deep roots to absorb maximum water from soil.

Succulent Stems:

Some xerophytes have special parenchyma cells in stems or roots in which they store large quantities of water. This makes their stems or roots wet and juicy, called succulent organs.

Example:

- Cacti (singular cactus)

3. Halophytes:

Occurrence:

Halophytes live in sea waters and are adapted to salty environments.

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Entry of Salts:

Salts enter in the bodies of such plants due to their higher concentration in sea water.

Removal of Water:

Water tends to move out of their cells into the hypertonic sea water.

Mechanism:

When salts enter into cells, plants carry out active transport to move and hold large amount of salt in vacuoles. Salts are not allowed to move out through the semi-permeable membranes of vacuoles. So the sap of vacuoles remains even more hypertonic than sea water. In this way, water does not move out of cells.

Example:

- Many sea grasses

Q. No. 6 Explain role of skin in homeostasis.

ROLE OF SKIN IN HOMEOSTASIS

Skin performs role in:

- The maintenance of body temperature
- Removal of excess water and salts

Structure of Skin:

Our skin consists of two layers:

1. Epidermis
2. Dermis

1. Epidermis:

The outer protective layer without blood vessels is called epidermis.

2. Dermis:

The inner layer of skin which consists of:

- Blood vessels
- Sensory nerve endings
- Sweat glands
- Oil glands
- Hairs
- Fat cells

FUNCTIONS

Role in Insulation:

Function of Fat Cells:

The thin layer of fat cells in the dermis insulates the body.

Function of Hairs:

- Contraction of small muscles attached to hairs forms 'Goosebumps'. It creates an insulation blanket of warm air.

Role in Cooling:

Skin helps in providing cooling effect when sweat is produced by sweat glands and excess body heat escapes through evaporation.

Removal of Metabolic Wastes:

The following metabolic wastes are also removed in sweat:

- Excess water
- Salts
- Urea
- Uric acid

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Q. No. 7 Describe the human urinary system.

HUMAN URINARY SYSTEM

The human urinary system consists of the following:

1. A pair of kidneys
2. A pair of ureters
3. A urinary bladder
4. Urethra

1. A Pair of Kidneys:

A pair of kidneys is present against the back wall of abdominal cavity just below diaphragm, one on either side of the vertebral column. The kidneys filter blood to produce urine.

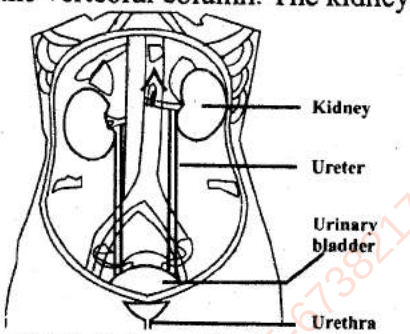

Figure: The Urinary System of Humans

2. A Pair of Ureters:

The ureters carry urine from kidneys to the urinary bladder.

3. A Urinary Bladder:

The urinary temporarily stores urine until it is released from the body.

4. Urethra:

Urethra is the tube that carries urine from urinary bladder to the out side of the body.

Q. No. 8 Describe the structure of kidney.

HUMAN KIDNEY

Colour:

The kidneys are dark-red in colour.

Shape:

The kidneys are bean shaped.

Dimensions:

Each kidney is 10 cm long, 5 cm wide and 4 cm thick.

Weight:

Each kidney weighs about 27 grams.

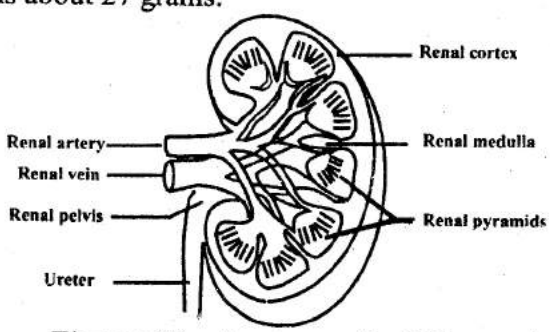

Figure: The Anatomy of a Kidney

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Location:

The kidneys are placed against the back wall of abdominal cavity just below diaphragm, one on either side of vertebral column. The left kidney is a little higher than the right. The concave side of the kidney faces vertebral column.

Protection:

The kidneys are protected by the last two ribs.

STRUCTURE OF KIDNEY

Each kidney consists of the following structures:

1. Hilus
2. Renal Cortex
3. Renal Medulla
4. Renal Pyramids
5. Renal Pelvis

1. Hilus:

There is a depression near the centre of the concave area of the kidney called hilus. This is the area of kidney through which ureter leaves kidney and the following structures enter and leave kidney:

- Blood vessels
- Lymphatic vessels
- Nerves

2. Renal Cortex:

Renal cortex is the outer part of kidney. It is dark red in colour.

3. Renal Medulla:

Renal medulla is the inner part of the kidney. It is pale red in colour.

4. Renal Pyramids:

Renal medulla consists of several cone shaped areas called renal pyramids.

5. Renal Pelvis:

Renal pyramids project into a funnel-shaped cavity called renal pelvis, which is the base of ureter

Q. No. 9 Describe the structure of nephron.

NEPHRON

Definition:

The functional unit of the kidneys is called nephron.

Number:

There are over one million nephrons in each kidney.

STRUCTURE OF NEPHRON

There are two parts of a nephron

1. Renal Corpuscle
2. Renal Tubule

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1. **Renal Corpuscle:**

The renal corpuscle is not tubular. It consists of two parts:

Glomerulus:

It is the network of capillaries. The capillaries of the glomerulus arise from the afferent arteriole and join to form the efferent arteriole.

Bowman's Capsule:

Bowman's capsule is a cup-shaped structure that encloses glomerulus.

2. **Renal Tubule:**

The renal tubule is the part of the nephron which starts after Bowman's capsule. It consists of three parts:

Proximal Convoluted Tubule:

The first portion of the renal tubule is called proximal convoluted tubule.

Loop of Henle:

Next portion of renal tubule is U-shaped and is called the Loop of Henle.

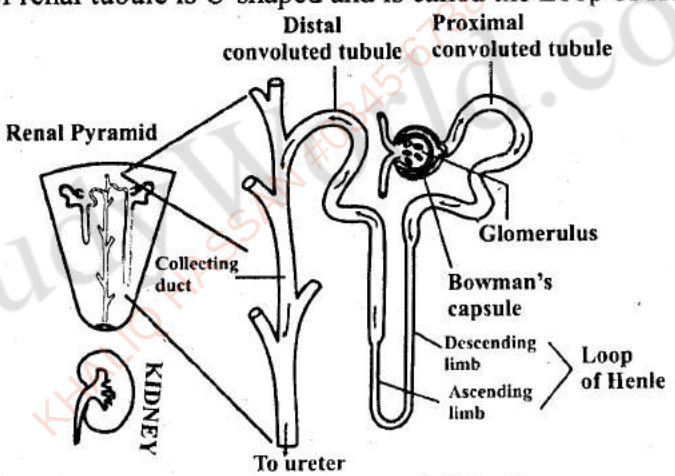

Figure: The Structure of a Nephron

Distal Convoluted Tubule:

The last portion of the renal tubule is the distal convoluted tubule.

Collecting Duct:

The distal convoluted tubules of many nephrons open in a single collecting duct.

Papillary Ducts:

Many collecting ducts join together to form several hundred papillary ducts which drain into renal pelvis.

Q. No. 10 Describe the functioning of kidney.

FUNCTIONING OF KIDNEY

The main function of kidney is urine formation, which takes place in the following three steps:

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1. **Pressure Filtration:**

This is the first step. When blood enters the kidney via the renal artery, it goes to many arterioles, and then to the glomerulus. The pressure of blood is very high and so most of the water, salts, glucose and urea of blood is forced out of glomerular capillaries.

Glomerular Filtrate:

The material that passes into the Bowman's capsule from the glomerulus after pressure filtration is called glomerular filtrate.

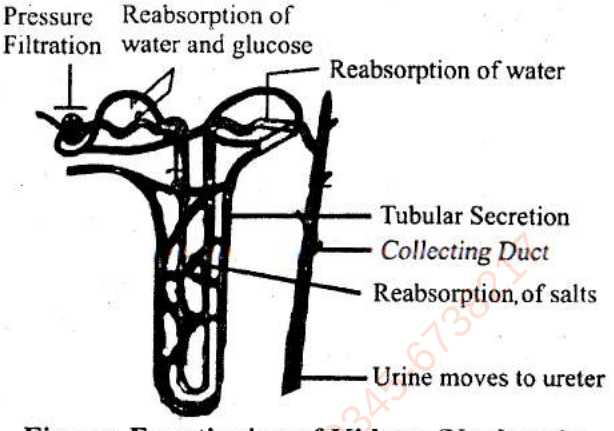

Figure: Functioning of Kidney (Nephron)

2. **Selective Re-absorption:**

The second step is the selective re-absorption. In this step about 99% of the glomerular filtrate is reabsorbed into the blood capillaries surrounding renal tubule. The selective re-absorption occurs through:

- Osmosis
- Diffusion
- Active transport

Proximal Convoluted Tubule:

Some water and most of the glucose is reabsorbed form the proximal convoluted tubule. Here, salts are reabsorbed by active transport and then water follows by osmosis.

Loop of Henle:

The descending limb of Loop of Henle allows the re-absorption of water while the ascending limb of Loop of Henle allows the re-absorption of salts.

Distal Convoluted Tubule:

The distal convoluted tubule again allows the re-absorption of water into the blood.

3. **Tubular Secretion:**

The third step is the tubular secretion. Different ions, creatinine, urea etc. are secreted from blood into the filtrate in renal tubule. This is done to maintain blood at a normal pH (7.35 to 7.45).

Urine:

After all these mentioned steps, the filtrate present in renal tubules is known as urine. It moves into collecting ducts and then into pelvis.

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Q. No. 11 Describe the normal chemical composition of urine.

NORMAL CHEMICAL COMPOSITION OF URINE

Ingredients	Quantity
Water	95%
Urea	9.3 g/l
Chloride ions	1.87 g/l
Sodium ions	1.17 g/l
Potassium ions	0.750 g/l
Other ions and compounds	Variable amounts

Q. No. 12 Describe the osmoregulatory function of kidney.

OSMOREGULATORY FUNCTION OF KIDNEY

Osmoregulation:

The regulation of the concentration of water and salts in blood and other body fluids is called osmoregulation.

Importance:

Kidneys play important role in osmoregulation by regulating the water contents of blood. It is an important process as excessive loss of water concentrates the body fluids whereas excess intake of water dilutes them.

Greater Water Potential:

When there is excess water in body fluids, kidneys form dilute (hypotonic) urine. For this purpose, kidneys filter more water from glomerular capillaries into Bowman's capsule. Similarly less water is reabsorbed and abundant dilute urine is produced. It brings down the volume of body fluids to normal.

Lesser Water Potential:

When there is shortage of water in body fluids, kidneys filter less water from glomerular capillaries and the rate of reabsorption of water is increased. Less filtration and more reabsorption produce small amount of concentrated (hypertonic) urine. It increases the volume of body fluids to normal.

Hormonal Control:

This whole osmoregulatory process of kidney is under hormone control.

Q. No. 13 Write a note on kidney stones.

KIDNEY STONES

Formation:

When urine becomes concentrated crystals of the following salts are formed:

- Calcium oxalate
- Calcium phosphate
- Ammonium phosphate
- Uric acid

Such large crystals can not pass in urine and form hard deposits called kidney stones.

Other Organs:

Most stones start in kidney, some may travel to ureter or urinary bladder.

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Causes:

The major causes of kidney stones are:

- Age
- Diet (containing more green vegetables, salts, vitamins C and D)
- Recurring urinary tract infections
- Less intake of water
- Alcohol consumption

Symptoms:

The symptoms of kidney stones include:

- Severe pain in kidney or in lower abdomen
- Vomiting
- Frequent urination
- Foul-smelling urine
- Urine with blood and pus

Treatment:

The treatment of kidney stones includes:

Excessive Water Intake:

About 90% of all kidney stones can pass through the urinary system by drinking plenty of water.

Surgical Treatment:

In surgical treatment, the affected area is opened and stone(s) are removed.

Lithotripsy:

Lithotripsy is another method for the removal of kidney stones. In this method, non-electrical shock waves from outside are bombarded on the stones in the urinary system. Waves hit the dense stones and break them. Stones become sand-like and are passed through urine.

Q. No. 14 Write a note on dialysis.

DIALYSIS

Definition:

The cleaning of blood by artificial ways is called dialysis.

METHODS OF DIALYSIS

There are two methods of dialysis

1. Peritoneal Dialysis
2. Haemodialysis

1. Peritoneal Dialysis:

Peritoneum:

In this type of dialysis, the dialysis fluid is pumped for a time into the peritoneal cavity which is the space around gut. This cavity is lined by peritoneum. Peritoneum contains blood vessels.

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Extraction of Waste Materials:

When we place dialysis fluid is peritoneal cavity, waste materials from peritoneal blood vessels diffuse into the dialysis fluid, which is then drained out.

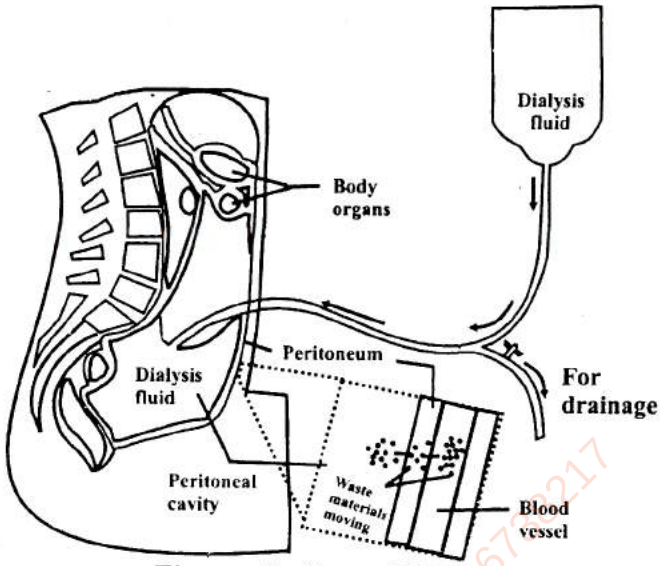


Figure: Peritoneal Dialysis

Duration:

This type of dialysis can be performed at home, but must be done every day.

2. Haemodialysis:

Dialyzer:

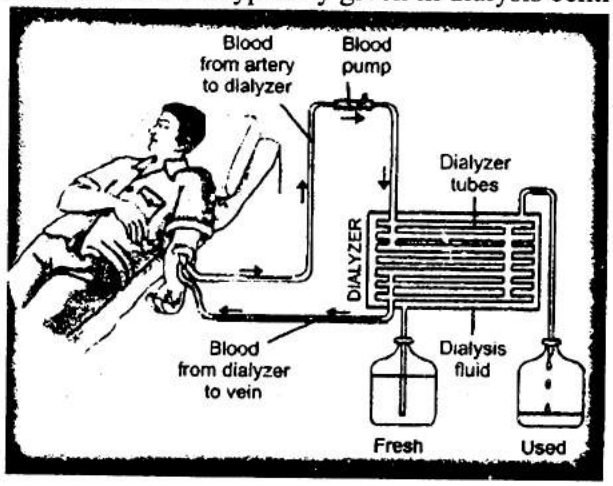
In haemodialysis, patient's blood is pumped through an apparatus called dialyzer. The dialyzer contains long tubes, the walls of which act as semi-permeable membranes.

Extraction of Waste Materials:

Blood flows through the tubes while the dialysis fluid flows around the tubes. Extra water and wastes move from blood into the dialysis fluid. The cleansed blood is then returned back to body.

Duration:

The haemodialysis treatments are typically given in dialysis centres three times per week.



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Q. No. 15 Write a note on kidney transplant.

KIDNEY TRANSPLANT

Problem with Dialysis:

Dialysis needs to be repeated after every few days and is unpleasant for patients and attendants.

End Stage Treatment:

Another treatment for the end-stage kidney failure is kidney transplantation.

Definition:

The replacement of patient's damaged kidney with a donor healthy kidney is called kidney transplant.

Donors:

Kidney may be donated by a deceased donor or living donor. The donor may or may not be a relative of the patient.

Matching:

Before transplant, the tissue proteins of donor and patient are matched.

Transplantation:

The donor's kidney is transplanted in patient's body and is connected to the patient's blood and urinary system.

Average Life Time:

The average lifetime for a donated kidney is ten to fifteen years.

Failure of Transplant:

When a transplant fails, the patient may be given a second kidney transplant. In this situation, the patient is treated through dialysis for some intermediary time.

Problems after Transplant:

Problems after a transplant may include:

- Transplant rejection
- Infections
- Imbalances in body salts
- Bone problems
- Ulcers

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REVIEW QUESTION:

MULTIPLE CHOICE QUESTIONS

Q.1 Each question has four possible answers. Circle the correct answer.

1. **The human urinary system consists of:**
   (a) Rectum, lungs, kidneys, ureters (b) Kidneys, ureters, urinary bladder
   (c) Skin, liver, lungs, kidneys (d) Kidneys, ureters, urinary bladder, urethra
2. **Which organ is responsible for filtering the blood?**
   (a) Intestine (b) Brain
   (c) Stomach (d) Kidney
3. **The tube between kidney and urinary bladder is the:**
   (a) Ureter (b) Urethra
   (c) Renal tubule (d) Nephron
4. **'Body balance' of water, salts, temperature and glucose is termed as:**
   (a) Excretion (b) Tubular secretion
   (c) Homeostasis (d) Re-absorption
5. **Which is the correct order for the path taken by urine after it leaves the kidneys?**
   (a) Urethra, bladder, ureters (b) Bladder, ureters, urethra
   (c) Ureters, bladder, urethra (d) Bladder, urethra, ureters
6. **What is the function of the ureter?**
   (a) To store urine (b) To carry urine from the kidney to the bladder
   (c) To carry urine out of the body (d) To remove waste from the blood
7. **What waste products are excreted by kidneys?**
   (a) Urea, water & salts (b) Salts, water and carbon dioxide
   (c) Urea & water (d) Urea & salts
8. **The two main functions of sweat are:**
   (a) To keep the body cool and to remove excess proteins
   (b) To keep the body warm and to filter the blood
   (c) To filter the blood and to remove waste products
   (d) To remove waste products and to cool the body
9. ***Which* would NOT be present in the filtrate entering the Bowman's capsule of nephron?**
   (a) Water (b) Calcium ions
   (c) Blood cells (d) Urea
10. **During peritoneal dialysis, the waste materials move from:**
    (a) The abdomen to the dialysis fluid
    (b) The dialysis fluid to the peritoneum blood vessels
    (c) The peritoneum blood vessels to the dialysis fluid
    (d) The dialysis fluid to the abdomen

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	d	3	a	4	c	5	c
6	b	7	a	8	d	9	c	10	c

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SHORT QUESTIONS

1. **What are the major organs involved in homeostasis in human body? State the roles of each of these organs.**

Consult Short Question No. 6

2. **Identify and label the following: diagram**

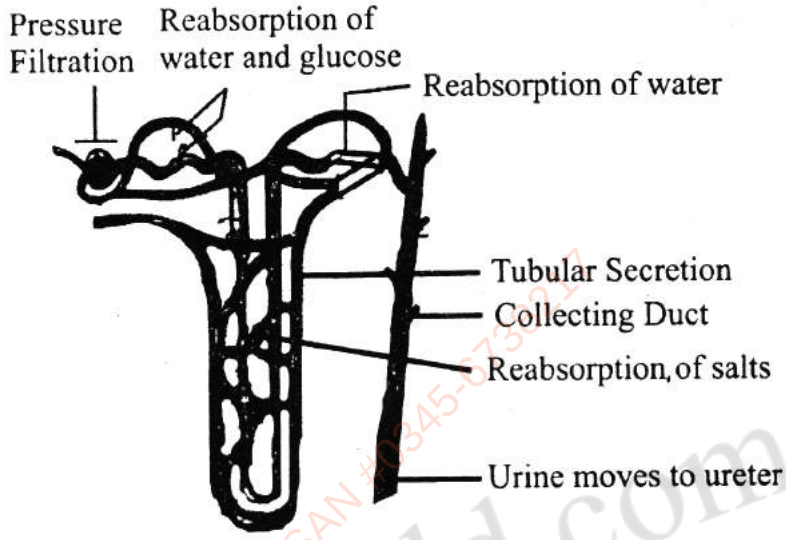

Figure: Functioning of Kidney

UNDERSTANDING THE CONCEPT

1. **Describe the process of selective re-absorption in the kidneys.**

Consult Long Question No. 10

2. **How do the plants excrete extra water and salts from their bodies?**

Consult Long Question No. 3 and 4

3. **What is the functional unit of the kidney? Describe its structure and draw labeled diagram.**

Consult Long Question No. 9

4. **What steps are involved in the formation of urine in the kidneys?**

Consult Long Question No. 10

5. **"Along with excretion, kidneys also play role in Osmoregulation, "Comment on this statement.**

Consult Long Question No. 12

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THE TERMS TO KNOW

Bowman's capsule:
Part of nephron; cup-shaped structure enclosing the glomerulus

Collecting duct:
The tube into which the renal tubule of nephron opens

Dialysis:
The cleaning of blood (removing nitrogenous wastes and extra water) by artificial ways

Dialyzer:
The apparatus used for haemodialysis

Distal convoluted tubule:
The last part of the nephron

Excretion:
The process by which the metabolic wastes are removed from the body

Glomerular filtrate:
The material that passes from glomerulus into the Bowman's capsule

Glomerulus:
The network of capillaries in the nephron of the kidney

Guttation:
Appearance of drops of xylem sap on the tips or edges of leaves

Haemodialysis:
The dialysis in which patient's blood is pumped through the apparatus called dialyzer for cleaning

Hilus:
A depression near the centre of the concave area of the kidney; the area through which the ureter, blood and lymphatic vessels and nerves enter/leave the kidney

Homeostasis:
The maintenance of a constant internal environment in response to environmental changes

Lithotripsy:
Treatment for removing kidney stones; non-electrical shock waves are bombarded on the stones to break them

Loop of Henle:
The U-shaped portion of the renal tubule of nephron

Nephron:
The functional unit of kidneys

Osmoregulation:
The regulation of water content in body fluids

Papillary ducts:
The ducts formed by the joining of many collecting ducts; open into renal pelvis

Peritoneal dialysis:
The dialysis in which the dialysis fluid is pumped into the abdominal peritoneal cavity; the wastes from the blood vessels of the peritoneum diffuse into the dialysis fluid which is then drained out

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Pressure filtration:

The first step in urine formation; the process in which most of the water, salts, glucose and urea of the blood is forced out of the glomerulus and passes into Bowman's capsule

Proximal convoluted tubule:

The part of the nephron between Bowman's capsule and the Loop of Henle

Renal corpuscle:

The collective name of the glomerulus and Bowman's capsule of the nephron

Renal pelvis:

The funnel-shaped cavity into which the renal pyramids of kidney project

Renal pyramid:

Cone-shaped areas in the renal medulla

Renal tubule:

The part of the nephron after Bowman's capsule; consists of proximal convoluted tubule, Loop of Henle and distal convoluted tubule

Selective reabsorption:

The second step in urine formation, in it about 99% of the glomerular filtrate is reabsorbed into the blood capillaries surrounding the renal tubule

Tubular secretion:

The third step in urine formation; different ions, creatinine, urea etc. are secreted from the blood into the filtrate in the renal tubule

Ureter:

A tube that carries urine from a kidney to the urinary bladder

Urethra:

The tube that carries urine from urinary bladder to the outside of the body

Urinary baldder:

A sac-like organ where urine is stored before being excreted

Urinary system:

The system responsible for the production and excretion of urine; includes kidneys, ureters, urinary bladder and urethra

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CHAPTER 12

COORDINATION AND CONTROL

MULTIPLE CHOICE QUESTIONS

Each question has four possible answers. Circle the correct answer.

(1) A wave of electrochemical changes that travels along the length of the neuron:
(a) Nerve (b) Nerve impulse (c) Dendrite (d) Axon

(2) The nucleus and most of the cytoplasm of the neuron is located in:
(a) Dendrite (b) Node of Ranvier (c) Schwann cell (d) Cell body

(3) The impulses that jump over the areas of myelin going from node to node:
(a) Saltatory (b) Stagnant (c) Stationery (d) Running

(4) The neurons that conduct impulses from receptors to central nervous system:
(a) Motor (b) Inter (c) Associative (d) Sensory

(5) Brain is situated inside:
(a) Sternum (b) Cranium (c) Scapula (d) Clavicle

(6) The part of forebrain that lies above midbrain:
(a) Hypothalamus (b) Thalamus (c) Cerebrum (d) Pons

(7) Which lobes contain sensory areas that receive impulses from skin:
(a) Frontal (b) Parietal (c) Occipital (d) Temporal

(8) The part of brain that lies on the top of spinal cord:
(a) Medulla oblongata (b) Cerebellum (c) Pons (d) Cerebrum

(9) Number of pairs of spinal nerves in human body:
(a) 30 (b) 31 (c) 32 (d) 33

(10) Length of spinal cord in an adult:
(a) 30 cm (b) 35 cm (c) 40 cm (d) 45 cm

(11) Number of pairs of cranial nerves in humans:
(a) 24 (b) 20 (c) 16 (d) 12

(12) The organs which are specifically built to detect particular type of stimuli are called:
(a) Neurons (b) Brain (c) Effectors (d) Receptors

(13) Which prevent fine particles entering into the eye:
(a) Eyelids (b) Eyelashes (c) Sclera (d) Cornea

(14) Round hole present in the centre of the iris:
(a) Cornea (b) Sclera (c) Choroid (d) Pupil

(15) Which are sensitive to dim light?
(a) Rods (b) Cones (c) Both a and b (d) None of these

(16) The part of the eye where there are no rods and cones:
(a) Fovea (b) Optic disc (c) Retina (d) Blind spot

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(17) **Date of death of Ali Ibn Isa:**
(a) 1010 (b) 1011 (c) 1012 (d) 1013

(18) **Date of birth of Ibn al-Haytham:**
(a) 960 (b) 965 (c) 970 (d) 975

(19) **The movable bones in middle ear include:**
(a) Malleus (b) Incus (c) Stapes (d) All of these

(20) **To which part of ear tympanum belongs:**
(a) Inner ear (b) Middle ear (c) External ear (d) None of the

(21) **The pituitary gland is attached to:**
(a) Cerebrum (b) Thalamus (c) Medulla (d) Hypothalan

(22) **If the production of somatotrophin is decreased, the person suffers from:**
(a) Dwarfism (b) Gigantism (c) Acromegaly (d) Diabetes

(23) **The hormone secreted by thyroid gland:**
(a) Parathormone (b) Oxytocin (c) Thyroxin (d) Vasopressin

(24) **Hormone secreted by adrenal medulla:**
(a) Epinephrine (b) Corticosteroid (c) Insulin (d) Glucagon

(25) **Blood glucose level after 8-10 hours fast for pre-diabetic:**
(a) From 70 to 99 mg/100ml *(b) From 100 to 125mg/100ml*
(c) 126 mg/100ml and above (d) From 10 to 20mg/100ml

(26) **Blood glucose 2 hours after a 75 gram glucose drink for diabetic:**
(a) Less than 140 mg/100ml (b) From 140 to 200 mg/100ml
(c) Over 200 mg/100ml (d) From 40 to 80 mg/100ml

(27) **Testosterone is secreted by:**
(a) Ovaries (b) Testes (c) Pancreas (d) Pituitary

(28) **By which test the amount of glucose in blood is measured?**
(a) CBG (b) GCB (c) BCG (d) BGC

(29) **The smallest bone of the human body:**
(a) Malleus (b) Stapes (c) Sternum (d) Clavicle

(30) **Malleus is attached with:**
(a) Ear drum (b) Cochlea (c) Vestibule (d) Oval window

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.
1	b	2	d	3	a	4	d	5
6	a	7	b	8	a	9	b	10
11	d	12	d	13	b	14	d	15
16	d	17	c	18	b	19	d	20
21	d	22	a	23	c	24	a	25
26	c	27	b	28	d	29	b	30

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SHORT QUESTIONS

Define coordination

COORDINATION

The process in which the body works as one unit, and its different organs and systems ...e and work in harmony with each other is called coordination.

What is the advantage of coordination?

ADVANTAGE OF COORDINATION

Coordination enables the organism to respond to happenings in the world around.

What are two types of coordination in living organisms?

TYPES OF COORDINATION

There are two types of coordination in organisms:

Nervous coordination brought about by nervous system.

Chemical coordination brought about by endocrine glands.

How does coordination take place in unicellular organisms?

COORDINATION IN UNICELLULAR ORGANISMS

Coordination takes place in unicellular organisms. The response to stimuli is brought through chemicals.

What is the difference between coordination systems in animals and plants?

DIFFERENCE IN COORDINATION SYSTEMS

Animals have both the nervous and chemical coordination systems in their bodies while and other organisms have only chemical coordination.

Write the five components of a coordinated action.

COMPONENTS OF A COORDINATED ACTION

The following are the components of a coordinated action:

Stimulus

Receptor

Coordinator

Effector

Response

What are the two major components of nervous system?

MAJOR COMPONENTS OF NERVOUS SYSTEM

The two major components of nervous system are as follow:

Central Nervous System:

It comprises of

Brain

Spinal cord

Peripheral Nervous System:

It consists of nerves that arise from central nervous system and spread in different parts body.

Define nerve impulse.

NERVE IMPULSE

A wave of electrochemical changes that travels along the length of neurons is called impulse.

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Q. No. 9 How regeneration of broken nerve cells take place?

REGENERATION OF BROKEN NERVE CELLS

The regeneration of broken nerve cells takes place by a protein called nerve-growth-factor.

Q. No. 10 How degenerating brain cells are repaired?

REPAIRING DEGENERATED BRAIN CELLS

The degenerating brain cells could be repaired by using embryonic stem cells.

Q. No. 11 Write the types of neurons on the basis of their function.

TYPES OF NEURONS

On the basis of their functions, neurons are of three types;

1. Sensory Neurons:

Sensory neurons conduct sensory information (nerve impulse) from receptors towards the CNS. Sensory neurons have one dendrite and one axon.

2. Interneurons:

Interneurons form brain and spinal cord. They receive information, interpret them and stimulate motor neurons. They have many dendrites and axons.

3. Motor Neurons:

Motor Neurons carry information from interneurons to muscle or glands (effectors). They have many dendrites but only one axon.

Q. No. 12 Define nerve.

NERVE

The union of several axons that are enveloped by a covering made of lipid form a nerve.

Q. No. 13 Write types of nerves based on the property of axons.

TYPES OF NERVES

Based on the property of axons, the nerves are classified into three types:

1. Sensory Nerves:

Sensory nerves contain the axons of sensory neurons only.

2. Motor Nerves:

Motor nerves contain the axons of motor neurons only.

3. Mixed Nerves:

Mixed nerves contain the axons of both sensory and motor neurons.

Q. No. 14 What is a ganglion?

GANGLION

In certain parts of the body, the cell bodies of many neurons form a group enveloped by a membrane. This is called ganglion.

Q. No. 15 What is hippocampus?

HIPPOCAMPUS

Location:

Hippocampus is a structure that is deep in the cerebrum.

Function:

It functions for the formation of new memories.

Damaged Hippocampus:

People with a damages hippocampus can not remember things that occurred after the damage but can remember things that occurred before damage.

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Q. No. 16 What is brain stem?

BRAIN STEM

The following three parts of brain collectively referred as brain stem because these connect the rest of the brain to spinal cord.

- Medulla oblongata
- Pons
- Midbrain

Q. No. 17 What is the length of spinal cord?

LENGTH OF SPINAL CORD

Spinal cord is roughly 40 cm long and about as wide as your thumb for most of its length.

Q. No. 18 What is the effect of too much light and too little light on retina?

EFFECT OF LIGHT INTENSITY ON RETINA

Too much light being let into the eye can damage the retina. Too little light makes sight difficult.

Q. No. 19 How many rods and cones are present in human eye?

Number of Rods: 125 Lakhs

Number of Cones: 7 Lakhs

Q. No. 20 Why the eyes of cats and dogs shine at night?

SHINING OF EYES

The eyes of cats and dogs shine at night because of the presence of tapetum behind the eyes which is a layer capable of reflecting light.

Q. No. 21 Why colour vision is essential for a pilot?

NECESSITY OF COLOUR VISION

For a pilot, colour vision is essential so that he/she can recognize:

- Aircraft position lights
- Light gun signals
- Airport beacon
- Approach slope indicators
- Chart symbols

A pilot must have the ability to perceive these colours necessary for the safe performance of his/her duties.

Q. No. 22 Which one is the most influential book written in the history of physics?

MOST INFLUENTIAL BOOK

Ibn al-Haytham's "Book of Optics" has been ranked alongside a book of Isaac Newton. It is one of the most influential books written in the history of physics.

Q. No. 23 Why owl is not able to see during day time?

VISION OF OWL IN DAY TIME

Owl is not able to see during day time. The reason for this is the deficiency of cones which receive and sense the bright light. But the presence of more rods gives it greater power of vision during night. All animals that search for prey during night have this characteristic.

Q. No. 24 Which is the smallest bone of the human body?

SMALLEST BONE OF HUMAN BODY

Stapes is the smallest bone of human body.

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Q. No. 25 What is thunderstorm and lightening?

THUNDERSTORM AND LIGHTENING

A thunderstorm is characterized by the presence of lightning and a thunder. The lightning is caused by an electrical charge due to the movement of water droplets or crystals carried by the wind. The sudden increase in pressure and temperature from lightning produces rapid expansion of the air. This expansion of air produces a sound of thunder. The flash of lightning is followed after some seconds by a roar of thunder. This time difference is due to the fact that sound travels slower than light.

Q. No. 26 Name some activities of the body which are required to be regulated.

ACTIVITIES REQUIRE REGULATION

Following are the some of the activities of the body which are required to be regulated:

- Growth
- Reproduction
- Maintenance of glucose concentration in blood
- Reabsorption of water in kidneys

Q. No. 27 Define Hormone.

HORMONE

A specific chemical messenger molecule synthesized and secreted by endocrine gland is called a hormone.

Q. No. 28 What is the difference between endocrine and exocrine glands?

DIFFERENCE BETWEEN ENDOCRINE AND EXOCRINE GLANDS

Endocrine Glands	Exocrine Glands
These glands are ductless and release their secretions called hormones directly into blood stream. ***Examples:*** • Pituitary gland • Thyroid gland	These glands have ducts for releasing their secretions into the target organs. ***Examples:*** • Digestive glands • Skin glands

Q. No. 29 What are the different roles of hormones?

ROLES OF HORMONES

- The stepwise process of metamorphosis in many animals is controlled by hormones.
- Life activities such as cell division in invertebrates are also regulated by hormones.
- Hormones also control activities like migration in birds.
- Hormones have been identified even in unicellular organisms.

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Why during summer, urine output is low?

LOW URINE OUTPUT DURING SUMMER

During summer, urine output is low. Due to increased sweating, the water level of blood ...ed. As a result, pituitary gland releases more antidiuretic hormone (ADH) into blood.

What is the relationship between calcitonin and parathormone?

RELATIONSHIP BETWEEN CALCITONIN AND PARATHORMONE

Calcitonin and parathormone complement each other and regulate the level of calcium ...blood.

Q. No. 32 What is tatany?

TETANY

Symptoms:

Tetany is marked by:

- Flexion of the wrist and ankle joints
- Muscle twitching
- Cramps
- Convulsions

Cause:

It is due to decreased calcium level which makes the nerves and muscles more excitable.

Q. No. 33 Which changes prepare the body to face any emergency situation?

FACING EMERGENCY SITUATIONS

When a person experiences fear, anger or anxiety, the following changes prepare the body to face any emergency situation:

- The rate and intensity of heartbeat increases
- Blood flow to the limbs increases
- Blood flow to alimentary canal and skin is reduced

Q. No. 34 At what rate the blood glucose concentration is maintained?

MAINTENANCE OF BLOOD GLUCOSE CONCENTRATION

The blood glucose concentration is maintained at the rate of 80 to 120 mg per 100 ml of blood.

Q. No. 35 What is BGC test?

BLOOD GLUCOSE CONCENTRATION TEST

The amount of glucose in blood is measured by this test. It is used to diagnose diabetes. Blood glucose may be measured on a fasting basis (collected after an 8 to 10 hour fast). Randomly (anytime) and after a meal.

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Q. No. 36 **Write the blood glucose concentration level of normal, pre-diabetic and diabetic persons after 8-10 hours fast.**

BLOOD GLUCOSE AFTER 8-10 HOURS FAST

Blood Glucose After 8-10 hours fast	
BGC	**Diagnosis**
From 70 to 99 mg/ 100ml	Normal
From 100 to 125 mg/ 100ml	Pre-diabetic
126 mg/ 100ml and above	Diabetic

Q. No. 37 **Write the blood glucose concentration level of normal, pre-diabetic and diabetic persons 2 hours after a 75 gram glucose drink.**

BLOOD GLUCOSE CONCENTRATION 2 HOURS AFTER A75 GRAM GLUCOSE DRINK

Blood Glucose 2 hours after a 75 gram Glucose drink	
BGC	**Diagnosis**
Less than 140 mg/ 100ml	Normal
From 140 to 200 mg/ 100ml	Pre-diabetic
Over 200 mg/ 100ml	Diabetic

Q. No. 38 **How knowledge of nervous system has helped man to treat nervous disorders?**

TREATMENT OF NERVOUS DISORDERS

The knowledge of the composition and functioning of nervous system has helped man in the diagnosis and treatment of nervous disorders including paralysis and epilepsy. Man has discovered the areas of brain that receive information from different sense organs and the areas that send messages to different effectors. Such knowledge helps a lot in identifying the malfunctioning areas of brain.

Q. No. 39 **Why objects should never be placed in a patient's mouth during a seizure attack?**

SEIZURE ATTACK

During a seizure attack, objects should never be placed in a patient's mouth as it can result in serious injury. It is possible that the patient will bite his/her own tongue.

Q. No. 40 **What is the contribution of Ali Ibn Isa?**

CONTRIBUTION OF ALI IBN ISA

Ali Ibn Isa was a famous Arab scientist.

Period: 950-1012

Contribution:

He wrote three books on ophthalmology (study of the disease and surgery of eyes). He described 130 eye diseases and prescribed 143 drugs to treat these diseases.

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Q. No. 41 What is the contribution of Ibn al-Haytham?

CONTRIBUTION OF IBN AL-HAYTHAM

Ibn al-Haytham was an Arab scientist.

Period: 965-1039

Contributions:

He made significant contributions to the principles of eye and vision. He is regarded as the father of optics (study of the behavior of light). His "Book of Optics" correctly explained and proved the modern theory of vision. He discussed the topics of medicine and eye surgery in his book. He made several improvements to eye surgery and accurately described the following:

- The process of sight
- The structure of eye
- Image formation in eye
- Visual system
- The principles of pinhole camera

Q. No. 42 What is deafness?

DEAFNESS

Deafness is a state in which hearing is not possible.

Cause:

The defect of ear drum, cochlea, middle ear ossicles, or auditory nerve may cause deafness. Infection in Eustachian tube may spread to middle ear too. Ear drum may be damaged by an infection in auditory canal.

Effect on Hearing:

Excessive noise, strong blows on cheek, pointed objects entering auditory canal and attack from insects, may also affect hearing.

Q. No. 43 How do ears maintain the balance of body?

MAINTENANCE OF BALANCE OF BODY

Semicircular canals and vestibule help to maintain the balance of body. Semicircular canals contain sensory nerves which can detect any movement of head. Vestibule can detect any changes in the posture of body. The neurons coming from these two receptors reach cerebellum through the auditory nerve.

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LONG QUESTIONS

Q. No. 1 Describe the components of a coordinated action.

COMPONENTS OF A COORDINATED ACTION

A coordinated action has following five components:

1. Stimulus
2. Receptor
3. Coordinator
4. Effector
5. Response

1. Stimuli:

Definition:

Any change in external and internal environment of an organism which can provoke a response is called a stimulus.

Examples:

- Touch
- Light
- Heat
- Cold
- Pressure
- Sound waves
- Presence of chemicals
- Microbial infections

2. Receptors:

Definition:

The organs, tissues or cells which are specifically built to detect particular type of stimuli are called receptors.

Examples:

- Sound waves are detected by ears
- Light is detected by eyes
- Chemicals in air are detected by nose
- Touch, heat, cold and pressure are detected by skin
- Taste is detected by tongue

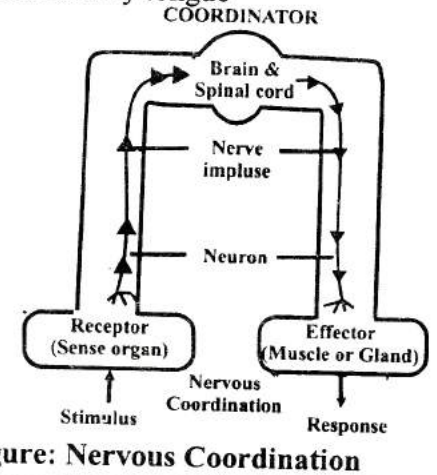

Figure: Nervous Coordination

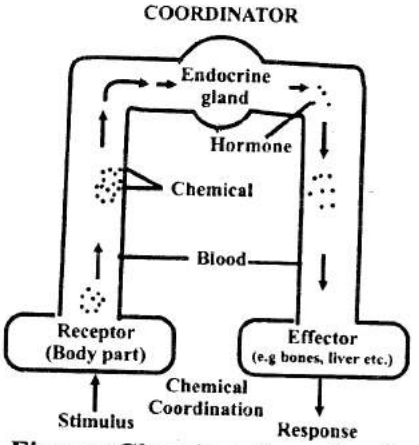

Figure: Chemical Coordination

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3. Coordinators:

Definition:

The organs that receive information from receptors and send messages to particular organs for proper action are called coordinators.

Nervous Coordination:

In nervous coordination, brain and spinal cord are coordinators. They receive information and send messages through neurons in the form of nerve impulses.

Chemical Coordination:

In chemical coordination, various endocrine glands play the role of coordinators. The receive information in the form of various chemicals and send messages by secreting particular hormones in blood.

4. Effectors:

Definition:

The parts of body which receive messages from coordinators and produce particular responses are called effectors.

Nervous Coordination:

In nervous coordination, neurons carry messages from coordinators (brain and spinal cord) to muscles and glands, which act as effectors.

Chemical Coordination:

In chemical coordination, particular hormones carry messages from coordinators (endocrine glands) to particular target tissues, which act as effectors. For some hormones, nephrons act as effectors. Similarly, bones and liver act as effectors for many hormones.

5. Response:

Definition:

The action performed by the effectors on receiving the message from coordinators is called response.

Examples:

- Pulling our hand away from something very hot
- The movement of the flower of sunflower towards light

Nervous Coordination:

Nervous coordination produces immediate but short-living responses.

Chemical Coordination:

Chemical coordination produces slow but long-living responses.

Q. No. 2 Describe the structure of a neuron.

NEURON

Introduction:

Neuron is also called as nerve cell.

Definition:

The unit of the nervous system is called neuron or nerve cell.

Number:

The human nervous system consists of billions of neurons plus supporting neuroglial cells.

Function:

Neurons are specialized cells that are able to conduct nerve impulses from receptors to coordinators and from coordinators to effectors. In this way they communicate with each other and with other types of body cells.

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PARTS OF A NEURON

A neuron consists of the following parts:

1. Cell Body
2. Dendrites
3. Axon
4. Schwann Cells
5. Myelin Sheath
6. Nodes of Ranvier

1. Cell Body:

The nucleus and most of the cytoplasm of a neuron is located in its cell body.

2. Dendrites:

Different processes extend out from cell body, these are called dendrites. Dendrites conduct impulses toward cell body.

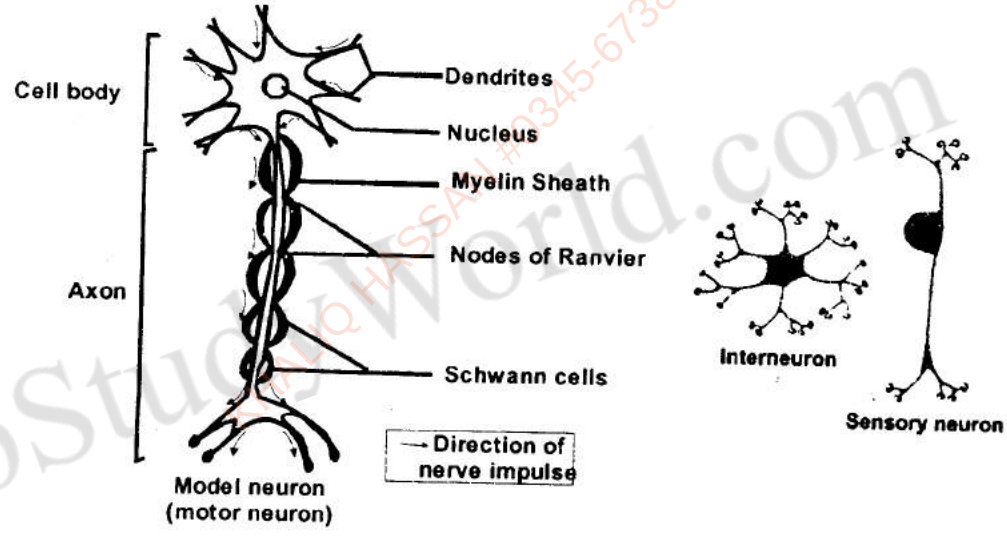

Figure: Neurons

3. Axon:

An axon arises from the cell body and conducts impulses away from cell body.

4. Schwann Cells:

Schwann cells are special neuroglial cells located at regular intervals long axons.

5. Myelin Sheath:

In some neurons, Schwann cells secrete a fatty layer called myelin sheath, over axons.

Saltatory Impulse:

Myelin sheath is an insulator so the membrane coated with this sheath does not conduct nerve impulse. In such a neuron, impulses 'jump' over the areas of myelin going from node to node. Such impulses are called saltatory ('jumping') impulses. This increases the speed of nerve impulse.

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6. Nodes of Ranvier:

Between the areas of myelin on an axon, there are non-myelinated points, called the nodes of Ranvier.

Q. No. 3 Describe the structure of brain.

BRAIN

Introduction:

In animals, all life activities are under the control of brain. The structure of brain is suitable to perform this function.

Cranium:

Brain is situated inside a bony cranium which is the part of skull.

Meninges:

Inside cranium, brain is covered by three layers called meninges. Meninges protect brain and also provide nutrients and oxygen to brain tissue through their capillaries.

Ventricles:

The brain contains fluid-filled ventricles that are continuous with the central canal of spinal cord.

Cerebrospinal Fluid:

Fluid within ventricles and central canal is called cerebrospinal fluid (CSF).

THE DIVISIONS OF BRAIN

There are three major regions in the brain of human and other vertebrates. These are:

1. Forebrain
2. Midbrain
3. Hindbrain

1. FOREBRAIN

Forebrain is the largest area of brain. It is most highly developed in humans. Following are the important parts of this region.

(i) Thalamus
(ii) Hypothalamus
(iii) Cerebrum

(i) Thalamus:

Location:

Thalamus lies just below cerebrum.

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Functions:

- Thalamus serves as a relay centre between various parts of brain and spinal cord.
- It also receives and modifies sensory impulses (except from nose) before they t cerebrum.
- Thalamus is also involved in pain perception and consciousness (sleep and awake

(ii) Hypothalamus:

Location:

Hypothalamus lies above midbrain and just below thalamus.

Size:

In humans, it is roughly the size of an almond.

Functions:

- One of the most important functions of hypothalamus is to link nervous syst endocrine system.
- It controls the secretions of pituitary gland.
- It also controls feelings such as rage, pain, pleasure and sorrow.

(iii) Cerebrum:

Cerebrum is the largest part of forebrain.

Functions:

It controls:

- Skeletal muscles
- Thinking
- Intelligence
- Emotions

Divisions of Cerebrum:

Cerebrum is divided into two cerebral hemispheres.

Olfactory Bulbs:

The anterior parts of cerebral hemispheres are called olfactory bulbs which impulses from olfactory nerves and create the sensation of smell.

Cerebral Cortex:

The upper layer of cerebral hemispheres is cerebral cortex that consists of gre The grey matter of nervous system consists of cell bodies and non-myelinated axons.

White Matter:

Beneath this layer is present the white matter. The white matter of nervou consists of myelinated axons.

Lobes of Cerebral Cortex:

Cerebral cortex has a large surface area and is folded in order to fit in skull. It into four lobes.

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(i) Frontal Lobe:

- Controls motor functions
- Permits conscious control of skeletal muscles
- Coordinates movements involved in speech

(ii) Parietal Lobe:

- Contains sensory areas that receive impulses from skin

(iii) Occipital Lobe:

- Receives and analyzes visual information

(iv) Temporal Lobe:

- Concerned with hearing and smell

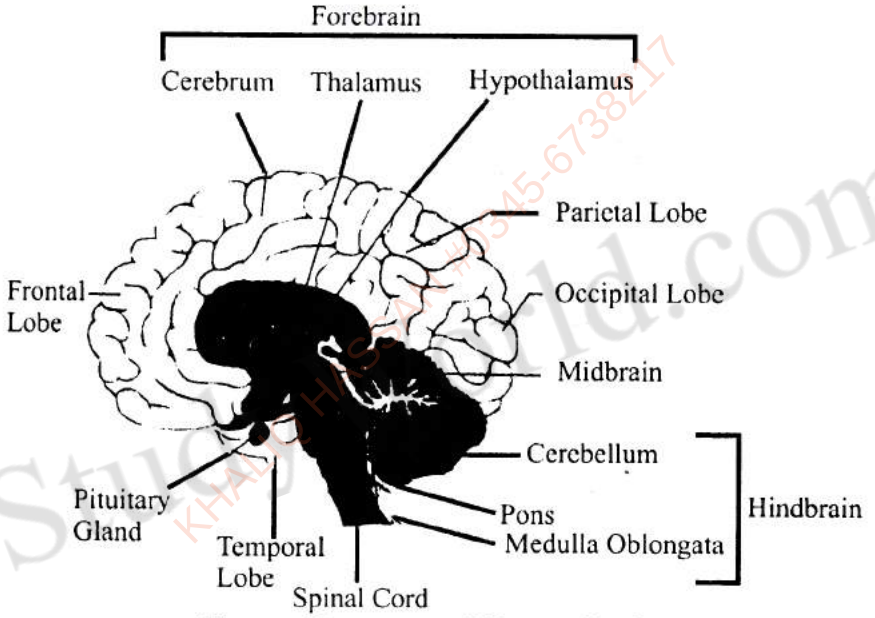

Figure: Structure of Human Brain

2. MIDBRAIN

Location:

Midbrain lies between hindbrain and forebrain and connects the two.

Functions:

- The midbrain receives sensory information and sends it to the appropriate part of forebrain.
- Midbrain also controls some auditory reflexes and posture.

3. HINDBRAIN

Hindbrain consists of three major parts:

(i) Medulla Oblongata
(ii) Cerebellum
(iii) Pons

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(i) Medulla Oblongata:

Location:

Medulla oblongata lies on the top of spinal cord. Information that passes between spinal cord and the rest of brain pass through medulla.

Functions:

It controls:

- Breathing
- Heart rate
- Blood pressure
- Reflexes such as vomiting, coughing, sneezing

(ii) Cerebellum:

Location:

Cerebellum lies behind medulla oblongata.

Function:

It coordinates muscle movements.

(iii) Pons:

Location:

Pons is present on top of medulla.

Functions:

- It assists medulla in controlling breathing.
- It also serves as a connection between cerebellum and spinal cord.

Q. No.4 Write a note on spinal cord.

SPINAL CORD

Introduction:

The spinal cord is in fact a tubular bundle of nerves.

Location:

It starts from brain stem and extends to lower back.

Meninges:

Like brain, spinal cord is also covered by meninges.

Protection:

The vertebral column surrounds and protects spinal cord.

Outer Region:

The outer region of spinal cord is made of white matter containing myelinated axons.

Central Region:

The central region is butterfly shaped that surrounds the central canal. It is made of grey matter containing neuron cell bodies.

Spinal Nerves:

31 pairs of spinal nerves arise along spinal cord. These are **"mixed"** nerves because each contains axons of both sensory and motor neurons.

Roots of Spinal Nerve:

At the point where a spinal nerve arises from spinal cord, there are two roots of spinal nerve. Both roots unite and form one mixed spinal nerve.

Dorsal Root:

The dorsal root contains sensory axons and a ganglion where cell bodies are located.

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Ventral Root:

The ventral root contains axons of motor neurons.

Functions:

Spinal cord performs two main functions:

- It serves as a link between body parts and brain. Spinal cord transmits nerve impulses from body parts to brain and from brain to body parts.
- Spinal cord also acts as a coordinator, responsible for some simple reflexes.

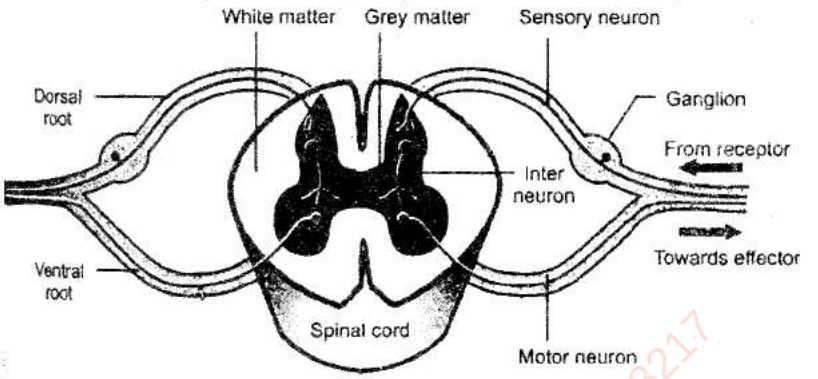

Figure: Spinal Cord and Spinal Nerves

Q. No. 5 Write a note on peripheral nervous system.

PERIPHERAL NERVOUS SYSTEM

Composition:

The peripheral nervous system (PNS) is composed of nerves and ganglia. Ganglia are the clusters of neuron cell bodies outside central nervous system.

Cranial Nerves:

The nerves that arise or lead to brain are called cranial nerves. Humans have 12 pairs of cranial nerves. Some cranial nerves are sensory, some are motor and some are mixed.

Spinal Nerves:

The nerves that arise or lead to spinal cord are called spinal nerves. Humans have 31 pairs of spinal nerves. All spinal nerves are mixed nerves.

Pathways:

The cranial and spinal nerves make two pathways:

(i) Sensory Pathway:

The sensory pathway conducts impulses from receptors to central nervous system.

(ii) Motor Pathway:

The motor pathway conducts impulses from central nervous system to effectors.

Motor pathway makes two systems:

1. Somatic Nervous System
2. Autonomic Nervous System

1. Somatic Nervous System:

Composition:

It includes all of the motor neurons that conduct impulses from central nervous system to skeletal muscles.

Function:

- It is responsible for the conscious and voluntary actions.

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2. Autonomic Nervous System:

Composition:

It consists of motor neurons that send impulses to:

- Cardiac muscles
- Smooth muscles
- Glands

Function:

- It is responsible for the activities, which are not under conscious control.

Types of Autonomic Nervous System:

Autonomic nervous system comprises of:

(i) Sympathetic Nervous System

(ii) Parasympathetic Nervous System.

(i) Sympathetic Nervous System:

Sympathetic nervous system prepares body to deal with emergency situations. This is often called the "fight or flight" response. During an emergency situation, this system takes necessary actions. It

- Dilates pupils
- Accelerates heartbeat
- Increases breathing rate
- Inhibits digestion

(ii) Parasympathetic Nervous System:

When stress ends, the parasympathetic nervous system takes action and normalizes all the functions. It

- Causes pupils to contract
- Promotes digestion
- Slows the rate of heartbeat and breathing rate

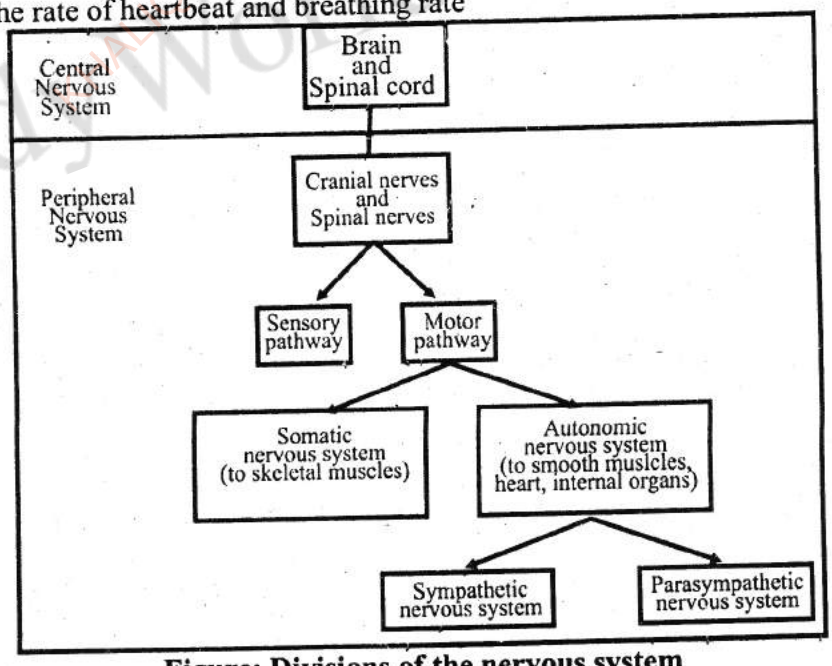

Figure: Divisions of the nervous system

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Q. No.6 Write a note on reflex action.

TYPES OF ACTIONS

When central nervous system sends impulses to muscles and glands, two types of actions (responses) result.

1. Voluntary Actions
2. Involuntary Actions

1. Voluntary Actions:

The higher centres of brain control the conscious action or voluntary actions.

2. Involuntary Actions:

When impulses are not passed to the higher centres of brain, it results in responses which are not under conscious control. Such responses are called involuntary actions.

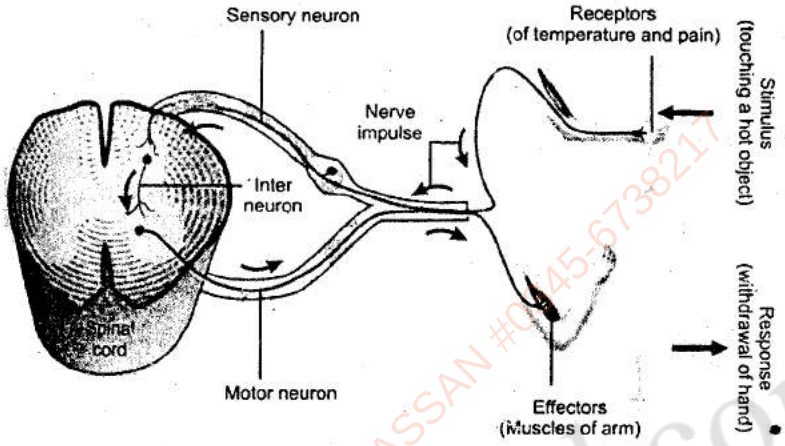

Figure: Reflex Arc in a Reflex Action

REFLEX ACTION

Definition:

A very quick involuntary response produced by the central nervous system is called reflex action.

Reflex Arc:

The pathway followed by the nerve impulses for producing a reflex action, is called reflex arc.

Example:

The most common example of reflex action is the withdrawal of hand after touching a hot object.

Mechanism:

In this reflex action, spinal cord acts as coordinator. Heat stimulates temperature and pain receptors in skin. A nerve impulse is generated which is carried by sensory neurons to the interneurons of spinal cord. From interneurons, the impulse is passed to motor neurons, which carry it to the muscles of arm. As a result, the muscles contract to withdraw hand. During it, other interneurons transmit nerve impulses up to brain so that the person becomes aware of pain and what happened.

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Q. No. 7 Describe the structure of eye.

EYE

Introduction:

Eyes are the sensory organs that are associated with the sense of vision.

Orbit:

Our eyes are located in small portions of skull known as the orbits or eye sockets.

Eyelids:

Eyelids wipe eyes and prevent dehydration. They spread tears on eyes, which contains substances for fighting bacterial infections.

Eyelashes:

Eyelashes prevent fine particles from entering eye.

STRUCTURE OF EYE

The structure of eye can be divide into three main layers.

1. Outer Layer
2. Middle Layer
3. Inner Layer

1. Outer Layer:

The outer layer of eyeball consists of sclera and cornea.

Sclera:

Sclera gives eye most of its white colour. It consists of dense connective tissue and protects the inner components of eye and maintains its shape.

Cornea:

In the front, sclera forms the transparent cornea. Cornea admits light to the interior of eye and bends light rays so that they can be brought to a focus.

2. Middle Layer:

The middle layer is called choroid.

Choroids:

The choroid contains blood vessels and gives the inner eye a dark colour. The dark colour prevents disruptive reflections within eye.

Iris:

Behind cornea, choroid bends to form a muscular ring, called iris.

Pupil:

There is round hole, called pupil, in the centre of iris. After striking the cornea, light passes through the pupil. The size of pupil is adjusted by the muscles of iris. Pupil constricts in bright light when the circular muscles of iris contract. Similarly, pupil dilates in dim light when the radial muscles of iris contract.

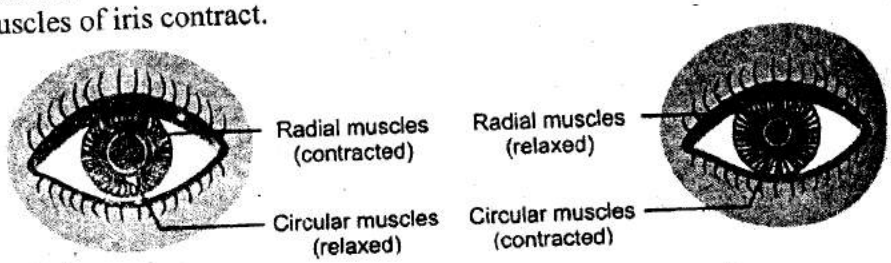

Figure: Contraction and Dilation of Pupil

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Lens:

Behind iris, there is a convex lens, which focuses light on retina.

Ciliary Muscles:

Lens is attached to ciliary muscles of eye via a ring of suspensory ligament. To clearly see an object far away, ciliary muscles are relaxed and lens becomes less convex. When ciliary muscles contract, lens becomes more convex and round.

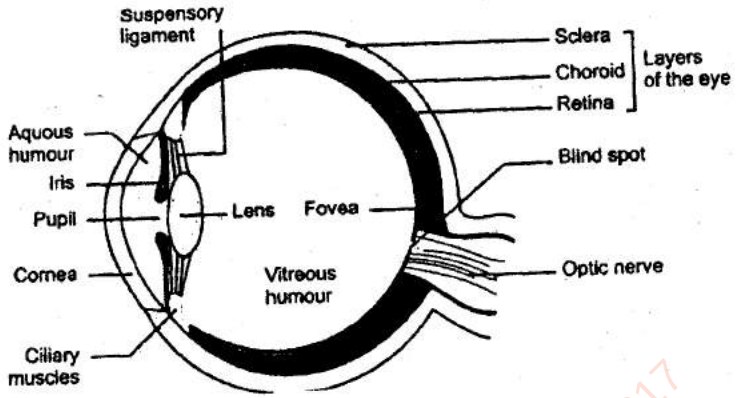

Figure: Structure of Human Eye

3. Inner Layer:

Retina:

The inner layer is sensory is called as retina. It contains the photosensitive cells called rods and cones and associated neurons.

Rods:

Rods are sensitive to dim light.

Cones:

The cones are sensitive to bright light and so distinguish different colours.

Fovea:

Fovea is a dip in retina, directly opposite to lens and is densely packed with cone cells. It is largely responsible for colour vision and sharpness.

Optic Disc:

Optic disc is a point on retina where the optic nerve enters retina.

Blind Spot:

There are no rods and cones at optic disc point that is why it is referred to as the blind spot.

Chambers of Eye:

The iris divides the cavity of eye into two chambers.

Anterior Chamber:

The anterior chamber is in front of iris i.e. between cornea and iris. The anterior chamber contains a clear fluid known as aqueous humour.

Posterior Chamber:

The posterior chamber is between iris and retina. The posterior chamber contains a jelly-like fluid known as vitreous humour. It helps maintain the shape of eye and suspends the delicate lens.

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Q. No. 8 Write a note on image formation.

IMAGE FORMATION

Light from objects enters eye and is refracted when it passes through cornea, aqueous humour, lens and vitreous humour. Lens also focuses light on retina. As a result, the image falls on retina. Rods and cones generate nerve impulses in the optic nerve. These impulses are carried to the brain, which makes the sensation of vision.

Role of Rhodopsin:

Rods contain a pigment called rhodopsin. When light falls on rhodopsin, it breaks for generating a nerve impulse. In the absence of light, the breakdown products are again converted into rhodopsin.

Night Blindness:

Body synthesizes rhodopsin from vitamin A and that is why the deficiency of vitamin A causes poor night vision. This problem is called night blindness.

Role of Iodopsin:

Cones also contain a pigment, known as iodopsin. There are three main types of cones and each type has a specific iodopsin. Each type of cones recognizes one of the three primary colours i.e. blue, green and red.

Colour Blindness:

If any type of cones is not working well, it becomes difficult to recognize that colour. Such person is also not able to distinguish different colours. This disease is called colour blindness and it is a genetic problem.

Q. No.9 Describe disorders of eye.

Disorders of the Eye

The working of eye is affected by the changes in the shape of eyeball.

1. Myopia
2. Hypermetropia

1. Myopia (short sight):

The image of a distant object is formed in front of retina.

Cause:

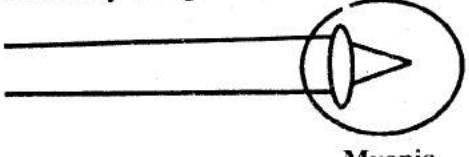
The elongation of eyeball results in myopia.

Effect:

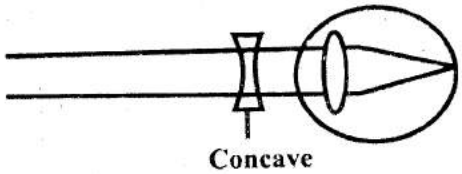
Such persons are not able to see distant objects clearly.

Treatment:

This problem can be rectified by using concave lens.



Myopia



Concave lens

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2. Hypermetropia (Long sight):

The image is formed behind retina

Cause:

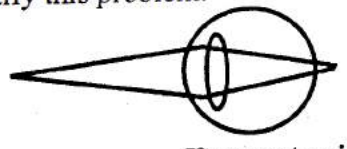
It happens when eyeball shortness.

Effect:

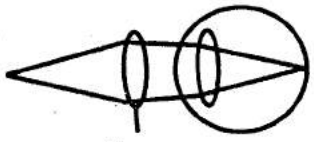
Such persons are not able to see near objects clearly.

Treatment:

Convex lens is used to rectify this problem.



Hypermetropia



Convex lens

Q. No. 10 Describe the structure of ear. Also explain the process of hearing.

STRUCTURE OF EAR

Introduction:

Hearing is an important as vision. Our ear helps us in hearing and also to maintain the balance of equilibrium of our body.

Ear has the following three main parts:

1. External ear
2. Middle ear
3. Internal ear.

1. External Ear:

External ear consists of pinna, auditory canal and ear drum (tympanum).

Pinna:

Pinna is the broad external part, made of cartilage and covered with skin. It helps to direct sound waves into auditory canal.

Auditory Canal:

There are special glands in the walls of auditory canal, which produce wax. The wax and the hairs in auditory canal protect ear from small insects, germs and dust. In additions to this, they help to maintain the temperature and dampness of auditory canal.

Ear Drum:

Auditory canal ends in ear drum. This membrane separates external ear from middle ear.

2. Middle Ear:

Middle ear is a chamber after external ear.

Ossicles:

Three small bones, called middle ear ossicles, are present in a chain in middle ear. These movable bones include malleus, incus and stapes. Malleus is attached with ear drum, then comes incus and finally stapes that is connected with a membrane called oval window.

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Oval Window:

Oval window separates middle ear from inner ear.

Eustachian Tube:

Middle ear also communicates with the nasal cavity through Eustachian tube. This tube regulates the air pressure on both sides of eardrum.

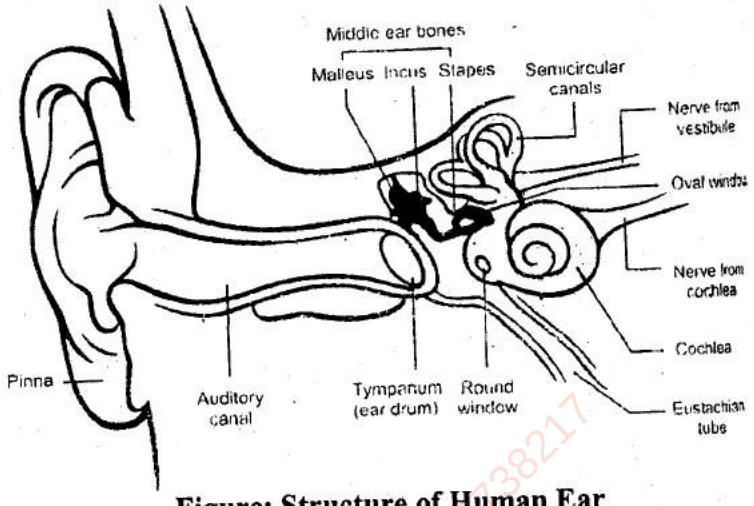

Figure: Structure of Human Ear

3. **Inner Ear:**

Inner ear consists of three parts i.e. vestibule, semicircular canals and cochlea.

Vestibule:

Vestibule is present in the centre of inner ear.

Semicircular Canals:

Three canals called semicircular canals are posterior to the vestibule.

Cochlea:

The cochlea is made of three ducts and wraps itself into a coiled tube. Sound receptor cells are present within the middle duct of cochlea.

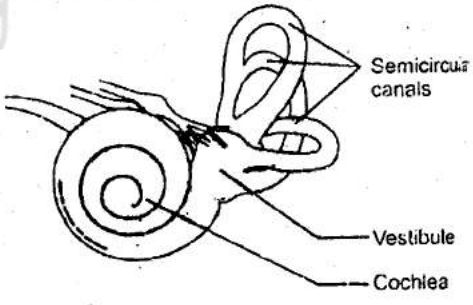

Figure: Structure of the Inner Ear

The Process of Hearing:

The pinna of the external ear focuses and directs sound waves into auditory canal. The sound waves strike ear drum and produce vibrations in it. From ear drum, the vibrations strike middle ear and produce further vibrations in malleus, incus and then stapes. From stapes, the vibrations strike the oval window and then reach the fluid-filled middle duct of cochlea. The fluid of cochlea is moved and receptor cells are stimulated. The receptor cells generate a nerve impulse, which travels to brain and is interpreted as sound.

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Q. No. 11 Write a note on pituitary gland.

PITUITARY GLAND

Shape:
It is a pea-shaped gland.

Location:
It is attached to the hypothalamus of brain.

Master Gland:
Many hormones (trophic hormones) of pituitary gland influence the secretions of other endocrine glands.

Direct Action:
Some hormones of this gland act directly on various tissues of body.

Lobes of Pituitary Gland:
There are two lobes of pituitary gland:

1. Anterior lobe
2. Posterior lobe

1. Anterior Lobe:
It produces many hormones.

(i) Somatotrophin:
One of its important hormones is somatotrophin. It is also known as growth hormone. It promotes the growth of body.

Dwarfism:
If the production of this hormone is diminished during growing age, the rate of growth decreases. This condition is called dwarfism.

Gigantism:
If this hormone is excessively produced during growing age, it leads to gigantism. The person becomes very tall and overweight.

Acromegaly:
If somatotrophin is excessively produced after growing age, internal organs and body extremities alone grow large. This condition is known as acromegaly. Such persons will have large hands, feet and jawbones.

(ii) Thyroid Stimulating Hormone (TSH):
Another important hormone secreted by the anterior lobe of pituitary gland is thyroid stimulating hormone (TSH). It stimulates thyroid gland to secrete its hormones.

Other Hormones:
The remaining hormones of anterior lobe influence reproductive organs and also control adrenal glands.

2. Posterior Lobe:
The posterior lobe of pituitary gland stores and secretes two hormones.

(i) Vasopressin (Antidiuretic Hormone)
(ii) Oxytocin

Production:
These hormones are produced by hypothalamus (a part of brain).

(i) Vasopressin:
Vasopressin increases the rate of reabsorption of water from nephrons.

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Lesser Water Potential:

When we have low amount of water in body fluids, pituitary gland secretes vasopressin and so more reabsorption of water occurs from nephrons into blood. In this way, body retains water and less amount of urine is produced.

Greater Water Potential:

On the other hand, when body fluids have more than normal water, there is a decline in the secretion of this hormone. If pituitary gland does not secrete this hormone in the required amount; less water is reabsorbed from nephrons and there is excessive loss of water through urine. This condition is known as **diabetes insipidus**.

(ii) Oxytocin:

The hormone, oxytocin stimulates the contraction of uterus walls in mothers for child birth. Moreover, this hormone is necessary for the ejection of milk from breast.

Q. No. 12 Write a note on thyroid gland.

THYROID GLAND

Largest Gland:

This is the largest endocrine gland in human body.

Location:

It is present in neck region, below larynx.

Hormones:

It produces two hormones.

(i) Thyroxin
(ii) Calcitonin

(i) Thyroxin:

Thyroxin increases the break down of food (oxidation) and release of energy in body. It is also responsible for the growth of body.

Goiter:

Iodine is required for the production of thyroxin. If a person lacks iodine in diet, thyroid gland cannot make its hormone. In this condition, thyroid gland enlarges. This disorder is called goiter.

Hypothyroidism:

Hypothyroidism is caused by the under-production of thyroxin. It is characterized by low energy production in body and slowing down of heart-beat.

Hyperthyroidism:

Hyperthyroidism is caused by over-production of thyroxin. Its symptoms are:

- Increase in energy production
- Increased heart-beat
- Frequent sweating
- Shivering of hands

(ii) Calcitonin:

The thyroid gland produces another hormone called calcitonin. It decreases the level of calcium ions in blood and promotes the absorption of calcium from blood into bones.

Q. No. 13 Write a note on parathyroid glands.

PARATHYROID GLANDS

Number:

These are four glands.

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Location:

These are situated on the posterior side of thyroid gland.

Parathormone:

They produce a hormone known as parathormone. It increases the level of calcium ions in blood.

Increased Production of Parathormone:

When there is increased production of parathormone, more than normal calcium salts are absorbed from the bones and added to blood. Consequently the bones become brittle.

Decreased Production of Parathormone:

If there is deficiency in the production of parathormone, blood calcium level falls. It leads to tetany, which affects the functioning of muscles.

Q. No. 14 Write a note on adrenal glands.

ADRENAL GLANDS

Number:

These are two in number.

Location:

These are situated above kidneys.

Structure:

Each adrenal gland consists of two parts:

1. The outer part is cortex.
2. The inner part is medulla.

1. Adrenal Cortex:

The adrenal cortex secretes many hormones called corticosteroids which maintain the balance of salts and water in blood.

2. Adrenal Medulla:

Adrenal medulla secretes a hormone called epinephrine or adrenaline in response to stress. It prepares our body to overcome emergency situations. Therefore, adrenaline is also termed as 'emergency hormone'.

Q. No. 15 Write a note on pancreas.

PANCREAS

This organ has two functions.

Exocrine Role:

The major part of pancreas is a ducted (exocrine) gland. This portion secretes digestive enzymes, through a duct, into the small intestine.

Endocrine Role:

Some portions of pancreas serve as ductless (endocrine) gland. This portion contains groups of endocrine cells referred to as islets of Langerhans. These islets secrete two hormones:

(i) Insulin
(ii) Glucagon

(i) Insulin:

Insulin influences the liver to take excess glucose from blood and so the blood glucose concentration falls.

(ii) Glucagon:

Glucagon influences the liver to release glucose in blood and so the blood glucose concentration rises.

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Diabetes Mellitus:

If a person's pancreas does not make normal quantity of insulin, the blood glucose concentration rises and we say that the person has diabetes mellitus.

Symptoms:

Persons with diabetes have:

- Loss of body weight
- Weakening of muscles
- Tiredness

Control:

The disease can be controlled by insulin administration. Formerly, insulin extracted from animals was used for this purpose. But now human insulin produced from bacteria through genetic engineering is available.

Normal Blood Glucose Concentration:

The blood glucose concentration is maintained at the rate of 80 to 120 mg per 100 ml of blood.

Q. No. 16 Write a note on gonads.

GONADS

The male and female reproductive organs are called as gonads. The male reproductive organs are called testes and female reproductive organs are called ovaries. In addition to producing gametes, gonads also secrete hormones, called sex hormones.

Testes:

The singular of testes is testis. Testes secrete hormones e.g. testosterone, which is responsible for the development of male secondary sex characters such as:

- Growth of hair on face
- Coarseness of voice

Ovaries:

Ovaries secrete the following hormones:

- Estrogen
- Progesterone

Function:

These both hormones are responsible for the development of female secondary characters such as the development of breast etc.

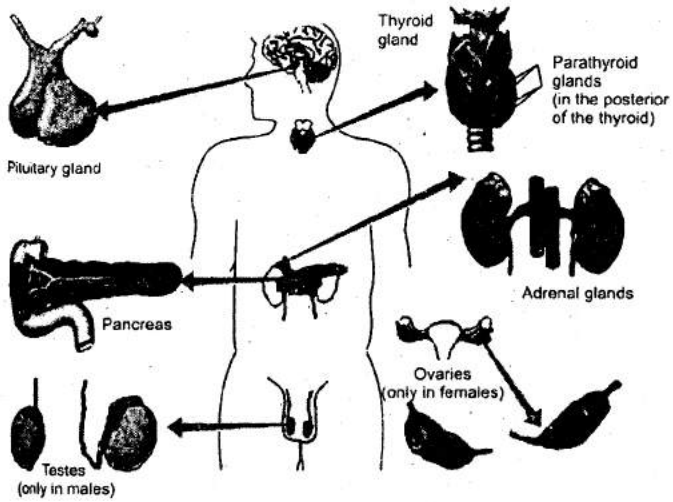

Figure: Endocrine Glands in Human Body

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Q. No. 17 Describe feed back mechanisms.

FEEDBACK MECHANISMS

Definition:

The regulation of a process by the output of the same process is called feedback mechanism.

Explanation:

Endocrine glands do not secrete their hormones at a constant rate. The rate varies with the needs of the body. Like many other functions in body, the secretion of hormones is also regulated by feed back mechanisms.

Types of Feedback Mechanisms:

Feedback mechanisms are of two types:

1. Negative Feedback
2. Positive Feedback

1. Negative Feedback:

Definition:

The feedback in which the output of a process decreases or inhibits the process is called negative feedback.

Need:

This mechanism works to return a condition towards its normal value.

Example:

Control of Blood Glucose Concentration (BGC):

The blood glucose concentration (output) controls the process i.e. the secretion of insulin and glucagon.

Rise in Blood Glucose Concentration (BGC):

When the blood glucose concentration rises, pancreas secretes insulin. It decreases the blood glucose concentration. Decline in the blood glucose concentration to a normal set-point inhibits the secretion of insulin.

Decline in Blood Glucose Concentration (BGC):

When blood glucose concentration drops below normal, pancreas secretes glucagon. It raises the blood glucose concentration. In this case, rise in the blood glucose concentration to a normal set-point inhibits the secretion of glucagon.

2. Positive Feedback:

Definition:

The feedback in which the changes resulting from a process increases the rate of process is called positive feedback.

Example:

Suckling Action of Infants:

Suckling action of an infant stimulates the production of a hormone in mother. This hormone works for the production of milk. More suckling leads to more hormone, which is turn leads to more milk production.

Q. No. 18 Describe disorders of nervous system.

NERVOUS DISORDERS

The disorders of the nervous system are called as nervous disorders.

Types of Nervous Disorders:

There are two main types of nervous disorders:

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1. **Vascular Disorder:**
The disorders which are due to any disturbance in the blood supply of nervous system are called vascular disorders.

Example:
Paralysis

2. **Functional Disorder:**
The disorders which are due to any disturbance in nerve impulse generation and transmission are called functional disorders.

Example:
- Epilepsy

PARALYSIS

Definition:
The complete loss of function by one or more muscle groups is called paralysis.

Causes:
It is most often caused by damage to the central nervous system (brain or spinal cord). The damage may be due to:
- Stroke (rupture in a blood vessel of brain or spinal cord)
- Blood clotting in these blood vessels
- Poison produced by polio viruses

Effected Areas:
Patient may have weak paralysis throughout his/her body or have paralysis in one side of the body. There may also be paralysis in the lower extremities or in all four limbs.

EPILEPSY

Definition:
The nervous disorder in which there is abnormal and excessive discharge of nerve impulses in brain is called epilepsy.

Symptom:
It causes unprovoked seizures in patient. A seizure of epilepsy is a temporary abnormal state of brain marked by convulsions.

Causes:
- In younger people, epilepsy may be due to genetic or developmental causes.
- In people over age 40 years, brain tumours are more likely to cause epilepsy.
- Head trauma and central nervous system infections may cause epilepsy at any age.

Cure:
There is no known cure of epilepsy but medicines can control seizures. Patients of epilepsy have to take medicines daily for the treatment as well as prevention of seizures. These are termed "anticonvulsant" or "antiepileptic" drugs.

Precaution:
During a seizure attack, objects should never be placed in a patient's mouth as it can result in serious injury. It is possible that the patient will bite his/her own tongue.

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MULTIPLE CHOICE QUESTIONS

1. **Processes that carry nerve impulses away from the cell body are called:**
   (a) Axons (b) Dendrites (c) Synapses (d) Myelin sheath
2. **The portion of the nervous system that is involuntary in action:**
   (a) Somatic nervous system (b) Motor nervous system
   (c) Autonomic nervous system (d) Sensory nervous system
3. **Which neurons are present inside the central nervous system?**
   (a) Sensory neurons only (b) Motor neurons only
   (c) Sensory and motor neurons both (d) Interneurons only
4. **The part of the brain responsible for muscle movement, interpretation of the senses and the memory is the:**
   (a) Pons (b) Medulla oblongata (c) Cerebrum (d) Cerebellum
5. **Apart from hearing, what other major body function is performed by the ear?**
   (a) Hormone secretion (b) Body balance
   (c) Reduction in nerve pressure (d) All of these
6. **The myelin sheath is formed by _____ , which wrap around the axons of some neurons.**
   (a) Nodes of Ranvier (b) Axons (c) Dendrites (d) Schwann cells
7. **This is not a part of the hindbrain:**
   (a) Pons (b) Medulla oblongata (c) Cerebrum (d) Cerebellum
8. **If you look at an intact human brain, what you see the most is a large, highly convoluted outer surface. This is the:**
   (a) Cerebrum (b) Cerebellum (c) Pons (d) Medulla oblongata
9. **Insulin and glucagon are produced in the:**
   (a) Hypothalamus (b) Anterior pituitary (c) Liver (d) Pancreas
10. **All of these are hormones except:**
    (a) Insulin (b) Thyroxin (c) Glucagon (d) Pepsinogen

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	a	2	c	3	d	4	c	5	b
6	d	7	c	8	a	9	d	10	d

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SHORT QUESTIONS

1. **Identify the two types of coordination in living organisms.**
   Consult Short Question No. 3
2. **Differentiate between the modes of nervous and chemical coordinations.**
   Consult Long Question No. 1
3. **What are the main components of coordination?**
   Consult Short Question No. 6
4. **Define reflex action and reflex arc.**
   Consult Long Question No. 6
5. **Trace the path of a nerve impulse in case of a reflex action.**
   Consult Long Question No. 6
6. **Describe the pupil reflex in dim and bright light.**
   Consult Long Question No. 7
7. **How would you associate the role of vitamin A with vision and effects of its deficiency on retina?**
   Consult Long Question No. 8
8. **Define the terms; hormone and endocrine system.**
   Consult Short Question No. 27 and 28

UNDERSTANDING THE CONCEPT

1. **Explain what can happen if there is no coordination in the activities of organisms.**
   If there is no coordination in the activities of organisms cannot perform there activities well.

Example:

When we are writing something, our hands and fingers work in collaboration with our muscles, eyes, thoughts etc. and then very intricate movements result.

Advantage:

Coordination also enables the organism to respond to happenings in the world around it.

2. **Explain the location and function of these parts of brain; cerebrum, cerebellum, pituitary gland, thalamus, hypothalamus, medulla oblongata.**
   Consult Long Question No. 3
3. **Define neuron and describe the structure of a general neuron.**
   Consult Long Question No. 2
4. **Describe the structure of human eye.**
   Consult Long Question No. 7
5. **How would you describe the structure of the external, middle and inner ear of man?**
   Consult Long Question No. 10
6. **What are short sight and long sight problems and how these can be treated?**
   Consult Long Question No. 9
7. **Explain the role of ear in the maintenance of balance.**
   Consult Short Question No. 43
8. **Relate the contribution of Ibn-al-Haitham and AI-Ibn-Isa with knowledge abou structure of eye and treatment of various ophthalmic diseases.**
   Consult Short Question No. 40 and 41

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9. **Outline the major glands of the endocrine system (pituitary, thyroid, pancreas, adrenal, gonads), with name of their hormones and their functions.**
   Consult Long Question No. 11,12, 14, 15 and 16
10. **Describe negative feedback with reference to insulin and glucagon.**
   Consult Long Question No. 17
11. **Explain how adrenaline may be involved in exercise and emergency conditions.**
   Consult Long Question No. 14
12. **Enlist the important symptoms and treatments of paralysis and epilepsy.**
   Consult Long Question No. 18

THE TERMS TO KNOW

Acromegaly:
Abnormal growth due to excessive production of growth hormone after growing age; the internal organs and body extremities alone grow large and affected persons have large hands, feet and jaw bones

Antidiuretic hormone:
The hormone of the posterior pituitary; promotes the reabsorption of water in renal tubules

Aqueous humour:
The fluid present in the anterior chamber of the eye i. e. between the cornea and the iris

Axon:
A long, thin fibre that carries nerve impulse away from the cell body of a neuron

Cell body:
The part of the nerve cell that contains nucleus

Cerebellum:
The part of the hindbrain; controls muscle movements

Cerebral hemisphere:
The divisions of the cerebrum of the brain

Cerebrospinal fluid:
The fluid in the ventricles of the brain and in the central canal of the spinal cord

Cerebrum:
The largest part of the forebrain; controls many sensory and motor functions

Cochlea:
The part of the inner ear; consists of three ducts wrapped in the form of a coiled tube; contains sound receptors

Colour blindness:
Genetic disorder in which person fails to recognize the basic colours

Cones:
The photosensitive cells in the retina of the eye; sensitive to bright light and so distinguish different colours

Cornea:
The transparent part of sclera that forms in the front of the eye through which light enters

Cranial nerve:
Nerves that arise from or lead to the brain

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Dendrites:
Short, branched projections of neuron's cell body; transmit nerve impulses towards cell body

Diabetes mellitus:
More than normal level of glucose in blood; a condition caused by insufficient concentration of insulin in blood

Dwarfism:
Less than normal body growth; a condition caused when growth hormone is insufficient during the growing age

Ear drum:
Tympanic membrane; a membrane stretched across the inner end of the auditory canal of the ear

Effectors:
The parts of the coordination system that respond when stimulated by nerve impulses or hormones

Endocrine gland:
A ductless gland; produces and secretes hormones

Epilepsy:
A nervous disorder characterized by recurrent unprovoked seizures (convulsions)

Estrogen:
A hormone secreted by the ovaries; promotes development of secondary sex characteristics and regulates the reproductive cycle

Eustachian tube:
A tube between middle ear and the nasal cavity that equalizes the pressure on both sides of the ear drum

Exocrine gland:
A gland that discharges its secretion into a duct

Ganglion:
The aggregation of the cell bodies of the neurons

Grey matter:
The nervous tissue containing cell bodies and non-myelinated processes of the neurons

Hormone:
A substance that is secreted by an endocrine gland directly into the blood and that produces a specific effect on a particular tissue

Hypermetropia:
The condition in which a person is not able to see near objects clearly; happens when the eye ball shortens and image is formed behind the retina

Hypothalamus:
The part of the forebrain below the thalamus; controls body temperature, blood pressure and emotion

Insulin:
The hormone produced by the Islets of Langerhans; lowers blood glucose level

Interneurons:
The neurons present in the brain and spinal cord

Iodopsin:
A pigment present in the cones of retina

Iris:
A muscular ring formed by the bending of the choroid behind the cornea of the eye

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Islets of Langerhans:
Groups of endocrine cells present in pancreas; secrete hoemones insulin and glucagon

Medulla oblongata:
Part of the hindbrain; on the top of the spinal cord; controls breathing, heart rate, blood pressure and many reflexes

Meninges:
Three layers around the brain and the spinal cord; protect them and provide nutrition and oxygen through their capillaries

Mixed nerve:
The nerve containing axons of both the sensory and motor neuron

Motor nerve:
The nerve containing axons of motor neuron only

Myelin sheath:
The insulating sheath around the axons of some neurons

Myopia:
The condition in which a person is not able to see distant objects clearly; happens due to the elongation of the eye ball and image is formed in front of the retina

Nerve:
The union of several axons that are enveloped by a covering made of lipid

Nodes of Ranvier:
The non-myelinated points where the areas of myelin on the axons of neurons

Optic disc:
Blind spot; a point on the retina of the eye where the optic nerve enters the retina; no photosensitive cells exist at this point

Oxytocin:
The hormone secreted by the posterior pituitary; stimulates the contraction of uterus walls in females for child birth; necessary for ejection of milk from the breasts.

Paralysis:
Complete loss of function by one or more muscle groups due to damage in the nervous system.

Parathormone:
Hormone of the parathyroid glands; increases the level of calcium ions in the blood

Parathyroid:
The endocrine glands located on the posterior sides of the thyroid gland; secrete parathormone

Pituitary:
The endocrine gland attached to the hypothalamus that controls many other endocrine glands in the body.

Pons:
Part of the hindbrain; present on the top of medulla; assists the medulla in controlling breathing and serves as a connection between the cerebellum and the spinal cord.

Progesterone:
A hormone secreted by the ovaries that maintains the uterus during pregnancy.

Pupil:
The opening in the centre of the iris of the aye.

Receptor:
The organ, tissues or cells which detect particular type of stimuli.

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Reflex arc:
The nerve pathway over which the nerve impulses travel in a reflex action.

Retina:
The inner most and the sensitive layer of the eye.

Rhodopsin:
A pigment present in the rods of the retina.

Rods:
The photosensitive cells present in the retina of the eye; sensitive to dim light.

Schwann cells:
The supporting cells around neurons; form the myelin sheath.

Sclera:
The tough, white outer layer of the eye

Semicircular canals:
The three canals present posterior to the vestibule in the inner ear

Sensory nerve:
The nerve which contain only the axons of the sensory neurons

Somatotrophin:
Growth hormone; a hormone of the anterior pituitary; promotes the growth of the body

Spinal nerve:
The nerve which arise from the spinal cord

Suspensory ligament:
The ring that attaches the lens of the eye to the ciliary muscles

Testosterone:
The male sex hormone secreted by testes; stimulates the development of male reproductive system and the male secondary sex characters.

Thalamus:
The part of the forebrain; serves as a relay centre between various parts of the brain and spinal cord

Thyroid:
The endocrine gland located in front of the trachea; secretes hormones thyroxin and calcitonin

Thyroxin:
The hormone of the thyroid gland; increases the breakdown of the food and release of energy; also responsible for the growth of the body

Tympanum:
Tympanic memebrane; ear drum

Vasopressin:
Antidiuretic hormone; the secreted by the posterior pituitary; responsible for the reabsorption of water from renal tubules of the nephron

Vestibule:
Part of the inner ear; helps to maintain balance of the body

Vitreous humour:
The fluid present in the posterior chamber of the eye i. e. between the iris and retina

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CHAPTER 13

SUPPORT AND MOVEMENT

MULTIPLE CHOICE QUESTIONS

Each question has four possible answers. Circle the correct answer

Act of changing place or position by entire body or its parts:

(a) Locomotion (b) Movement (c) Support (d) Coordination

Skeleton provides:

(a) Physical support (b) Attachment for muscles

(c) Protection for the bodies of animals (d) All of these

The cells of cartilage are called:

(a) Osteocytes (b) Chondrocytes (c) Erythrocytes (d) Leucocytes

Type of cartilage in inter-vertebral discs:

(a) Hyaline (b) Elastic (c) Fibrous (d) None of these

Hardest connective tissue in the body:

(a) Bone (b) Cartilage (c) Tendon (d) Ligament

The interior part of bone which is soft and porous is called:

(a) Compact bone (b) Spongy bone (c) Tendon (d) Cartilage

Spongy bone contains:

(a) Blood vessels (b) Bone marrow (c) Both a & b (d) None of these

Number of bones in a baby:

(a) 190 (b) 206 (c) 300 (d) 256

Mature bone cells are called:

(a) Thrombocytes (b) Chondrocytes (c) Erythrocytes (d) Osteocytes

Date of birth of Andreas Vesalius:

(a) 1510 (b) 1512 (c) 1514 (d) 1516

Total number of bones in axial skeleton:

(a) 60 (b) 80 (c) 70 (d) 90

Number of cranial bones:

(a) 8 (b) 10 (c) 12 (d) 14

Number of facial bones:

(a) 12 (b) 14 (c) 16 (d) 18

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(14) Total middle ear ossicles in human body:

(a) 3 (b) 6 (c) 9 (d) 12

(15) Number of pairs of ribs in human skeleton:

(a) 6 (b) 12 (c) 18 (d) 24

(16) Total number of bones in both hands:

(a) 14 (b) 28 (c) 42 (d) 56

(17) Number of bones in each foot:

(a) 56 (b) 42 (c) 28 (d) 14

(18) The longest bone in our body:

(a) Thigh bone (b) Hand bone (c) Foot bone (d) Hip bone

(19) The upper jaw is fixed with:

(a) Vertebral column (b) Skull (c) Both a & b (d) None of these

(20) Example of ball-and-socket joint:

(a) Skull (b) Vertebrae (c) Elbow (d) Hip

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	b	2	d	3	b	4	c	5	a
6	b	7	c	8	c	9	d	10	c
11	b	12	a	13	b	14	b	15	b
16	d	17	c	18	a	19	b	20	d

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SHORT QUESTIONS

Q. No. 1 Why organisms need support?

NEED OF SUPPORT

The organisms with greater sizes need support to keep their body mass as one unit. This is particularly true for organisms that live on land.

Q. No. 2 What is movement?

MOVEMENT

The act of changing place or position by entire body or its parts of an organism is called movement.

Types of movement:

There are two types of movement:

- Movements of body parts
- Locomotion

Q. No. 3 Define locomotion.

LOCOMOTION

The movement of an animal as a whole from one place to another is called locomotion.

Q. No. 4 Define skeleton.

SKELETON

The framework of hard articulated structures that provides physical support, attachment for skeletal muscles and protection for the bodies of animals is called skeleton.

Q. No. 5 What are the advantages of skeleton? Or
What does skeleton provide?

ADVANTAGES OF SKELETON

The skeleton provides:

- Physical support
- Attachment for skeletal muscles
- Protection for the bodies of animals

Q. No. 6 What is the difference between endoskeleton and exoskeleton?

DIFFERENCE BETWEEN ENDOSKELETON AND EXOSKELETON

Endoskeleton	Exoskeleton
Definition: The skeleton which is inside of the body is called endoskeleton. ***Example:*** • Man	***Definition:*** The skeleton which is on the outside of the body is called exoskeleton. ***Example:*** • Arthropods

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Q. No. 7 What is the role of skeletal system?

ROLE OF SKELETAL SYSTEM

The big functions of the skeletal system are:

Protection:

Skeleton provides protection to many internal organs. For example:

- Skull protects brain
- Vertebral column protects spinal cord
- Ribs protect most of the internal organs

Support:

Vertebral column provides the main support to the body mass.

Movements:

In our body, skeleton works very closely with the muscular system to help us move.

Q. No. 8 What are tendons and ligaments?

TENDONS AND LIGAMENTS

Tendons and ligaments are a type of connective tissues that contain tightly packed collagen fibres.

Q. No. 9 How many bones are present in a baby and an adult?

NUMBER OF BONES

The babies are born with about 300 soft bones. Some of these bones later fuse together, so that the adult skeleton has 206 hard bones.

Q. No. 10 What is the contribution of Andreas Vesalius?

CONTRIBUTION OF ANDREAS VESALIUS

Period: 1514 – 1564

Place of Birth: He was born in Brussels, Belgium.

Contribution:

He is honoured for developing modern anatomical studies. He made many discoveries in anatomy, based on studies made by dissection of human dead bodies.

Book Contents:

His book contained the most accurate depictions of the whole skeleton and muscles of the human body.

Q. No. 11 Which one is the longest bone in the body?

LONGEST BONE

The thigh bone is the longest bone in the body.

Q. No. 12 Discuss the evolution of ear bones and jaws in mammals.

EVOLUTION OF EAR BONES AND JAWS IN MAMMALS

The upper jaw is fixed with the skull and is composed of two bones. The lower jaw is mobile and articulates with the skull. In lower vertebrates, the lower jaw is made up of more than one bone while in mammals, it is made up of single bone.

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During evolution, mammals modified the lower jaw bones and incorporated four of them into the middle ear (in the form of malleus and incus in both ears). This adaptation proved beneficial for mammals. Lower jaw with single bone is stronger and the malleus and incus also improve hearing

Q. No. 13 What do you know about the movement of neck joint?

MOVEMENT OF NECK JOINT

The neck joint between vertebral column and head allows movements side to side.

Q. No. 14 Can muscles push?

No, muscles can not push. The muscles can only pull or contract.

Q. No. 15 Write some activities that require combined action of several muscles.

Most activities in our body require combined action of several muscles, like:

- Walking
- Running
- Playing

Q. No. 16 Why incidence of osteoporosis is more common in old females?

OSTEOPOROSIS IN OLD FEMALES

It is one of the functions of estrogen to deposit minerals in bones. In old age, when the reproductive cycle stops in females, not enough estrogen is secreted.

Q. No. 17 Which point of attachment is pulled when a muscle contracts?

Insertion is pulled when a muscle contracts.

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LONG QUESTIONS

Q. No. 1 Write a note on cartilage.

CARTILAGE

Introduction:
Cartilage is a dense, clear, blue-white firm connective tissue.

Strength:
Cartilage is less strong than bone.

Chondrocytes:
The cells of cartilage are called chondrocytes.

Lacuna:
Each chondrocyte lies in a fluid space called lacuna present in the matrix of cartilage.

Collagen Fibres:
The matrix of the cartilage contains collagen fibres.

Blood Circulation:
Blood vessels do not enter into the cartilage.

TYPES OF CARTILAGE

There are three types of cartilage:

1. Hyaline Cartilage
2. Elastic Cartilage
3. Fibrous Cartilage

1. Hyaline Cartilage:
The hyaline cartilage is strong yet flexible.

Presence:
It is found in:

- Nose
- Larynx
- Trachea
- Bronchial tubes
- Covering the ends of the long bones

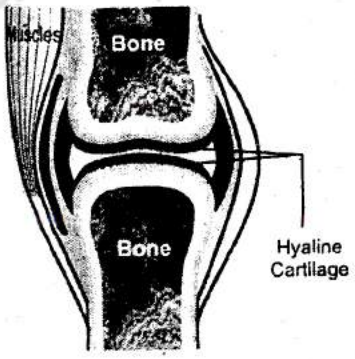

Figure: Hyaline Cartilage

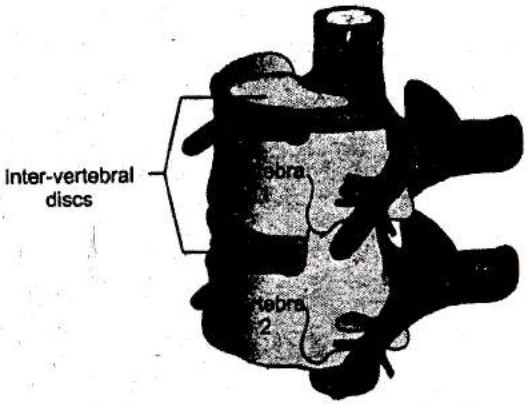

Figure: Fibrous Cartilage

2. Elastic Cartilage:
Elastic cartilage is similar in structure to the hyaline cartilage. It is also quite strong but has elasticity due to the network of elastic fibres in addition to collagen fibres.

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Presence:

It is found in:

- Epiglottis
- Pinna

3. **Fibrous Cartilage:**

Fibrous cartilage is very tough and less flexible due to the large number of thick collagen fibres present in knitted form.

Presence:

It is found in:

- Intervertebral discs

Q. No. 2 Write a note on bone.

BONE

Bone is the hardest connective tissue in the body.

Compact Bone:

The hard outer layer of the bone is called compact bone.

Spongy Bone:

The interior of the bone is soft and porous and is called spongy bone. Spongy bone contains blood vessels and bone marrow.

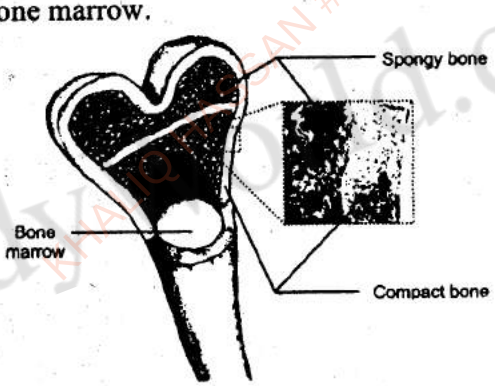

Figure: Compact and Spongy Bone

Composition of Matrix:

The matrix of the bones contains:

- Collagen
- Calcium
- Phosphate

Osteocytes:

Bone contains different types of cells. The mature bone cells are called osteocytes.

Number:

The babies are born with about 300 soft bones. Some of these bones later fuse together, so that the adult skeleton has 206 hard bones.

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Functions:

The bones perform the following functions:

- Movement
- Support
- Protection
- Storage of minerals
- Production of red and white blood cells

Q. No. 3 Write a note on human skeleton.

HUMAN SKELETON

The 206 bones in the adult human skeleton are organized into the longitudinal axis i.e. axial skeleton, to which appendicular skeleton is attached.

AXIAL SKELETON

Number of Bones:

Axial skeleton consists of the 80 bones in the head and trunk of the body.

Parts of Axial Skeleton:

It is composed of five parts:

1. **Skull:**

Skull contains 22 bones out of which 8 are cranial bones (enclosing the brain) and 14 are facial bones.

2. **Middle Ear Ossicles:**

There are 6 middle ear ossicles (3 in each ear).

3. **Hyoid Bone:**

There is a hyoid bone in the neck.

4. **Vertebral Column:**

Vertebral column contains 33 bones (vertebrae).

5. **Chest:**

The chest is made up of a chest bone calles sternum and 24 (12 pairs) ribs.

APPENDICULAR SKELETON

Number of Bones:

Appendicular skeleton is composed of 126 bones.

Pectoral Girdle:

Pectoral (shoulder) girdle is made up of 4 bones.

Arms:

Arms have 6 bones.

Hands:

Both hands have 56 bones.

Pelvic Girdle:

Pelvic girdle (hips) has 2 bones.

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Legs:

Legs have 8 bones.

Feet:

Both feet have 56 bones.

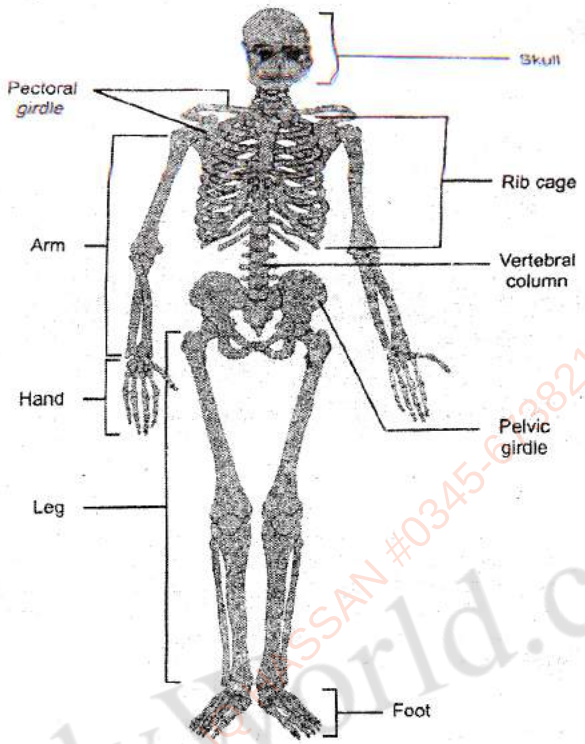

Figure: Human Skeleton

Q. No. 4 What is a joint? Explain different types of joints.

JOINT

Definition:

The location at which two or more bones make a contact is called as joint.

Functions:

The joints:

- Allow movement
- Provide mechanical support

Basis of classification:

The joints can be classified on the basis of the degree of movement they allow.

Types of joints

Following are the three main types of joints:

1. Immovable (fixed) Joints
2. Slightly Movable Joints
3. Movable Joints

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1. **Immovable (fixed) Joints:**
Such joints allow no movement.
Example:
- Joints between the skull bones

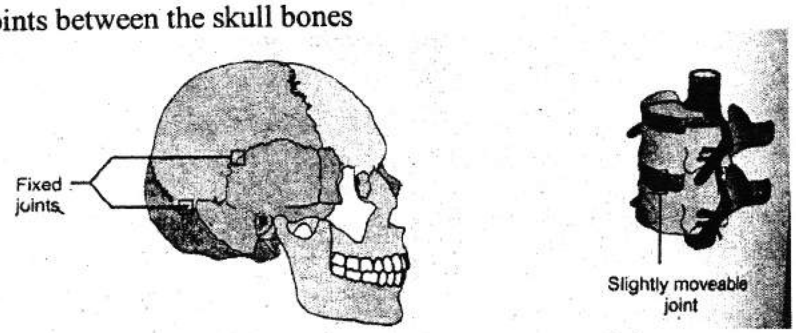
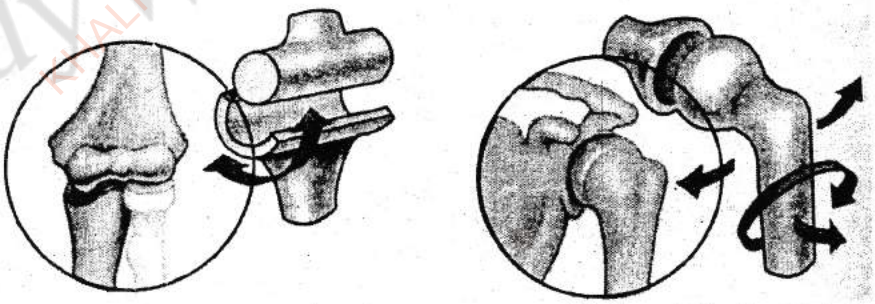


Figure: Fixed and Slightly Moveable Joints

2. **Slightly Movable Joints:**
Such joints allow slight movements.
Example:
- Joints between the vertebrae

3. **Movable Joints:**
They allow a variety of movements. There are many types of movable joints in the body. The main types are as follow:

Hinge Joints:
These joints move back and forth like the hinge on the door and allow movements in one plane only.
Examples:
- Knee joint
- Elbow joint



Hinge Joint **Ball-and-Socket Joint**

Figure: Two Types of Moveable Joints

Ball-and-Socket Joints:
These joints allow movements in all directions.
Examples:
- Hip joint
- Shoulder joint

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Q. No. 5 Explain the roles of tendons and ligaments.

ROLE OF TENDONS AND LIGAMENTS

Introduction:

Tendons and ligaments are bands of connective tissues.

Composition:

These are made of collagen.

Tendons:

Tendons are tough bands and attach muscles to bones.

Function:

When a muscle contracts tendon exerts a pulling force on the attached bone, which moves as a result.

Ligaments:

Ligaments are strong but flexible bands and join one bone to another at joints.

Function:

They prevent dislocation of bones at joints.

Q. No. 6 **Explain the action of antagonist muscles at elbow joint.**

ACTION OF ANTAGONIST MUSCLES AT ELBOW JOINT

Movements in Bones:

The movements in bones are brought about by the contractions of skeletal muscles, which are attached with them by tendons.

Origin:

One end of a skeletal muscle is always attached with some immovable bone. This end of muscle is called the origin.

Insertion:

The other end of the muscle is attached with a movable bone and is called the insertion.

Contraction of Muscle:

When a muscle is stimulated by a nerve impulse, it contracts to become shorter and thicker. Due to its contraction, it pulls the movable bone.

Antagonism:

Skeletal muscles are usually in pairs of antagonists. In the antagonist pair, both muscles do opposite jobs. When one muscle contracts the other relaxes and this phenomenon is known as antagonism.

Flexor:

When a muscle contracts and bends the joint, it is known as flexor muscle.

Flexion:

This bending movement done by flexor muscle is called flexion.

Extensor:

When a muscle contracts and straightens the joint, it is known as extensor muscle.

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Extension:

This straightening movement done by extensor muscle is called extension.

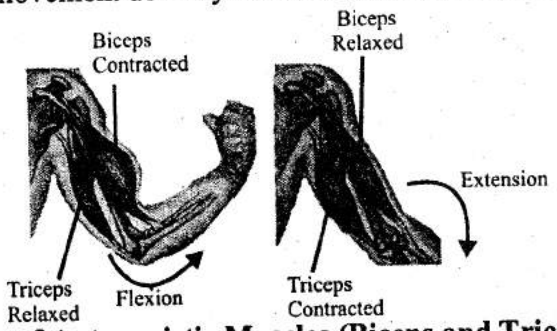

Figure: Action of Antagonistic Muscles (Biceps and Triceps) at Elbow

Biceps:

Biceps is a flexor muscle on the front of the upper arm bone.

Triceps:

Triceps is an extensor muscle on the back of arm.

Origin and Insertion:

Both these muscles have their origin at pectoral girdle and insertion at one of the two bones of forearm.

Contraction of Biceps:

When biceps contracts, the forearm (insertion end) is pulled upward. It is the flexion of elbow joint. During this flexion, triceps muscle relaxes.

Contraction of Triceps:

When triceps muscle contracts, forearm is pulled down. It is the extension at elbow joint. During this, biceps muscle relaxes.

Antagonistic Pair of Muscles:

In this way, biceps and triceps make up an antagonist pair of muscles. Similar pairs, working antagonistically across other joints, provide for almost all the movements of skeleton.

Q. No. 7 Write a note on osteoporosis.

OSTEOPOROSIS

Introduction:

Osteoporosis is the bone disease in adults, especially in old people. It is more common in old women.

Symptom:

In osteoporosis, there is a decrease in the density of bones due to the loss of calcium and phosphorus.

Causes:

It may be due to:

- Malnutrition
- Lack of proteins and vitamin C
- Lack of physical activities
- Deficiency of estrogen hormone

Old Age:

In old age, there is decreased secretion of growth hormones and it also leads to decreased deposition of minerals in bone matrix.

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Q. No. 8 Write a note on arthritis.

ARTHRITIS

Meaning:

Arthritis means "inflammation of joints".

Victims:

It is very common in old age and in women.

Symptoms:

It is characterized by:

- Pain
- Stiffness in joints

Most Affected Joints:

The most affected joints are the weight bearing joints. For example:

- Hip joint
- Ankle joint

TYPES OF ARTHRITIS

There are many types of arthritis. Some are as follow:

1. Osteo-arthritis
2. Rheumatoid Arthritis
3. Gout

1. Osteo-arthritis:

Causes:

It is due to:

- Degeneration in the cartilage present at joints
- Decreased lubricant production at joints

Outcomes:

In this arthritis, fusion of bones at joints may occur and joints may become totally immovable.

2. Rheumatoid Arthritis:

It involves the inflammation of the membranes at joints.

Symptoms:

Its symptoms include:

- Fatigue
- Low grade fever
- Pain in joints
- Stiffness in joints

3. Gout:

It is characterized by the accumulation of uric acid crystals in movable joints. It generally attacks the toe joints.

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MULTIPLE CHOICE QUESTIONS

1. **Find the ball-and-socket joint.**
   (a) Joint in the finger bones
   (b) Joint of neck and skull bones
   (c) Joint at elbow
   (d) Joint at pelvic girdle and leg bones
2. **All these are the parts of axial skeleton of humans except:**
   (a) Ribs
   (b) Sternum
   (c) Shoulder girdle
   (d) Vertebral column
3. **The disorders in which there is an accumulation of uric acid in joints:**
   (a) Gout
   (b) Rheumatoid arthritis
   (c) Osteoporosis
   (d) Osteo-arthritis
4. **What is correct about tendons?**
   (a) Tendons are flexible and they join muscles with bones
   (b) Tendons are non-elastic and they join bones with bones
   (c) Tendons are non-elastic and they join muscles with bones
   (d) Tendons are flexible and they join muscles with muscles
5. **How many bones make our skull?**
   (a) 14
   (b) 22
   (c) 24
   (d) 26
6. **What are the main components of a bone?**
   (a),Marrow, spongy bone, wax
   (b) Marrow, compact bone, wax
   (c) Compact bone and marrow
   (d) Compact bone, spongy bone, marrow
7. **What do some bones produce?**
   (a) Mucous
   (b) Hormones
   (c) Oxygen
   (d) Blood cells
8. **How would you define skeletal system?**
   (a) All the bones in body
   (b) All the muscles and tendons
   (c) All the body's organs, both soft and hard tissues
   (d) All the bones in body and the tissues that connect them
9. **Find the INCORRECT statement.**
   (a) Bone is where 'most blood cells are made
   (b) Bone serves as a storehouse for various minerals
   (c) Bone is a dry and non-living supporting structure
   (d) Bone protects and supports the body and its organs
10. **The purpose of rib cage is to:**
    (a) Protect the stomach
    (b) Protect the spinal cord
    (c) Protect the heart and lungs
    (d) Provide an object to which the lungs can attach

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	c	3	a	4	c	5	b
6	d	7	d	8	d	9	c	10	c

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SHORT QUESTIONS

1. Differentiate between cartilage and bone

Cartilage	Bone
Introduction: Cartilage is a dense, clear, blue-white firm connective tissue.	***Introduction:*** Bone is the hardest connective tissue in the body.
Strength: Cartilage is less strong than bone.	***Strength:*** Bone is more strong than cartilage.
Chondrocytes: The cells of cartilage are called chondrocytes.	***Osteocytes:*** Bone contains different types of cells. The mature bone cells are called osteocytes.
Types: • Hyaline Cartilage • Elastic Cartilage • Fibrous Cartilage	***Types:*** • Compact Bone • Spongy Bone

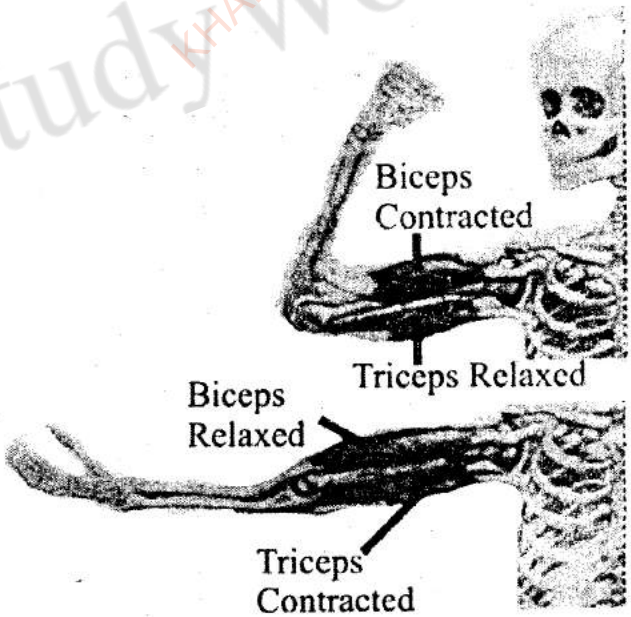
2. What is the role of skeleton in support and movement?

Consult Short Question No. 7

3. How would you differentiate between osteoporosis and arthritis?

Consult Long Question No. 7 and 8

4. Label the biceps and triceps in the following diagrams and also mention their contracted or relaxed states.



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UNDERSTANDING THE CONCEPT

1. **What are the main components of the axial skeleton and the appendicular skeleton of human?**

Consult Long Question No. 3

2. **Describe the types of joints and give examples.**

Consult Long Question No. 4

3. **What are ligaments and tendons? What function do they perform?**

Consult Long Question No. 5

4. **Explain antagonism in muscle action selecting biceps and triceps as example.**

Consult Long Question No. 6

THE TERMS TO KNOW

Antagonism:
Skeletal muscles are usually in pairs of antagonists. In the antagonist pair, both muscles do opposite jobs. When one muscle contracts the other relaxes and this phenomenon is known as antagonism.

Appendicular Skeleton:
The division of the skeleton that includes arms, hands, legs, feet, pectoral girdle and pelvic girdle

Arthritis:
Term used for the inflammation of the joints

Axial Skeleton:
The division of the skeleton that includes the skull, vertebral column, ribs and breast bone

Ball-and-Socket Joint:
The joint that allows movement in all directions e. g. hip and shoulder joints

Biceps:
A flexor muscle on the front of the upper arm bone

Bone:
Hard connective tissue; moves, supports and protects the various organs of the body

Cartilage:
The connective tissue that makes part of the human skeleton

Chondrocytes:
The cells present in the cartilage

Compact Bone:
The hard outer layer of bones

Cranial Bones:
The bones in the cranium

Extensor:
A muscle that extends a joint

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Fibrous Cartilage:
The cartilage that has large number of fibres in the matrix e. g. the cartilage in intervertebral discs

Flexor:
The muscle that bends a joint

Gout:
A type of arthritis, characterized by the accumulation of uric acid crystals in the movable joints

Hinge Joint:
A joint that permits movement of bones in one plane e. g. elbow and knee joints

Hyaline Cartilage:
The cartilage that has collagen fibres in its matrix; found covering the ends of long bones, in the nose, larynx, trachea and bronchial tubes

Insertion:
The end of the muscle that is attached with a movable bone

Joint:
The location at which two or more bones make contact

Lacuna:
The fluid filled space in bone and cartilage, where the cells are present

Ligament:
Strong but flexible connective tissue that joins one bone to bone at the joints

Origin:
The end of the muscle that is attached with the immovable bone

Osteoarthritis:
Inflammation in joints due to the degeneration of the cartilage present at the joints or due to decreased lubricant production at the joint

Osteocyte:
The mature bone cell

Osteoporosis:
A bone disease in adults, especially in old age, there is a decrease in the density of bones due to the loss of calcium and phosphorus

Rheumatoid Arthritis:
Painful inflammation of the membranes at the joints

Skeleton:
The framework of hard, articulated structures that provide physical support, attachment for skeletal muscles, and protection for the bodies of the animals

Spongy Bone:
The soft and porous interior of the bone; contains blood vessels and bone marrow

Sternum:
The chest bone

Tendon:
Tough connective tissue that attaches muscles to bones

Triceps:
Triceps is an extensor muscle on the back of arm.

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CHAPTER 14

REPRODUCTION

MULTIPLE CHOICE QUESTIONS

Each question has four possible answers. Circle the correct answer

(1) If an organism divides into two organisms, this division is called:
(a) Multiple fission (b) Binary fission (c) Fragmentation (d) Buddding

(2) In which organism the bud is not detached from the parent body?
(a) Yeast (b) Hydra (c) Coral (d) Amoeba

(3) Spore formation takes place in:
(a) Planaria (b) Amoeba (c) Hydra (d) Rhizopus

(4) Vegetative propagation in onion:
(a) Corm (b) Stem tuber (c) Rhizome (d) Bulb

(5) Which plant has vegetative propagation through rhizome?
(a) Ginger (b) Colocacia (c) Potato (d) Onion

(6) An example of stem tubers:
(a) Mint (b) *Chrysanthemum* (c) *Bryophyllum* (d) Yams

(7) An example of suckers:
(a) Potato (b) Yams (c) Onion (d) Mint

(8) The sporophyte generation is:
(a) Haploid (b) Diploid (c) Triploid (d) Tetraploid

(9) The generation which produces gametes:
(a) Sporophyte (b) Gametophyte (c) Both a & b (d) None of these

(10) Sporophyte generation produces spores by:
(a) Meiosis (b) Mitosis (c) Budding (d) Fragmentation

(11) Which one is the reproductive part of the plant?
(a) Stem (b) Root (c) Flower (d) Leaf

(12) The outermost whorl of flower:
(a) Androecium (b) Gynoecium (c) Corolla (d) Calyx

(13) Male reproductive part of the flower:
(a) Sepal (b) Patel (c) Carpe. (d) Stamen

(14) The transfer of pollens from anther of stamen to stigma of carpel:
(a) Reproduction (b) Pollination (c) Budding (d) Respiration

(15) The leaflets of corolla are called:
(a) Stamens (b) Carpels (c) Sepals (d) Petals

(16) The carpel consists of:
(a) Stigma (b) Style (c) Ovary (d) All of these

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(17) **The endosperm is:**
(a) Haploid (b) Diploid (c) Triploid (d) Tetraploid

(18) **Which part of carpel develops into fruit?**
(a) Stigma (b) Style (c) Ovary (d) None of these

(19) **Insect pollinated flowers:**
(a) Generally large (b) Generally small (c) Produce nectar (d) Both a & c

(20) **The scar on the seed coat where the seed is attached to the ovary wall:**
(a) Hilum (b) Hilus (c) Radicle (d) Plumule

(21) **Which one is an example of wind pollinated flower?**
(a) Buttercup (b) Rose (c) Sunflower (d) Willow

(22) **Angiosperm seeds consist of how many distinct parts:**
(a) 1 (b) 2 (c) 3 (d) 4

(23) **Which one part of the seed develops into root?**
(a) Plumule (b) Radicle (c) Endosperm (d) Testa

(24) **Which one is an example of epigeal germination?**
(a) Pea (b) Maize (c) Coconut (d) Cotton

(25) **Optimum temperature for the germination of seed:**
(a) 10-15°C (b) 25-30°C (c) 20-30°C (d) 15-25°C

(26) **The production of sperms in testes:**
(a) Spermatogenesis (b) Oogenesis (c) Reproduction (d) Fertilization

(27) **In which group of vertebrates external fertilization occurs?**
(a) Reptiles (b) Mammals (c) Amphibians (d) Birds

(28) **Testes are located in a bag of skin called:**
(a) Epididimys (b) Urethra (c) Scrotum (d) Semen

(29) **Pakistan's population in the year 2007-2008:**
(a) 163,000,000 (b) 163,775,000 (c) 163,075,000 (d) 173,775,000

(30) **In how many countries UNFPA works:**
(a) 130 (b) 135 (c) 140 (d) 145

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ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	b	2	c	3	d	4	d	5	a
6	d	7	d	8	b	9	b	10	a
11	c	12	d	13	d	14	b	15	d
16	d	17	c	18	c	19	c	20	a
21	d	22	c	23	b	24	d	25	b
26	a	27	c	28	c	29	b	30	c

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SHORT QUESTIONS

Q. No. 1 Define reproduction

REPRODUCTION

The production of individuals of the same species i. e. the next generation of species is called reproduction.

Q. No. 2 Is reproduction essential for life processes?

REPRODUCTION FOR LIFE PROCESSES

Reproduction is not essential for life processes, it is one of the fundamental characteristics of the living things.

Q. No. 3 What is the difference of survival between individual and species?

An individual can live without reproduction, but a species can not survive without reproduction.

Q. No. 4 What is the importance of reproduction?

IMPORTANCE OF REPRODUCTION

- Reproduction is essential for the continuation of species.
- It ensures that the genetic material of one generation is transmitted to the next.

Q. No. 5 How advantageous characteristics are transmitted to the next generation?

TRANSMISSION OF ADVANTAGEOUS CHARACTERISTICS

Each generation produces more offsprings than the next generation. Many individuals die before reaching the reproductive age due to various reasons like:

- Diseases
- Competition
- Genetic factors

Survival of the Fittest:

Only the fittest and the best survive reach the reproductive age. In this way the advantageous characteristics are transmitted to the next generation.

Q. No. 6 Differentiate between sexual and asexual reproduction.

DIFFERENCE BETWEEN SEXUAL AND ASEXUAL REPRODUCTION

Sexual Reproduction	Asexual Reproduction
• The simple cell division that produces an exact duplicate of an organism is called asexual reproduction. • It does not involve the fusion of gametes.	• The process of reproduction that involves the fusion of male and female gametes is called sexual reproduction. • The male gametes are called sperms and female gametes are called eggs.

Q. No. 7 Write names of major plant groups.

MAJOR PLANT GROUPS

The major plants groups are:

1. Mosses
2. Ferns
3. Seed plants

The seed plants include:
Gymnosperms (naked seeds)
Angiosperms (flowering plants)

Q. No. 8 Why mosses and ferns require water for their sexual reproduction?

REQUIREMENT OF WATER FOR REPRODUCTION

In mosses and ferns sperms are motile and can swim to egg cells. Therefore, these plants require water in the form of dew or rain for sexual reproduction.

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Q. No. 9 Why the method of reproduction is also called micro-propagation?

MICRO-PROPAGATION

The method of reproduction is also called micro-propagation sine it uses only a small part of plant.

Q. No. 10 What is a whorl of the flower?

WHORL OF FLOWER

All the structures present at one are collectively called the whorl of the flower.

Q. No. 11 What is the contribution of Theophrastus?

CONTRIBUTION OF THEOPHRASTUS

He was the successor of Aristotle. He was great philosopher.

- He laid a solid foundation of botany including the morphology and functions of the flowers.
- He recognized the male and female sex parts of the flowers and described the pollination and fertilization in flowers.

Q. No. 12 What is parthenocarpy?

PARTHENOCARPY

In some plants, ovaries develop into fruit without the fertilization inside their ovules. This process is known as parthenocarpy and it results in seedless fruits.

Examples:

- Bananas
- Seedless varieties of grapes

Q. No. 13 What is the reason of success of seed plants?

SUCCESS OF SEED PLANTS

The evolution of seed has been proved as an important step in the success and spread of flowering plants, as compared to the seed-less plants like mosses and ferns.

Q. No. 14 What is dormancy?

DORMANCY

Most seeds go through a period, during which there is no growth. This period is called the dormancy of the seed. Dormant seeds are ripe seeds but do not germinate. Under favourable conditions, the seeds break dormancy and begin to germinate.

Q. No. 15 What is the future of ovule and ovary after fertilization in flower?

FUTURE OF OVULE AND OVARY

After fertilization, the ovule develops into seed while ovary develops into fruit.

Q. No. 16 Why do rabbits reingest their faeces?

REINGESTION OF FAECES

Rabbits reingest their own pallet-like faeces to digest their food further and extract sufficient nutrients.

Q. No. 17 What do you know about The United Nations Population Fund?

THE UNITED NATIONS POPULATION FUND

UNFPA began operations in 1969. It is the largest international organization funding for population and health programmes. The UNFPA works in over 140 countries, for awareness about the consequences of overpopulation.

Q. No. 18 What is the incidence of HIV in Pakistan?

INCIDENCE OF HIV IN PAKISTAN

According to the United Nations Programme on AIDS i.e. UNAIDS estimates, some 70,000 to 80,000 persons, or 0.1 percent of the adult population in Pakistan, are infected with HIV.

Q. No. 19 What is the number of drug addicts in Pakistan?

DRUG ADDICTS IN PAKISTAN

The number of drug addicts in Pakistan is currently estimated to be about 500,000 of whom 60,000 inject drugs.

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LONG QUESTIONS

Q. No. 1 Write a note on binary fission.

BINARY FISSION

Meaning:

Binary fission means "division into two".

Definition:

The simplest and most common method of asexual reproduction in which an organism divides into two by simple cell division is called binary fission.

Examples:

- It occurs in prokaryotes (bacteria)
- Many unicellular eukaryotes e.g. protozoa
- Some invertebrates

Binary Fission in Bacteria:

During binary fission in bacteria, the DNA is duplicated and so two copies of DNA are formed. The two copies move towards the opposite poles of cell. The cell membrane invaginates in centre and divides the cytoplasm into two. New cell wall is deposited between two cross membranes. It results in the formation of two daughter bacteria, which grow in size and divide again.

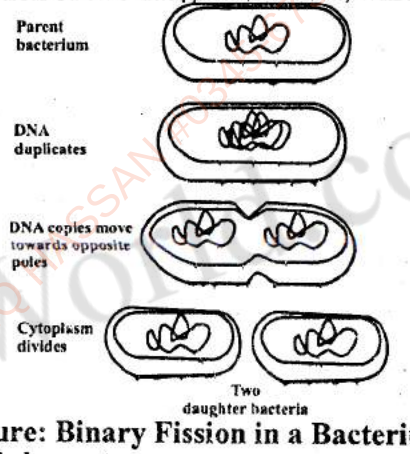

Figure: Binary Fission in a Bacterium

Binary Fission in Unicellular Eukaryotes:

During binary fission in unicellular eukaryotes, the nucleus of parent organism divides into two by mitosis. It is followed by the division of cytoplasm. So two daughter cells of almost equal size, are formed. Daughter cells grow in size and then divide again.

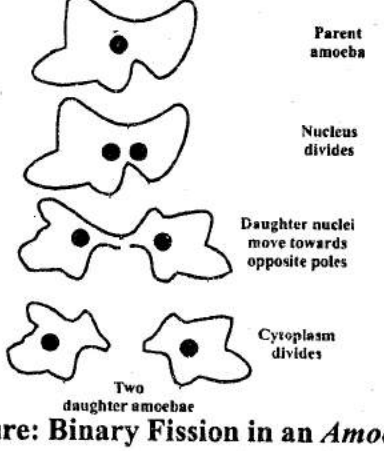

Figure: Binary Fission in an *Amoeba*

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Binary Fission in Invertebrates:

Some invertebrates also reproduce asexually through binary fission. During this reproduction, body is cut into two halves (fission) and the missing body parts are regenerated in both halves. This type of asexual reproduction is common in:

- Planarian
- *Many echinoderms*

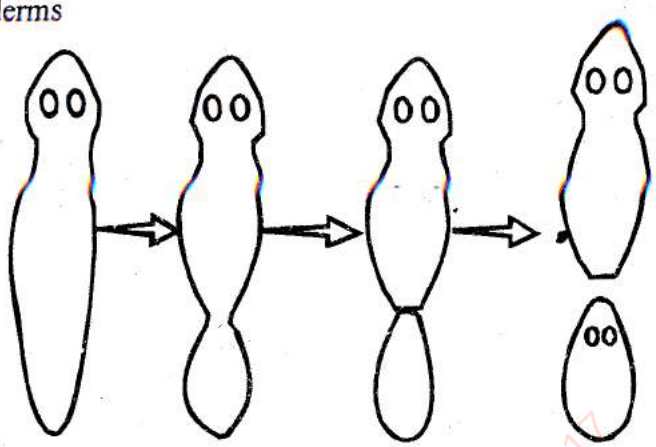

Figure: Binary Fission in a Planarian

Q. No. 2 Write a note on multiple fission.

MULTIPLE FISSION

Definition:

The kind of asexual reproduction in which one unicellular organism divided into many daughter cells is called multiple fission.

Example:

- Amoeba

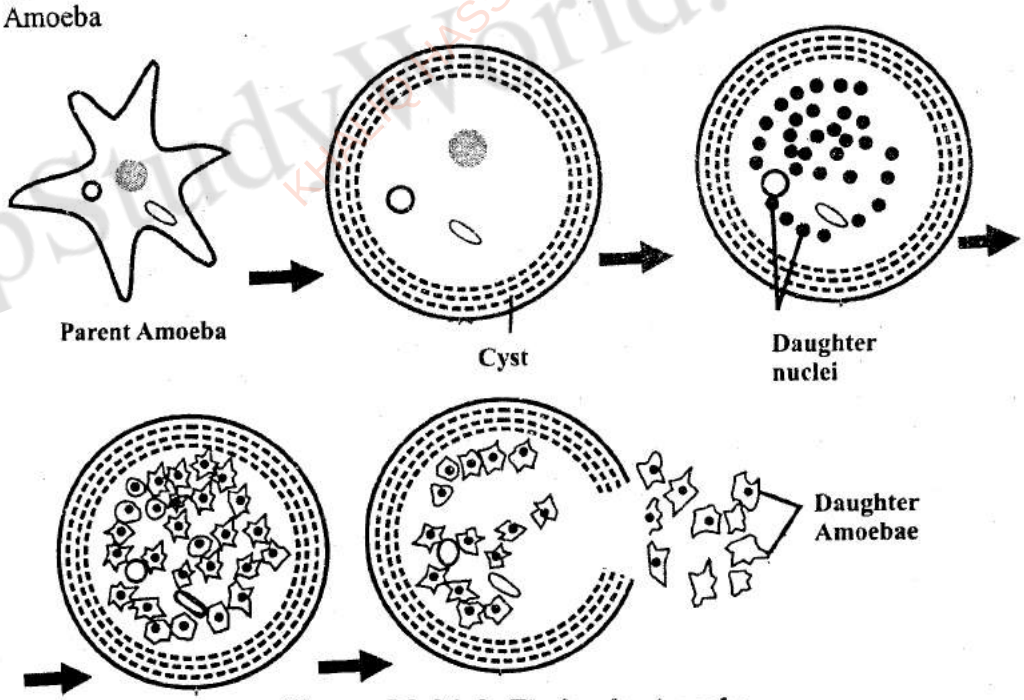

Figure: Multiple Fission in *Amoeba*

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Mechanism:

Formation of Cyst:

Some unicellular organisms form hard walls called cysts around them, under unfavourable conditions.

Division of Nucleus:

When favourable conditions return, the nucleus of parent divides into many daughter *nuclei by repeated divisions.*

Division of Cytoplasm:

This is followed by the division of cytoplasm into several parts. Each new part of cytoplasm encloses one nucleus.

Formation of Daughter Cells:

So a number of daughter cells are formed from a single parent at the same time. This kind of fission in known as multiple fission.

Q. No. 3 Write a note on fragmentation.

FRAGMENTATION

Definition:

A kind of asexual reproduction in which the detached parts of the organism develop into new organism is called fragmentation:

Examples:

Worms:

As certain worms grow to full size, they spontaneously break up into 8 or 9 pieces. Each piece (fragment) develops into a mature worm, and the process is repeated.

Planaria:

If a planarian breaks into many pieces instead of two, it will also be called as fragmentation.

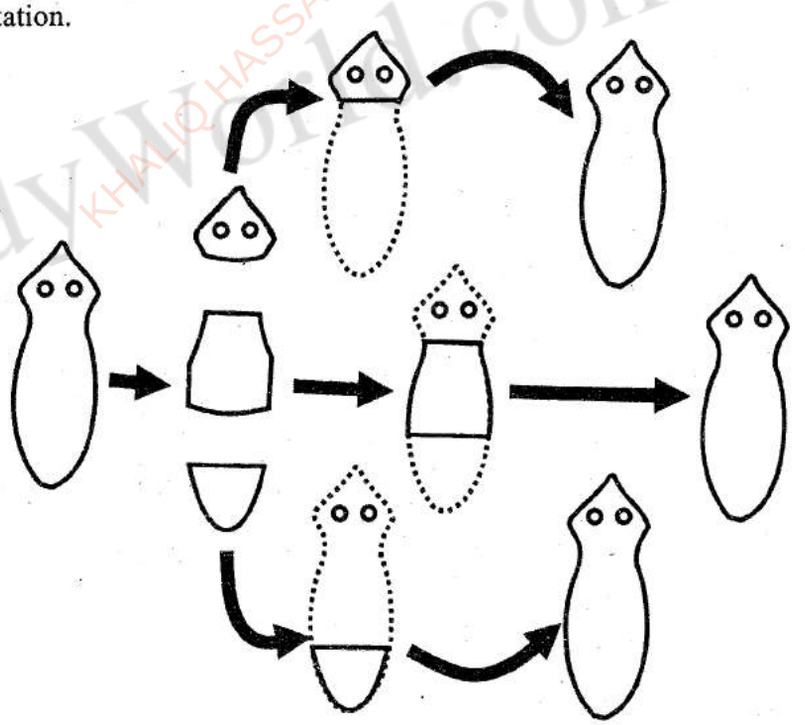

Figure: Fragmentation in a Planarian

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Q. No. 4 Write a note on budding.

BUDDING

Definition:

A type of asexual reproduction in which an outgrowth on the body of an organism develops into a new complete organism is called budding.

Bud:

The developed outgrowth on the body of an organism is called bud.

Explanation:

In this type of asexual reproduction, a bud develops as a small outgrowth on parent's body. The bud may separate from parent body. In some cases, the buds never separate and as a result, colonies of individuals are formed.

Examples:

- Yeasts
- Sponges
- Hydra
- Corals

Budding in Yeast:

Yeast is a unicellular fungus. A small bud is formed on one side of cell. The nucleus of cell divides and one of the daughter nuclei is passed into the bud. Parent cell may form more than one bud at a time. Each bud enlarges and develops the characteristics of parent organism.

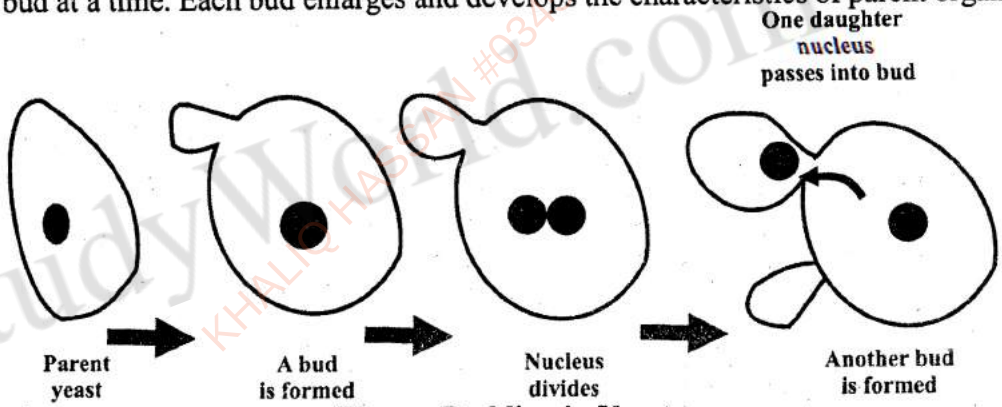

Figure: Budding in Yeast

Budding in Hydra:

A small bud is formed on the side of body of hydra, by mitosis. This bud enlarges by the formation of more cells. It then detaches from the parent body and grows into new organism.

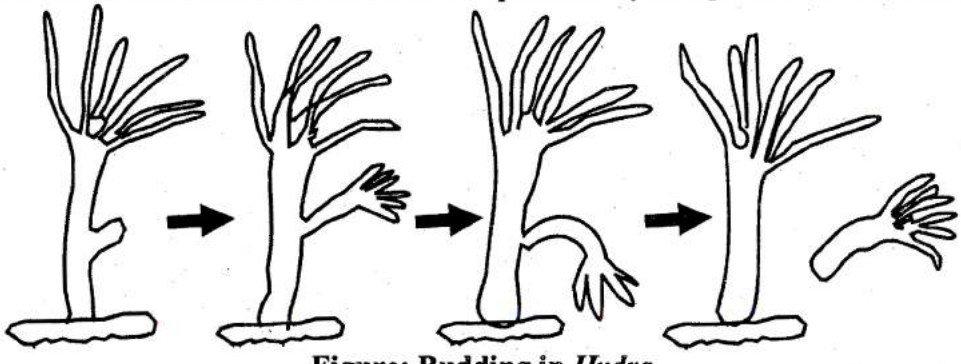

Figure: Budding in *Hydra*

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Budding in Corals:

In corals, the buds do not detach from the parent body. Corals form big colonies, because the buds grow into new organisms by remaining attached to the parent body.

Q. No. 5 Write a note on spore formation.

SPORE FORMATION

Spore:

An asexual reproductive structure that gives rise to new plant body directly is called as spore.

Example:

- It is generally seen in most fungi e.g. *Rhizopus*.

Spore Formation in *Rhizopus*:

When *Rhizopus* reaches reproductive age, its body cells form thick walled spore sacs called sporangia (sing. Sporangium). Inside each sporangium, a cell divides many times and forms many daughter cells called spores. Each spore is covered with a thick wall called cyst and it can survive unfavourable conditions. When sporangia are mature, they burst and release spores. Under favourable conditions, the spores germinate and develop into new *Rhizopus*.

Endospore Formation in Bacteria:

Under unfavourable conditions, some species of bacteria reproduce by forming spores. The bacterial spores are also thick walled. They are formed inside bacterial cells, so are called endospores. For example, the bacteria of the following species form endospores:

- *Clostridium*
- *Bacillus*

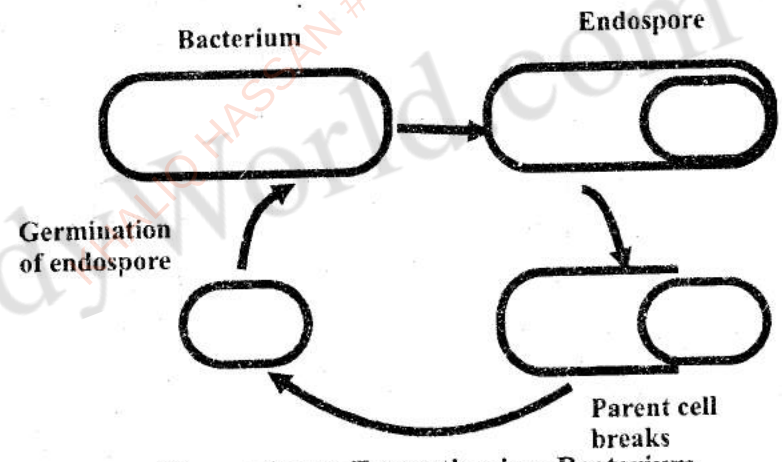

Figure: Spore Formation in a Bacterium

Q. No. 6 Write a note on parthenogenesis.

PARTHENOGENESIS

Definition:

A type of asexual reproduction in which an unfertilized egg develops into a new individual is called parthenogenesis.

Examples:

The phenomenon of parthenogenesis is observed in some:

- Fishes
- Frogs
- Insects

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Parthenogenesis in Honey Bees:

Queen honeybee lays eggs in the cells of honeycomb. Many eggs remain unfertilized and develop into haploid males called drones by parthenogenesis. At the same time, some eggs are fertilized by male bees and these develop into diploid females and new queen and worker bees are formed.

Q. No. 7 Write a note on natural vegetative propagation.

VEGETATIVE PROPAGATION

Definition:

The process in which vegetative parts of plants i.e. roots, stems or leaves give rise to new plants, is called vegetative propagation.

- It is also called vegetative reproduction.

NATURAL VEGETATIVE PROPAGATION:

If the process of vegetative propagation occurs naturally then it is called natural vegetative propagation.

Types of Natural Vegetative Propagation:

Vegetative propagation occurs naturally is several ways:

1. Bulbs
2. Corms
3. Rhizomes
4. Stem Tubers
5. Suckers
6. Leaves

1. Bulbs:

Bulbs are short underground stems surrounded by thick, fleshy leaves that contain stored food. Adventitious roots emerge under the base of bulb while shoots emerge from the top of the base.

Examples:

- Tulips
- Onions
- Lilies

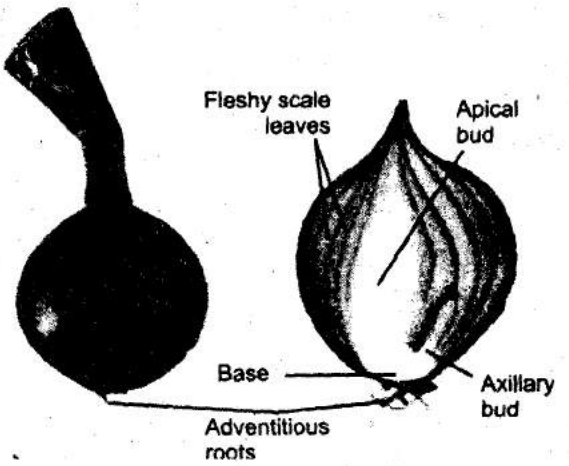

Figure: Bulb of Onion

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2. **Corms:**

Corms are short and swollen underground stems containing stored food. Buds are present at the top of corm. From a bud, shoot grows and forms a new plant.

Examples:

- Dasheen
- Garlic

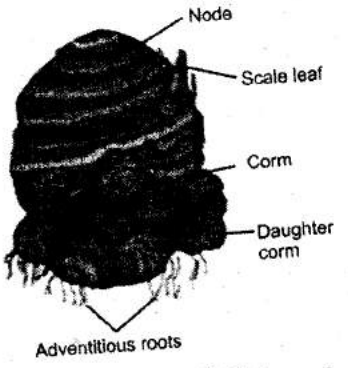

Figure: Corm of *Colocacia*

3. **Rhizomes:**

Rhizomes are horizontal underground stems with scale leaves. There are enlarged portions called nodes on rhizome. Buds are produced at nodes. The buds present on the upper surface of rhizome give rise to shoot. The lower surface of rhizome produces adventitious roots.

Examples:

- Ginger
- Ferns
- Water lilies

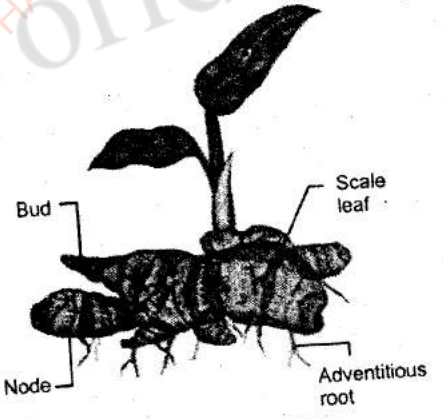

Figure: Rhizome of Ginger

4. **Stem Tubers:**

Stem Tubers are the enlarged portions of an underground stem (rhizome). There are aggregations of tiny buds in the form of "eyes" along the surface of tuber. Each bud develops into shoot that grows upward and also produces roots.

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Examples:

- Potatoes
- Yams

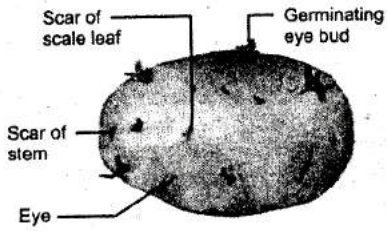

Figure: Stem Tuber of Potato

5. **Suckers:**

Suckers are lateral stems close to ground level. A sucker grows underground form some distance and then turns up, producing the new plant.

Examples:

- Mint
- *Chrysanthemum*

6. **Leaves:**

Vegetative propagation by leaves is not common and is seen in plants such as:

- *Bryophyllum* (Pather chut)

This plant has fleshy leaves and adventitious buds are present at the margins of leaves. When leaf falls on ground, the buds grow into new plants

Figure: *Bryophyllum* Leaf with Buds

Q. No. 8 Write a note on artificial vegetative propagation.

VEGETATIVE PROPAGATION

Definition:

The process in which vegetative parts of plants i.e. roots, stems or leaves give rise to new plants, is called vegetative propagation.

- It is also called vegetative reproduction.

ARTIFICIAL VEGETATIVE PROPAGATION

If the process of vegetative propagation occurs artificially by gardeners and farmers then it is called artificial vegetative propagation.

Types of Artificial Vegetative Propagation:

The following two are the most common methods of artificial vegetative propagation:

1. Cuttings
2. Graftin

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1. **Cuttings:**

In this method, cuttings may be taken mainly from the stems or roots of parent plant. These cuttings must have a meristematic region from which growth can occur. The cuttings will form roots and shoots when they are placed in a suitable soil and under right conditions like:

- Sufficient nutrients
- Water
- Sunlight

Roots and shoots grow and develop into a plant identical to the parent plant from which the cuttings were taken.

Stem Cuttings:

The following plants are artificially propagated through stem cuttings:

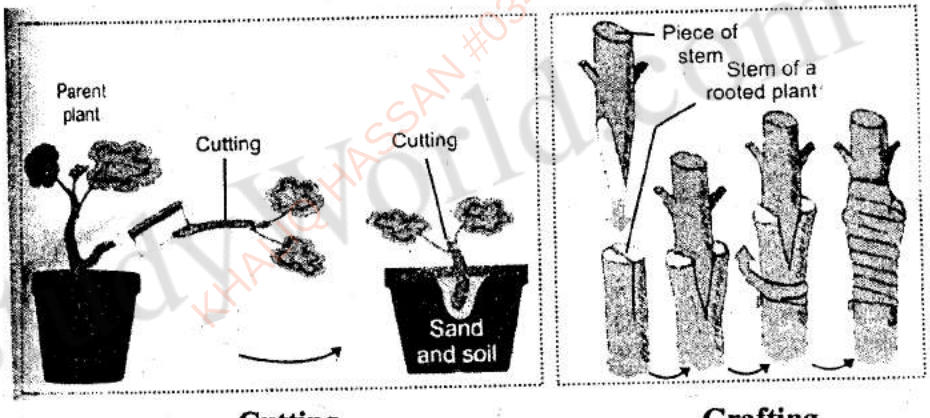
- Roses
- Ivy
- Grapevines

Root Cuttings:

Sweet potato is an enlarged root. Farmers place it in moist sand or soil until it produces several plantlets. Then the plantlets are removed and planted.

Advantages:

- This process is used to produce many plants from a single plant.
- All new plants are exactly the same.
- This artificial vegetative propagation has been very beneficial on sugar cane plantation.



Cutting **Grafting**

Figure: Artificial Vegetative propagation

2. **Grafting:**

In grafting, a piece of stem is cut from the plant and is attached with another plant with established root system. After a while, the vascular bundles of the attached stem piece and the host plant are connected to each other. The stem piece and the plant begin to grow together.

Examples:

This method is used to propagate:

- Many roses
- Peach trees
- Plum trees
- Various seedless fruits including grapes

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Q. No. 9 Describe the advantages and disadvantages of vegetative propagation of plants.

VEGETATIVE PROPAGATION

Definition:

The process in which vegetative parts of plants i.e. roots, stems or leaves give rise to new plants, is called vegetative propagation.

- It is also called vegetative reproduction.

ADVANTAGES OF VEGETATIVE PROPAGATION

Genetically Identical Offsprings:

The offsprings produced through vegetative propagation are genetically identical. Therefore beneficial characteristics can be preserved.

No Need of Pollination:

In vegetative propagation, there is no need of any mechanism of pollination. It helps to increase number of plants at a rapid rate.

Survival in Unfavourable Conditions:

The organs of vegetative propagation enable many plants to pass over unfavourable conditions.

Reproduction of Seedless Plants:

Plants bearing seedless fruits can be grown only by vegetative propagation.

DISADVANTAGES OF VEGETATIVE PROPAGATION

No Genetic Variations:

The plants do not have genetic variations.

Diseases Attack:

Species specific diseases can attack and this can result in the destruction of an entire crop.

Q. No. 10 Write a note on tissue culture and cloning.

TISSUE CULTURE AND CLONING

Cloning:

The process in which identical offsprings are produced from a single parent using its vegetative tissue or cell is called cloning.

Latest Method:

Cloning is the latest method of vegetative propagation.

Technique:

Tissue culture is the technique applied in this method.

Mechanism:

Extraction of Tissues:

Tissues are taken from any part of plant and are put in a suitable nutrient medium.

Formation of Calluses:

The tissue cells start mitosis and produce masses of cells called calluses are transferred to other medium that contains different hormones for the formation of:

- Roots
- Stem
- Leaves

Formation of New Plant:

Calluses make these structures and grow into new small plants. The small plants are then planted in pots and then in fields.

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Q. No. 11 Explain the phenomenon of alternation of generations.

ALTERNATION OF GENERATIONS

Definition:

The phenomenon in which two different generations alternate with each other during life cycle is known as alternation of generations.

Explanation:

In the life cycle of plants, two different generations alternate with each other.

Sporophyte Generation:

One generation is diploid and produces spores. It is called sporophyte generation. In most plants, sporophyte generation is dominant. It means that it is big in size and is independent.

Gametophyte Generation:

The other generation is haploid and produces gametes. It is called gametophyte generation. It is small in size and depends upon sporophyte. It produces gametes by mitosis.

Formation of Gametophyte:

Sporophyte produces haploid spores by meiosis. The spores develop into gametophyte.

Formation of Sporophyte:

The male and female gametes fuse and form diploid zygote. The zygote undergoes repeated mitosis and develops into a new diploid sporophyte.

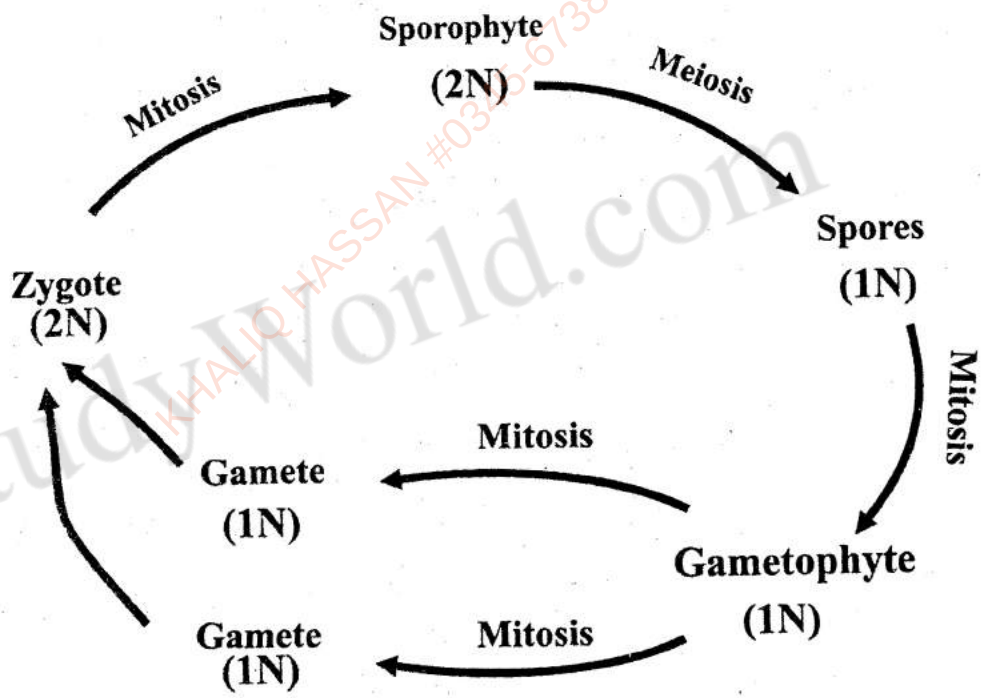

Figure: An overview of Alternation of Generations in Plants

Q. No. 12 Describe sexual reproduction in flowering plants.

SEXUAL REPRODUCTION IN FLOWERING PLANTS

In angiosperms, parent plant is diploid sporophyte generation. Flower is the reproductive structure in this generation. The flower components are arranged in the form of whorls. The outer two whorls in a flower are the non-reproductive whorls while the inner two whorls are the reproductive whorls.

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STRUCTURE OF A FLOWER

The following four whorls are present in the flower:

1. Calyx
2. Corolla
3. Androecium
4. Gynoecium

1. **Calyx:**

Calyx is the outermost whorl.

Colour:

It usually green in colour.

Sepals:

Its individual units (leaflets) are called sepals.

Function:

Sepals protect the inner whorls at bud stage.

2. **Corolla:**

Corolla is the next inner whorl.

Colour:

It is brightly coloured.

Petals:

Its individual units (leaflets) are called petals.

Function:

They serve to attract bees, birds etc. which are the agents of pollination.

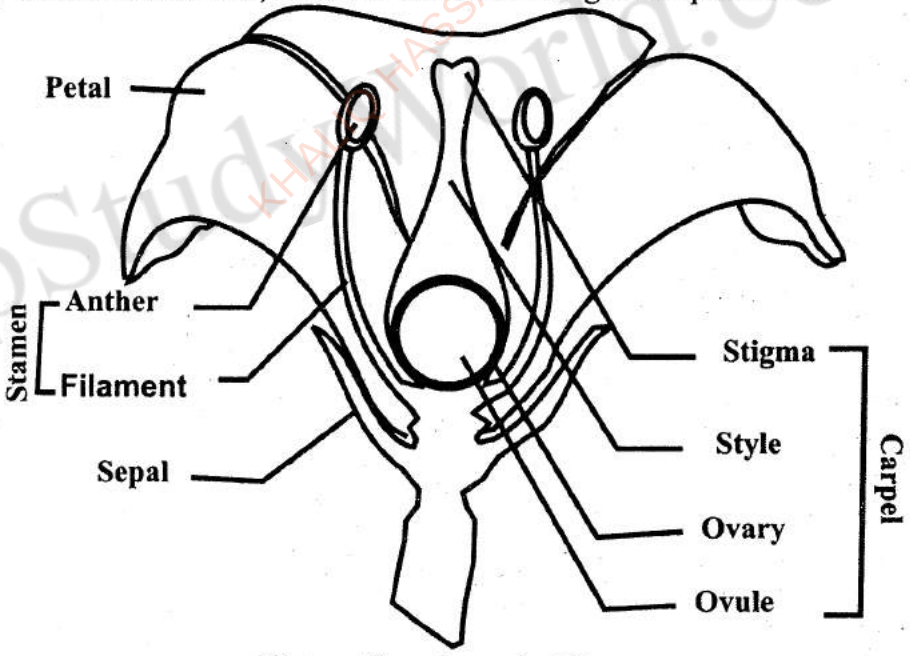

Figure: Structure of a Flower

3. **Androecium:**

Androecium is the third whorl and the male reproductive part of flower.

Stamens:

Its units are called stamens.

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Filament:

Each stamen has a thread-like filament at the free end of which anther is attached.

Anther:

Anther has pollen-sacs in which haploid microspores (pollen grains) are produced through meiosis. Each microspore germinates into the male gametophyte generation.

Formation of Male Gametophyte:

The nucleus of microspore undergoes mitosis and produces two nuclei:

- Tube nucleus
- Generative nucleus

The generative nucleus again undergoes mitosis and produces two sperms. So, a germinated microspore has a tube nucleus and two sperms. All these structures are the male gametophyte generation of plant.

4. Gynoecium:

Fourth whorl i.e. gynoecium is the female reproductive part of flower.

Carpels:

Its units are called carpels or pistils.

Structure of Carpel:

Each carpel is made up of:

- Basal ovary
- Middle style
- Upper stigma

Ovules:

Inside ovary, there are one to many ovules. Inside each ovule, one haploid macrospore produced through meiosis. Macrospore germinates into the female gametophyte generation.

Formation of Female Gametophyte:

The macrospore undergoes mitosis and produces an egg cell and some associate structures e.g. fusion nucleus. Egg cell and associated structures are the female gametophyt generation of plant.

Pollination:

When pollen grains mature, they are transferred to stigma. It is called pollination.

Formation of Pollen Tube:

On reaching the stigma, the tube nucleus of pollen grain constructs a pollen tube. Th pollen tube contains a tube nucleus and two sperms. The tube grows through style and ovary an enters ovule. Here, it bursts and releases the sperms. Both sperms enter the female gametophyte.

Formation of Zygote:

One sperm fuses with egg and forms a diploid (2N) zygote.

Formation of Endosperm Nucleus:

The other sperm fuses with diploid fusion nucleus and forms a triploid (3N) nucleu called endosperm nucleus.

Double Fertilization:

The process of fertilization involves two fusions, it is called double fertilization.

Developmental Process:

Zygote develops into embryo and endosperm nucleus develops into endosperm tissu (food of the growing embryo). Ovule then becomes seed and ovary changes into fruit.

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Dispersal of Seeds:

When seeds mature, they are dispersed. If seeds get suitable conditions, their embryos develop into new plants i. e. the diploid saprophytes of the next generation.

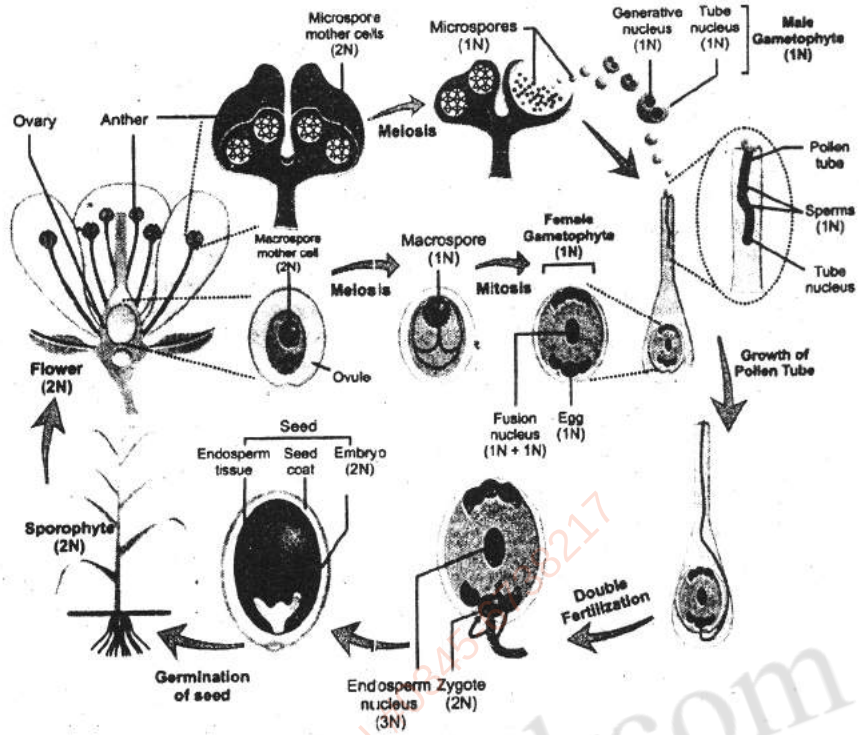

Figure: Life Cycle of a Flowering Plant

Q. No. 13 Define pollination. Explain its types. Discuss the adaptations in insect and wind pollinated flowers.

POLLINATION

Definition:

The transfer of pollen grains from anther of the stamen to stigma of carpel is called pollination.

Types of Pollination:

The following two types of pollination are recognized:

1. Self Pollination
2. Cross Pollination

1. Self Pollination:

The transfer of pollen grains from the anther of stamen to the stigma of the carpel of the same flower or other flower of the same plant is called self pollination.

2. Cross Pollination:

The transfer of pollen grains from anther of stamen to the stigma of the carpel of one plant to the flower on other plant of the same species is called cross pollination.

Cross pollination is brought about by various agencies like:

- Wind
- Water
- Bees
- Birds
- Bats
- Other animals including man

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ADAPTATIONS IN INSECT-POLLINATED AND WIND-POLLINATED FLOWERS

Feature	Insect Pollinated Flowers	Wind Pollinated Flowers
Flowers	Generally large	Generally small
Colour	Petals brightly coloured	Petals green or dull in colour
Nectar	Produce nectar	Do not produce nectar
Floral Arrangement	Flowers face upwards	Flowers hang down for easy shaking
Stamens and Stigmas	Enclosed inside ring of petals	Hang out of ring of petals
Pollen Grains	Small number produced /heavy and sticky	Large number produced / light with smooth surface
Stigma	Pinhead shaped with no branches	Feathery branches for catching pollen
Examples	• Buttercup • Rose • Wallflower • Sunflower • Orchid	• Grasses • Hazel • Willow • Corn

Q. No. 14 Describe the development and structure of seed.

DEVELOPMENT OF SEED

After fertilization in the female gametophyte, zygote divides repeatedly by mitosis and develops into an embryo. At this stage (in gymnosperms and angiosperms), ovule changes into seed. The formation of seed completes the process of sexual reproduction in seed plants.

STRUCTURE OF SEED

Angiosperm seeds consist of three distinct parts:

- The embryo formed from zygote
- The endosperm tissue formed from endosperm nucleus
- The seed coat which develops from the wall of ovule called integument

Seed Coat:

The other name of seed coat is testa. Seed coat develops from the integument, originally surrounding the ovule.

Thickness:

It may be a paper-thin layer, for example:

- Peanut

It may be thick and hard, for example

- Coconut

Function:

Seed coat protects embryo from mechanical injury and from drying out.

Hilum:

There is a scar on seed coat, called hilum. It is where the seed is attached to ovary wall (fruit).

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Micropyle:

At one end of hilum, there is micropyle. This is the same opening through which the pollen tube entered ovule.

Function:

Seed uses it for the absorption of water.

Embryo:

Embryo is actually an immature plant. It consists of:

- A radicle
- A plumule
- One or two cotyledons

Radicle:

The radicle of embryo develops into new root.

Plumule:

The plumule develops into new shoot.

Epicotyl:

The embryonic stem above the point of attachment of cotyledons is called epicotyl.

Hypocotyl:

The embryonic stem below the point of attachment of cotyledons is called hypocotyl.

Endosperm:

Within seed, there is a store of nutrients for the seedling that will grow from embryo. In angiosperms, the stored food is derived from the endosperm tissue. This tissue is rich in oil or starch and protein. In many seeds, the food of the endosperm is absorbed and stored by cotyledons.

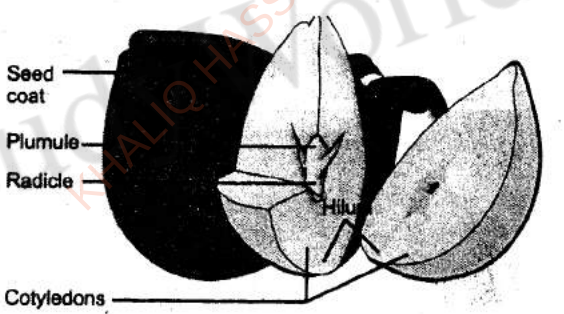

Figure: The Structure of a Dicot Seed

Q. No. 15 Define germination. Explain its types. What conditions are necessary for seed germination?

GERMINATION OF SEED

Definition:

The process by which a. seed embryo develops into a seedling is called seed germination.

Mechanism:

Absorption of water:

During germination, embryo soaks up water which causes it to swell, splitting the seed coat.

Formation of Root:

Root is the first structure that emerges from the radicle present in seed. It grows rapidly and absorbs water and nutrients from soil.

Formation of Shoot:

In the next phase, plumule develops into tiny shoot which elongates and comes out of soil.

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TYPES OF GERMINATION

On the basis of the elongation of hypocotyl and epicotyl, there are two types of germination:

1. Epigeal Germination
2. Hypogeal Germination

1. Epigeal Germination:

In epigeal germination, the hypocotyl elongates and forms a hook, pulling the cotyledons above ground. The examples of seeds that germinate this way are:

- Beans
- Cotton
- Papaya

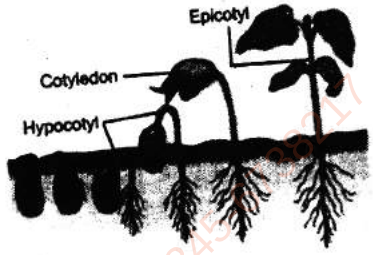

Figure: Epigeal Germination

2. Hypogeal Germination:

In hypogeal germination, the epicotyl elongates and forms the hook. In this type of germination, the cotyledons stay underground. The examples of seeds that germinate this way are:

- Pea
- Maize
- Coconut

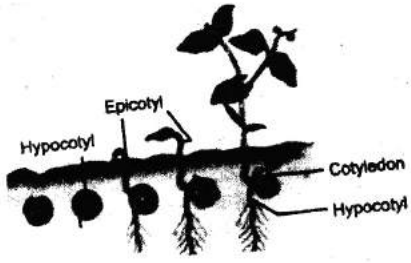

Figure: Hypogeal Germination

CONDITIONS FOR SEED GERMINATION

Seed germination depends on both internal and external conditions.

Internal Conditions:

The internal conditions include:

- A live embryo
- Sufficient food storage

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External Conditions:

The most important external conditions include:

1. Water
2. Oxygen
3. Favorable temperature

1. Water (moisture):

Seeds of most plants have low water content, and germination cannot occur until seed coat or other tissues have imbibed (taken in) water. The absorbed water is used in the digestion of the stored food and it also helps in elongation of hypocotyl and epicotyl.

2. Oxygen:

Oxygen is essential for the respiration in the cells of embryo.

3. Temperature:

Seeds differ greatly in their temperature requirements for germination. The optimum temperature for the germination of the seeds of most plants ranges from 25-30°C.

Q. No. 16 Write a note on gametogenesis in animals.

GAMETOGENESIS

Definition:

The formation of gametes is called gametogenesis.

Cell Division:

In this process, diploid (2N) gamete-mother cells undergo meiosis and form haploid (1N) gametes.

Gonads:

The male gametes (sperms) and female gametes (egg cells or ova) are produced in specialized organs called gonads.

Testes:

Male gonads are called testes. The singular of testes is testis.

Ovaries:

Female gonads are called ovaries.

SPERMATOGENESIS

Definition:

The process of production of sperms in testes is called spermatogenesis

Formation of Spermatogonia:

Some cells present in the walls of the seminiferous tubules of testes divide repeatedly by mitosis to form large number of diploid spermatogonia.

Formation of Primary Spermatocyte:

Some spermatogonia produce primary spermatocytes.

Formation of Secondary Spermatocyte:

Each primary spermatocyte undergoes meiosis-I and produces two haploid daughter cells called secondary spermatocytes.

Formation of Spermatids:

These cells undergo meiosis-II. In this way four haploid spermatids are produced from each primary spermatocyte. The spermatids are non-motile and many changes occur in them to convert them into motile cells.

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Formation of Sperms:

Their nuclei shrink and the following structures are formed:

- A corner called acrosome
- A tail
- A mitochondrial ring

After these changes, the spermatids are called sperms.

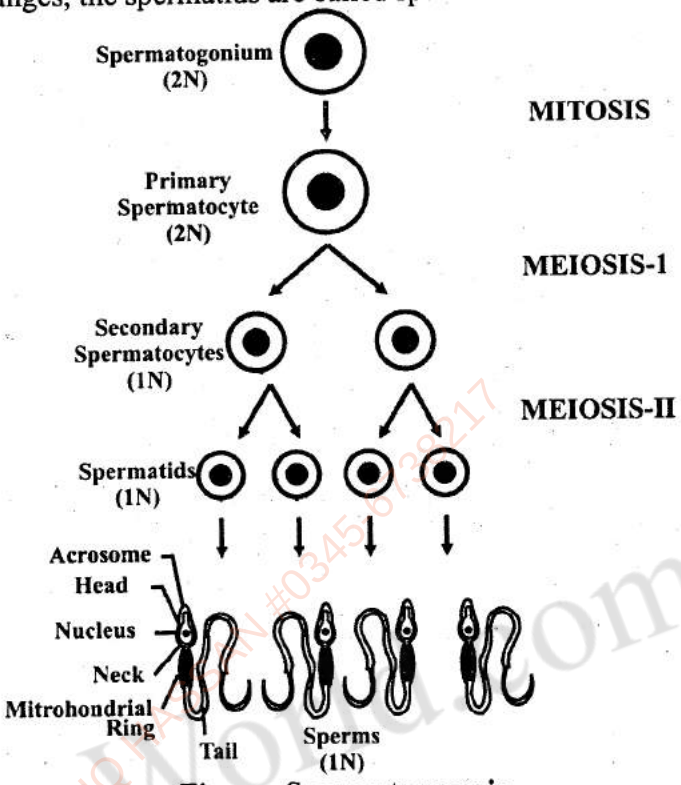

Figure: Spermatogenesis

OOGENESIS

Definition:

The process of production of egg cells in ovaries is called oogenesis.

Formation of Follicles:

Some cells of ovary prepare structures called follicles, in which many diploid oogonia are present. Some oogonia produce diploid primary oocytes.

Meiosis-I:

One of the primary oocytes completes meiosis-I and produces two haploid cells. The smaller cell is called first polar body and the larger one is called secondary oocyte.

Meiosis-II:

The secondary oocyte completes meiosis-II and produces two haploid cells i.e. a second polar body and an egg cell.

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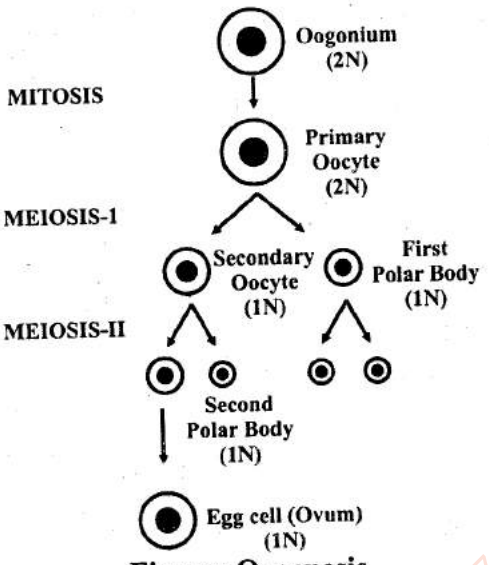

Figure: Oogenesis

Q. No. 17 Define fertilization. Explain its mechanisms.

FERTILIZATION

After the formation of gametes, fertilization occurs.

Definition:

The fusion of male gamete (sperm) and female gamete (egg or ovum) to form diploid zygote is called fertilization.

MECHANISMS OF FERTILIZATION

There are two mechanisms by which fertilization can take place:

1. External Fertilization
2. Internal Fertilization

1. External Fertilization:

In external fertilization, egg cells are fertilized outside of body. External fertilization occurs mostly in aquatic environment.

Release of Gametes:

It requires both the male and the female animals to release their gametes into their surroundings at almost the same time. For external fertilization, the animals have to release great number of gametes.

Disadvantage:

In external fertilization, there is risk of loss of gametes due to environmental hazards such as predators.

Examples:

External fertilization occurs in many invertebrates and the first two groups of vertebrates:

- Fishes
- Amphibians

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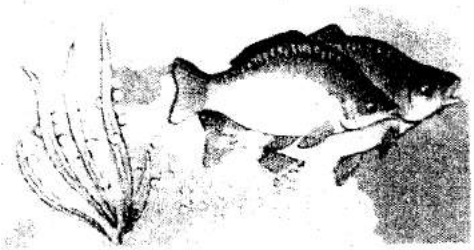

Figure: External Fertilization in Fish

2. Internal Fertilization:

In internal fertilization, egg cells are fertilized within the reproductive tract of female. Such animals provide protection to the developing embryo.

Examples:

It occurs in:

- Reptiles
- Birds
- Mammals

Reptiles and Birds:

After fertilization, reptiles and birds make protective shells around their egg cells and then lay them. The shell is resistant to water loss and damage.

Mammals:

In mammals (with the exception of egg-laying mammals) the development of fertilized egg into new baby takes place within mother body. In this case, there is extra protection to the embryo and mother also supplies everything that embryo needs.

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Figure: Reptiles and Bird's Egg provides Protection and Food to Embryo

Q. No. 18 Describe male reproductive system of rabbit.

MALE REPRODUCTIVE SYSTEM OF RABBIT

Introduction:

Rabbits are small mammals found in several parts of the world. They are used in research as experimental animals.

Parts:

The male reproductive system of rabbit consists of a pair of testes that produce sperms, the associated ducts that transport sperms to external genitalia and glands that add secretions to sperms.

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Scrotum:
Testes are located in a bag of skin called the scrotum that hangs below the body.

Seminiferous Tubules:
Each testis consists of a mass of coiled tubes called the seminiferous tubules. In these tubules, the sperms are formed.

Epididimys:
When sperm are mature, they accumulate in the collecting ducts of testes and then pass to epididimys.

Vas Deferens:
From epididimys, sperms move to a sperm duct called vas deferens.

Sperm Duct:
Both sperm ducts join urethra just below urinary bladder.

Urethra:
The urethra transports both sperm and urine.

Semen:
Semen is the material containing sperms in a fluid. It consists of 10% sperms and 90% fluid.

Associated Glands:
As the sperms pass down the ducts from testes to urethra, the associated glands add various secretions.

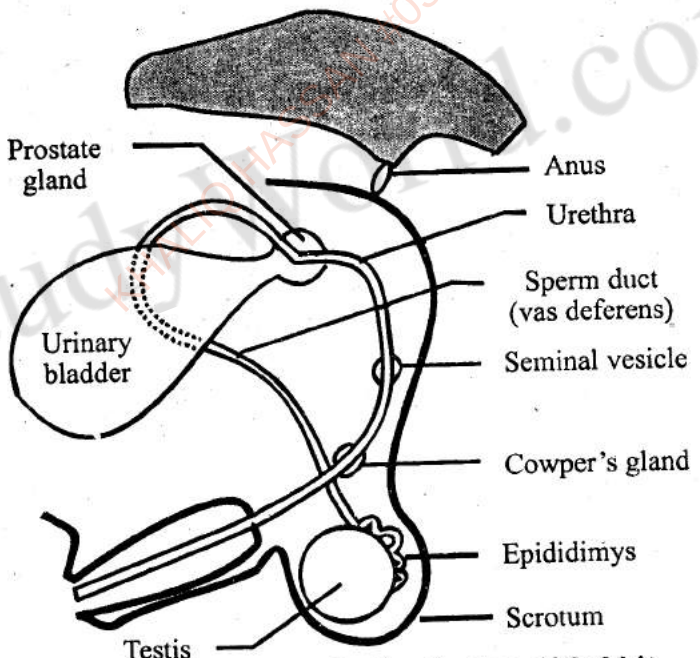

Figure: Male Reproductive System of Rabbit

Seminal Vesicles:
Seminal vesicles produce secretions that provide nutrients for sperms.

Prostate Gland:
Prostate gland produces a secretion that neutralizes the acidity of the fluid.

Cowper's Gland:
Cowper's glands produce secretions that lubricate the ducts.

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Q. No. 19 Describe female reproductive system of rabbit.

FEMALE REPRODUCTIVE SYSTEM OF RABBIT

Introduction:

Rabbits are small mammals found in several parts of the world. They are used in research as experimental animals.

Parts:

The female reproductive system of rabbit consists of ovaries and associated ducts.

Ovaries:

Ovaries are small oval organs situated in abdominal cavity just ventral to kidneys. Like most animals, female rabbits have a pair of ovaries. The outer region of ovary produces egg cells.

Follicles:

A cluster of specialized cells called follicle surrounds and nourishes each egg cell. From ovaries, egg cells are released in fallopian tubes.

Fallopian Tube:

The opening of fallopian tube lies close to ovary. Fertilization occurs in fallopian tubes.

Uterus:

The fertilized egg (zygote) is carried to uterus. The uterus of rabbit is divided into two separate parts or horns.

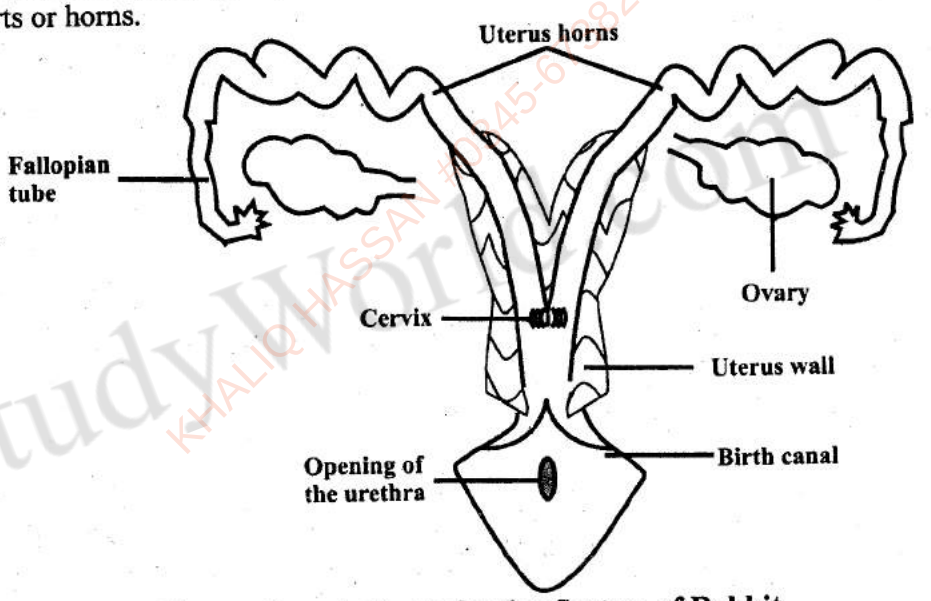

Figure: Female Reproductive System of Rabbit

Birth Canal:

The uterus horns join and open into vagina or birth canal.

Cervix:

Cervix is the portion of uterus, which separates it from birth canal, where sperms of male are deposited.

Q. No. 20 Describe fertilization and development in rabbit.

FERTILIZATION AND DEVELOPMENT IN RABBIT

Breeding Season:

Rabbits can breed throughout the year but male rabbits are commonly sterile during the summer months.

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Ejaculation of Sperms:

Male rabbit deposits its sperms in the vagina (birth canal) of female.

Fertilization:

Sperms swim through cervix and uterus to fallopian tubes where they fertilize the egg cells, released from ovary.

Embryo Formation:

After fertilization, zygote is carried to uterus. By this time, the zygote has started dividing and is now called embryo.

Placenta:

The embryo is implanted in uterus walls. A connection, called placenta, is established between embryo and uterus wall.

Gestation Period:

Embryo develops into new offspring (rabbit kit) in 30-32 days, after which it is born

Q. No. 21 Discuss growth in human population and its consequences.

GROWTH IN HUMAN POPULATION

Population Growth:

Pakistan's population in the year 2007-2008 was 163,775,000. By the end of this decade, our population is expected to exceed 176 million. Pakistan's population had a relatively high growth rate in past.

Overpopulation:

When population growth exceeds the carrying capacity of an area or environment, it results in overpopulation.

CONSEQUENCES OF OVERPOPULATION

Many problems are associated with human overpopulation.

Shortage of Resources:

The overpopulated areas face severe shortage of fresh water and natural resources.

Loss of Ecosystems:

Overpopulation results in deforestation and loss of ecosystems.

Pollution:

It leads to more pollution and global warming.

Mortality Rate:

There is high infant and child mortality rate in overpopulated areas due to poverty.

Increased Demands:

Overpopulation raises demands for more:

- Housing units
- Hospitals
- Jobs
- Educational institutions
- Food crops

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CHECKING OVERPOPULATION

The overpopulation should be checked otherwise we will have to face huge problems because of our limited resources.

Education:

People should be educated about the problems of overpopulation.

Role of Ministry of Population Welfare:

Pakistan's Ministry of Population Welfare has taken a number of steps to make people aware of the hazards of overpopulation and to stabilize the population to match our resource:

Q. No. 22 Write a note on AIDS.

AIDS: A Sexually Transmitted Disease

Sexually Transmitted Diseases (STDs):

The diseases that are transmitted through sexual act are called sexually transmitted diseases.

Leading Problem:

The most serious and challenging health problem faced by the world today is AIDS. It is also a sexually transmitted disease.

Abbreviation:

AIDS stands for Acquired Immune Deficiency Syndrome.

Causative Pathogen:

It is caused by Human Immuno Deficiency Virus (HIV).

Mode of Action:

The virus destroys white blood cells, which results in loss of resistance against infections.

Fatal Disease:

It is a fatal disease.

Spread of Disease:

It spreads through transfer of body fluids such as blood and semen.

Causes:

The main causes are:

- Unprotected sexual activities
- Use of infected needles
- Transfusion of infected blood

Q. No. 23 Describe the role of National AIDS Control Programme (NACP) and Non-Government Organizations in controlling AIDS.

ROLE OF NATIONAL AIDS CONTROL PROGRAMME (NACP)

Introduction:

Pakistan's Federal Ministry of Health established NACP in 1987.

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Objective:

The main objective of this programme is to help the public for the prevention of HIV transmission, safe blood transfusions and reduction of STDs.

Incidence:

The frequency of HIV infection in Pakistan is still low. But, the country is at risk of epidemic due to various risk factors:

- Exposure to infected blood or blood products
- Homo-sex
- Injecting drug users

Campaign:

For improved prevention by the general public, the NACP started services through TV and radio channels and print media in 2005. The objectives of this activity were to:

- Change public attitude for safe sexual activities
- Create demand for information on HIV and AIDS
- Improve attitudes and behaviour among healthcare workers

ROLE OF NON GOVERNMENT ORGANIZATIONS

According to the latest data by the World Bank, at least 54 NGOs are working in Pakistan for HIV/AIDS public awareness and for the care and support of persons living with HIV/AIDS.

- These NGOs also work on AIDS education and prevention for sex workers and other high-risk groups.
- NGOs serve as members of the Provincial Consortium on H IV/AIDS, which has been set up in all the provinces of Pakistan.

Access to Victims:

Although NGOs are very busy in HIV/AIDS prevention activities, it is believed that they are reaching less than 5 percent of the vulnerable population.

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REVIEW QUESTION:

MULTIPLE CHOICE QUESITONS

1. **Growing an entire new plant from part of the original plant is called:**
   (a) Budding (b) Regeneration
   (c) Fragmentation (d) Vegetative propagation
2. ***Rhizopus* reproduces asexually by:**
   (a) Binary fission (b) Budding
   (c) Spore formation (d) Endospore formation
3. **A corm develops into new garlic plant. This is the process of:**
   (a) Vegetative propagation (b) Regeneration
   (c) Meiosis (d) Gametogenesis
4. **Which is not an advantage of grafting?**
   (a) The graft is identical to the parent plant
   (b) Grafting allows the propagation of seedless fruits
   (c) The graft combines the characteristics of two plants
   (d) Grafting may allow for the faster production of desirable fruits
5. **Pollination is the transfer of pollens from:**
   (a) Anther to stigma (b) Stigma to anther
   (c) Sepal to petal (d) Petal to sepal
6. **Double fertilization in plants means:**
   (a) Fusion of two sperms with two egg cells
   (b) Fusion of one sperm with egg cell and other sperm with fusion nucleus
   (c) Fusion of two sperms with a single egg cell
   (d) Fusion of tube nucleus with fusion nucleus and sperm with egg cell
7. **After fertilization in plants, the fruit develops from:**
   (a) Ovule wall (b) Ovary wall
   (c) Petals (d) Anther
8. **Which part of the female reproductive system receives egg cells from the ovary?**
   (a) Fallopian tube (b) Uterus
   (c) Vagina (d) Cervix
9. **Inside testes, the sperms are produced in:**
   (a) Vas deferens (d) Sperm duct
   (c) Seminiferous tubules (d) Collecting ducts
10. **Which of these cells has haploid number of chromosomes?**
    (a) Spermatogonium (b) Primary spermatocyte
    (c) Secondary spermatocyte (d) AII of these

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	c	3	a	4	b	5	a
6	b	7	b	8	a	9	c	10	c

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SHORT QUESTIONS

1. How are the natural and artificial vegetative propagations the methods of asexual reproduction in plants?

Consult Long Question No. 7 and 8

2. Why do gardeners use the methods of cutting and grafting?

Consult Long Question No. 8

3. "Parthenogenesis is a type of asexual reproduction". Give comments on this statement.

Consult Long Question No. 6

4. Outline the life cycle of a flowering plant.

Consult Long Question No. 12

5. What structural adaptations will you find in a wind-pollinated flower?

Consult Long Question No. 13

6. Give an introduction of Pakistan's National AIDS Control Program.

Consult Long Question No. 23

UNDERSTANDING THE CONCEPT

1. What are the different ways by which prokaryotes, protozoans and fungi reproduce asexually?

Consult Long Question No. 1 and 4

2. Explain the different parts of the plant that help in natural vegetative propagation.

Consult Long Question No. 7

3. Explain, how the epigeal and hypogeal germinations are different?

Consult Long Question No. 15

4. What conditions are necessary for the germination of seeds?

Consult Long Question No. 15

5. Outline the methods of asexual reproduction in animals.
6. Write a note on the male and female reproductive systems of rabbit.

Consult Long Question No. 18 and 19

7. Describe the processes of spermatogenesis and oogenesis.

Consult Long Question No. 16

8. Why do we consider that overpopulation is a global problem?

Consult Long Question No. 21

THE TERMS TO KNOW

Acrosome:
Cap-like head of sperm cell which helps it in penetrating the egg cell

Alternation of generations:
In plants, the phenomenon in which the sporophyte and gametophyte generations alternate with each other.

Androecium:
The male reproductive whorl of the flower; consists of stamens.

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Anther:
The sac-like structure of a stamen in which pollen grains are produced.

Binary fission:
Division into two; the simplest method of asexual reproduction in prokaryotes and many unicellular eukaryotes.

Budding:
A type of asexual reproduction in which a bud develops as a small outgrowth on parent's body and forms the new individual.

Bulb:
Underground vertical shots which have modified leaves.

Calyx:
The outer whorl of flowers; consists of sepals

Carpel:
Part of the gynoecium of the flower; consists of stigma, style and ovary

Cervix:
In female reproductive system. The part which separates uterus from the vagina

Cloning:
Method of asexual reproduction in which identical off springs are produced from a vegetative tissue or cell of the parent.

Corm:
Short, swollen underground stem; has bud (s) at the top; gives rise to new plants by vegetative propagation.

Corolla:
The second whorl of flower, consisting of petals.

Cotyledon:
A modified leaf present in seeds; often gives nourishment to the developing seeding.

Cowperm gland:
An accessory gland in rabbits male reproductive system; provides lubrication to the ducts.

Endosperm nucleus:
In the female gametophyte, the triploid nucleus formed by the fusion of sperm and the fusion nucleus.

Endosperm tissue:
The tissues that develops from endosperm nucleus; often serves as a food supply for developing embryo.

Endospore:
The spore formed inside the bacterial cell

Epicotyl:
The embryonic stem above the point of attachment of the cotyledon(s)

Epididimys:
A storage area for sperms on the upper part of the testes.

Epigeal germination:
A type of seed germination in which the hypocotyle elongates and forms a hook, pulling the cotyledons above the ground.

Fallopian tube:
A part of the female reproductive system; receives egg cell discharged from the ovary.

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Follicle:
A structure in the ovary in which the mature egg develops.

Fragmentation:
A type of asexual reproduction in which the animal breaks up into many pieces and each piece develops into a mature animal

Fusion nucleus:
A part of the female gametophyte in plant: formed by the fusion of two nuclei; gives rise to endosperm nucleus when fertilized by a sperm

Gametogenesis:
The process of the formation of gametes

Gametophyte:
The haploid generation in plant life cycle; produces gametes

Grafting:
A type of artificial vegetative propagation in which a piece of stem in cut from the plant and is inserted into another plant with established root system.

Gynoecium:
The central whorl in the flower; consists of carpels.

Hilum:
A scar on the seed coat; the point where the seed is attached to the ovary wall.

Hypocotyl:
The embroyonic stem below the pint of attachment of cotyledon

Hypogeal germination:
A type of seed germination in which the epicotyle elongates and forms the hook while the cotyledons stay underground

Macrospore:
Haploid cell produced in the ovule; divides mitotically and produces the female gametophyte

Micropyle:
The opening in the ovule through which the pollen tube enters; seed uses it for the absorption of water.

Microspore:
Pollen grain; the haploid cells produced in the pollen sac; divi9de mitotically to produce male gametophyte.

Multiple fission:
Division into many; a method of asexual reproduction used by many unicellular paganisms organisms.

Oogonium:
The formation of ovum (egg cell)

Ovary:
The female gonad; produces egg cells and female sex hormones.

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Spermatogenesis:
The formation of sperms.

Spermatogonia:
The diploid cells in somniferous tubules of the testes; divide mitotically and produced primary spermatocytes.

Sporophyte:
The diploid generation in plant life cycle; produces spores.

Stamen:
The part of the androcium; consists of anther and filament.

Stigma:
The upper part of the carpel

Style:
The middle portion of the carpel

Testa:
See Seed coat

Testis:
The male gonads; produces sperms and male sex hormones.

Tuber:
The swollen ends of slender rhizomes (underground stem); new plants develop from buds on the stem tubers (vegetative propagation).

Uterus horn:
The two separate parts of the uterus in the female rabbit.

Vas deferens:
The tubes that carry sperms from each testis to the urethra

Vegetative propagation:
A type of asexual reproduction in which the vegetative parts of the plants i.e. roots, stems or leaves produce new generation.

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Ovule:

In seed plants, a structure present in the ovary; constains macrospore that develops into female gametophyte; ovule develops into seed after fertilization.

Parthenocarpy:

The process in which ovaries develop into fruit without the fertilization in the ovules present in them, results in seedless fruits e.g. bananas.

Parthenogenesis:

A form of asexual reproduction in which an unfertilized egg develops into new offspring.

Plumule:

The part of the plant embryo that develops into new shoot.

Pollen grain:

See Microspore

Pollen tube:

A tube formed by the tube nucleus of the pollen grain; carries sperms to the ovule.

Pollination:

The transfer of pollen grains from flower's anther to stigma.

Prostate gland:

An accessory gland in the male reproductive system; produces a secretion that neutralizes the acidity.

Radicle:

The part of the plant embryo that develops into new root.

Rhizomes:

Horizontal underground stems; have scale leaves with buds; shoots of the new plant develop and grow form buds (vegetative propagation)

Seed dormancy:

A period, during which there is no growth in the seed; seeds in dormancy are ripe seeds but do not germinate; under favorable conditions, the seeds break dormancy and begin to germinate.

Semen:

The material containing sperms in a fluid.

Seminal vesicle:

The associated gland in male reproductive system; produces secretions having nutrients for the sperms.

Seminiferous tubule:

The coiled tubes present in testes; sperms are formed in these tubules.

Sperm:

The male gamete

Spermatid:

The immature non-motile forms of sperms; are converted into sperms after many changes.

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CHAPTER 15

INHERITANCE

MULTIPLE CHOICE QUESTIONS

Each question has four possible answers. Circle the correct answer

(1) The branch of Biology that deals with inheritance:

(a) Physiology (b) Ecology (c) Pharmacology (d) Genetics

(2) The chromosomes carry the units of inheritance called:

(a) Loci (b) Dominance (c) Genes (d) Traits

(3) In humans, the number of pairs of homologous chromosomes:

(a) 21 (b) 22 (c) 23 (d) 24

(4) DNA wraps around histone proteins and forms round structures, called:

(a) Chromatin (b) Chromosome (c) Nucleolus (d) nucleosomes

(5) Watson and Crick proposed the DNA model in:

(a) 1951 (b) 1953 (c) 1955 (d) 1957

(6) In DNA molecule, adenine always pairs with:

(a) Guanine (b) Cytosine (c) Thymine (d) Uracil

(7) How many hydrogen bonds are present between cytosine and guanine?

(a) One (b) Two (c) Three (d) Four

(8) The locations or positions of genes on chromosomes:

(a) Loci (b) Traits (c) Inheritance (d) Nucleotides

(9) How many types of nucleotides are present in DNA?

(a) 1 (b) 2 (c) 3 (d) 4

(10) The specific combination of genes in an individual:

(a) Genotype (b) Phenotype (c) Dominance (d) Recessive

(11) How many plants were used by Mendel in his experiments?

(a) 26,000 (b) 28,000 (c) 30,000 (d) 32,000

(12) The term true breeding means:

(a) Heterozygous (b) Genotype (c) Phenotype (d) Homozygous

(13) Mendel obtained how many round seeds in monohybrid cross?

(a) 4784 (b) 5474 (c) 7454 (d) 4555

(14) Phenotypic ratio of monohybrid cross:

(a) 3:1 (b) 2:1 (c) 9:3:3:1 (d) 1:2:1

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(15) **Mendel obtained how many round yellow seed plants in dihybrid cross?**
(a) 32 (b) 101 (c) 108 (d) 315

(16) **The situation where two different alleles of a gene pair express themselves completely is called:**
(a) Co-dominance (b) Over dominance
(c) Incomplete dominance (d) None of these

(17) **Which genotype represents blood group A?**
(a) I^AI^A (b) I^Ai (c) Both a and b (d) None of these

(18) **Date of birth of C. de Buffon:**
(a) 1707 (b) 1710 (c) 1713 (d) 1716

(19) **Date of death of J. de Lamarck:**
(a) 1740 (b) 1742 (c) 1744 (d) 1829

(20) **Date of birth of Charles Darwin:**
(a) 1802 (b) 1808 (c) 1809 (d) 1810

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	c	3	c	4	d	5	b
6	c	7	c	8	a	9	d	10	a
11	b	12	d	13	b	14	a	15	d
16	a	17	c	18	a	19	d	20	c

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SHORT QUESTIONS

Q. No. 1 Define genetics.

GENETICS

The branch of biology which deals with the study of inheritance is called genetics.

Q. No. 2 What do you mean by inheritance?

INHERITANCE

Inheritance means the transmission of characteristics from parents to offspring.

Q. No. 3 What are traits?

TRAITS

The characteristics which are transferred from parents to offsprings are called traits.

Examples:

In man following are all inheritable traits:

- Height
- Colour of the eyes
- Intelligence

Q. No. 4 How do parents inherit characteristics>

INHERITANCE OF CHARACTERISTICS

Parents pass characteristics to their young through gene transmission. Equal numbers of chromosomes from each parent are combined during fertilization. The chromosomes carry the units of inheritance called the genes)

Q. No. 5 What are genes?

GENES

The genes are called units of inheritance that transmit traits from parents to offsprings.

Q. No. 6 Does the dominant allele affect the nature of recessive allele?

EFFECT ON RECESSIVE ALLELE

The dominant allele only suppresses the expression of recessive allele. It does not affect its nature.

Q. No. 7 How many pea plants were used by Mendel in his experiments?

NUMBER OF PEA PLANTS

Mendel used 28,000 pea plants in his experiments.

Q. No. 8 What does the term true breeding mean?

TRUE BREEDING

The term true breeding means homozygous.

Q. No. 9 How do variations caused by combinations of chromosomes?

VARIATIONS BY COMBINATIONS OF CHROMOSOMES

Variations are caused by different combinations of chromosomes in gametes and then in zygote.

Example:

In the case of humans, the possible number of chromosomal combinations at fertilization is 70,368,744,177,664. In other words, a couple can produce more than 70 trillion genetically different children.

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LONG QUESTIONS

Q. No. 1 Write a note on chromosomes.

CHROMOSOMES

Genes consist of DNA. They contain specific instructions for protein synthesis.

Introduction:

Chromosomes are thread like structures that are present in the nucleus of the cell and become visible only at the time of cell division.

Specific Number:

The body cells have a constant number of paired chromosomes.

Homologous Chromosomes:

The two chromosomes of a pair are known as homologous chromosomes. In human body cells, there are 23 pairs of homologous chromosomes for a total of 46 chromosomes.

Gametes:

During meiosis, the two members of each chromosome pair separate and each of them enters one gamete.

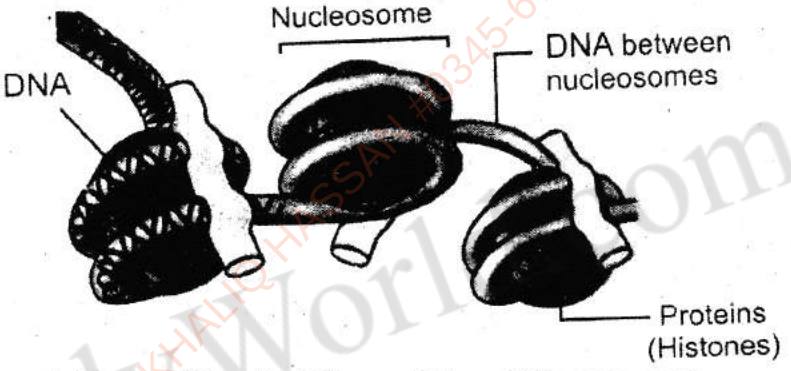

Figure: Chemical Composition of Chromosome

Chemical Composition of Chromosomes:

Chromatin:

Chromosome is made of chromatin material. Chromatin is a complex material, made of:

- DNA
- Proteins (mainly histone proteins)

Nucleosomes:

DNA wraps around histone proteins and forms round structures, called nucleosomes. DNA is also present between nucleosomes. In this way, the nucleosomes and the DNA between them look like "beads on a string".

Typical Structure:

The fibres consisting of nucleosomes condense into compact forms and get the structure of chromosomes.

Q. No. 2 Describe Watson and Crick model of DNA

WATSON AND CRICK MODEL OF DNA

Introduction:

In 1953, James Watson and Francis Crick proposed the structure for DNA.

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Double Helix Structure:

According to the Watson-Crick model, a DNA molecule consists of two polynucleotide strands. These strands are coiled around each other in the form of a double helix.

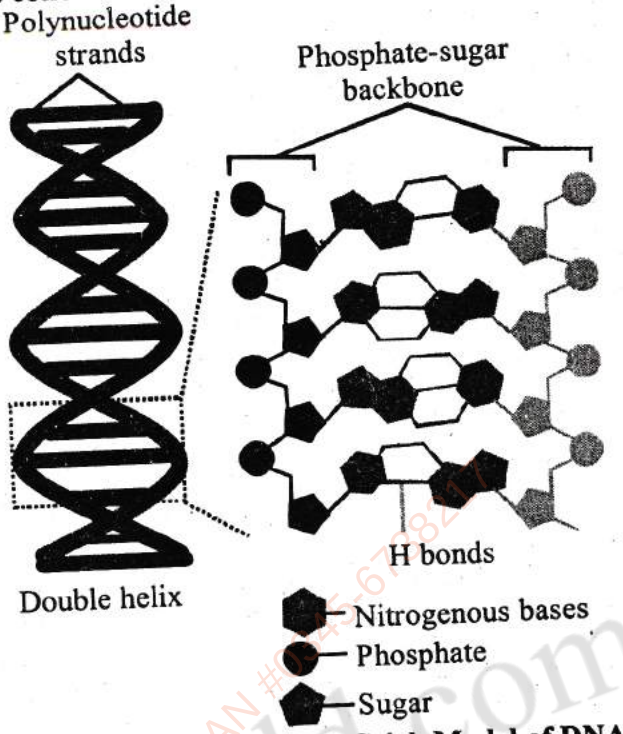

Figure: The Watson and Crick Model of DNA

Phosphate-Sugar Backbone:

There is a phosphate-sugar backbone on the outside of double helix.

Nitrogenous Bases:

The nitrogenous bases are on the inside of the double helix. In double helix, the nitrogenous bases of opposite nucleotides form pairs through hydrogen bonds.

Specific Pairing:

The pairing of nucleotides is highly specific. The nitrogenous base adenine of one nucleotide forms pair with the thymine of opposing nucleotide, while cytosine forms pair with guanine.

Hydrogen Bonds:

There are two hydrogen bonds between adenine and thymine while there are three hydrogen bonds between cytosine and guanine.

Q. No. 3 Write a note on DNA replication.

DNA REPLICATION

Introduction:

Before a cell divides, its DNA is replicated (duplicated). It is done to make the copies of the chromatids of chromosomes.

Mechanism of Replication:

Uncoiling Of DNA:

During replication, the DNA double helix is unwound and the two strands are separated, much like the two sides of a zipper.

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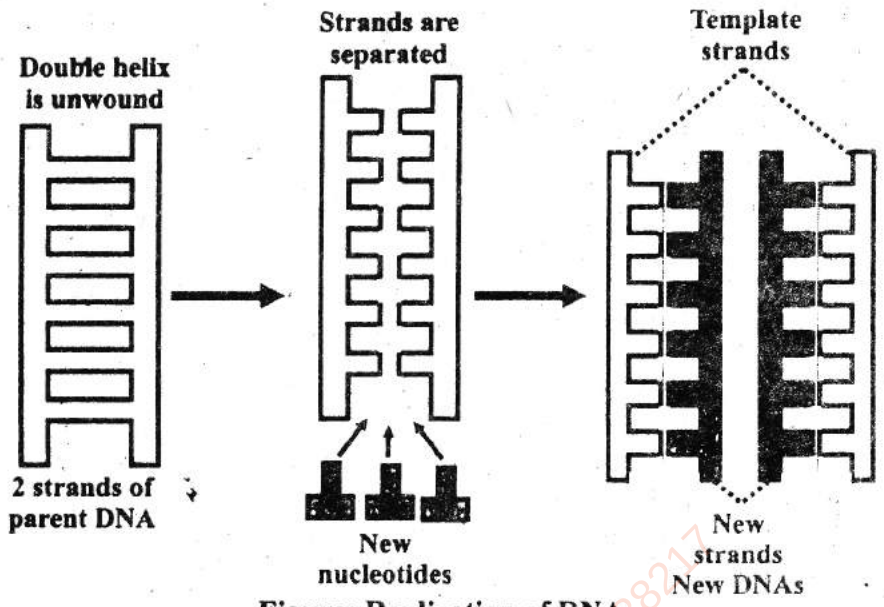

Figure: Replication of DNA

Formation of Template:

Each strand acts as a template to produce another strand.

Pairing of Nucleotides:

The DNA template nitrogenous bases make pairs with the nitrogenous bases of new nucleotides. In this way, both template strands make new polynucleotide strands in front of them.

New DNA Molecule:

Each template and its new strand together then form a new DNA double helix, identical to the original.

Q. No. 4 Explain, how does the DNA of chromosome work?

WORKING OF CHROMOSOMAL DNA

Genetic Material:

DNA is the genetic material. It contains the instructions to direct all the functions of cells. It performs its role by giving instructions for the synthesis of specific proteins.

Role of Proteins:

Some proteins perform structural roles while the others act as enzymes to control all biochemical reactions of cells.

DNA Control:

In this way, whatever a cell does, is actually controlled by its DNA. In other words, DNA makes the characteristic or trait of cell or organism.

Expression of Trait:

The traits are made by specific proteins. Specific proteins have specific number and sequence of their amino acids. DNA controls this sequence of amino acids by the sequence of its nucleotides. During protein synthesis, the sequence of DNA nucleotides decides that what will be the sequence of amino acids.

Transcription:

The specific sequence of DNA nucleotides is copied in the form of messenger RNA (mRNA) nucleotides. This process is called transcription.

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Translation:

The mRNA carries the sequence of its nucleotides to ribosome. The ribosome reads this sequence and joins specific amino acids, according to it, to form protein. This step is known as translation.

Gene:

The part of DNA (sequence of nucleotides) that contains the instructions for the synthesis of a particular protein is known as a gene.

Number:

DNA of each chromosome contains thousands of genes.

Pairing:

Like chromosomes, genes also occur in pairs, one on each homologous chromosome.

Loci:

The locations or positions of genes on chromosomes are known as loci (Singular. locus).

Gene-Trait Relationship:

Each gene determines a particular trait in an organism. Each individual carries at least one pair of genes for each trait.

Representation:

For convenience, pairs of genes are represented by a letter or symbol. Both members of a gene pair may be the same in some individuals (a condition which we may represent as AA or aa or BB) and different in others (Aa or Bb).

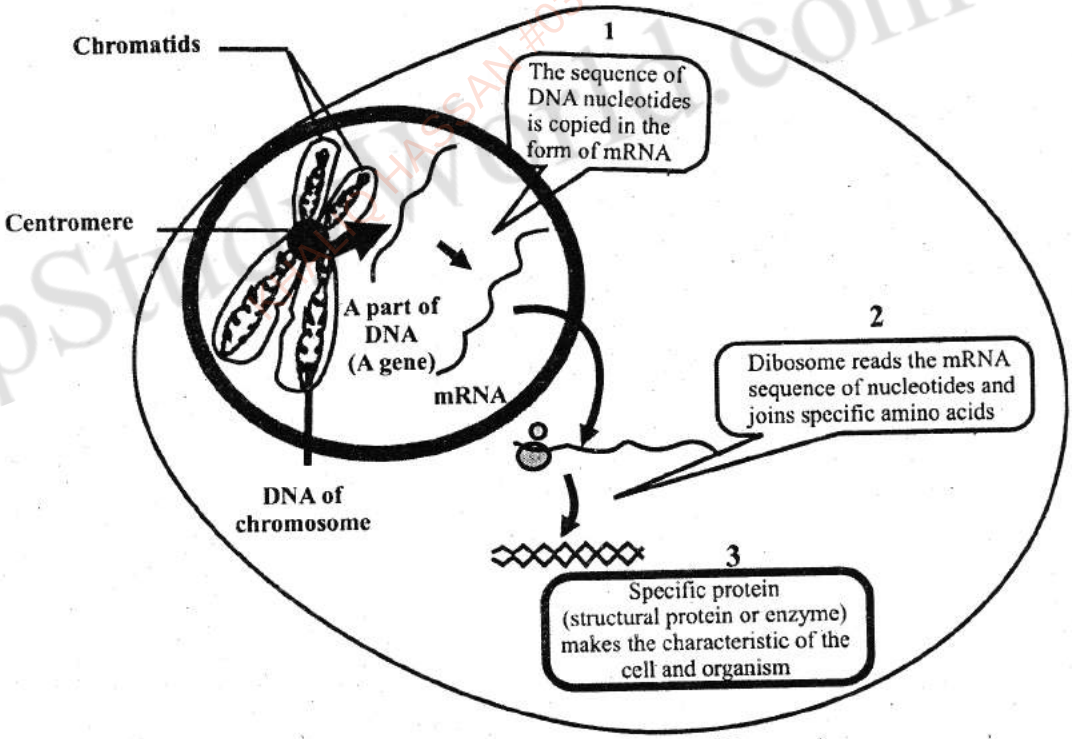

Figure: Working of DNA (Central Dogma)

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Alleles:

It means that a gene exists in more than one alternate forms. In the above example, 'A' and 'a' are the two alternate forms of a gene and 'B' and 'b' are the alternate forms of another gene. The alternate forms of a gene are called alleles.

Location of Alleles:

If an individual has Aa gene pair, 'A' and 'a' are the alleles of one another. In this individual, allele 'A' is located on one of the two homologous chromosomes and the allele 'a' is on the other chromosome.

Separation of Alleles:

When chromosomes separate during meiosis, alleles also separate and each gamete gets one of the two alleles. When gametes of both parents unite, the zygote (and the offspring also) receives one allele from each parent.

GENOTYPE AND ITS TYPES

Genotype:

The specific combination of genes in an individual is known as genotype.

Types of Genotype:

It is of two types:

1. Homozygous
2. Heterozygous

Example:

Albinism:

It is a condition in which normal body pigments are absent in an individual.

Alleles:

Like other traits, it is also controlled by one pair of genes. We can represent the two alleles of the pair as 'A' and 'a'.

Possible Combinations:

Three combinations i.e. genotypes are possible for these two alleles:

- AA
- Aa
- aa

Grouping of Genotypes:

These genotypes can be grouped into two types.

Homozygous Genotype:

The genotype in which the gene pair contains two identical alleles (AAor aa), is called homozygous genotype.

Heterozygous Genotype:

The genotype in which the gene pair contains two different alleles (Aa), is called heterozygous genotype.

Dominant Allele:

When in the heterozygous condition one allele masks or prevents the expression of the other, it is called the dominant allele. The dominant alleles are represented by capital letters.

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Recessive Allele:

The allele which is not expressed is called recessive. The recessive alleles are represented by lower case letters.

Albinism; A recessive Trait:

Albinism is a recessive trait. It is produced when both alleles are recessive.

In humans, allele 'A' produces normal body pigments while allele 'a' does not produce pigments.

- If genotype is AA or Aa, the individual will produce pigments.
- If genotype is aa, no pigments will be produced and the individual will be albino.

The allele 'A dominates over 'a', because in Aa individual pigments are produced and the effect of 'a' is suppressed by 'A'.

Phenotype:

The expression of this genotype in the form of trait is known as the phenotype.

Example:

Being albino or having normal pigmentation

Q. No. 5 Why did Mendel select pea plant.

SELECTION OF PEA PLANT

Introduction:

Gregor Mendel was a monk (priest) in Austria. He developed the fundamental principles of genetics.

Pea Plant:

Mendel selected pea plant (*Pisum sativum*) to carry out a large number of experiments. He argued that an organism for genetic experiments should have the following features:

Different Traits:

There should be a number of different traits that can be studied.

Contrasting Traits:

The organism should have contrasting traits. Each trait studied in pea plant had two distinct forms.

For example:

The trait of height there should be only two very different phenotypes i.e. tallness and dwarfness.

Different Traits and Their Phenotypes

Traits	Phenoytpes
Seed Shape	Round and Wrinkled
Seed Colour	Yellow and Green
Flower Colour	Purple and White
Pod Shape	Flat and Constricted
Pod Colour	Green and Yellow
Flower position	Axial and Terminal
Stem Length	Long and Short

Self-fertilizing Plant:

The organism (if it is a plant) should be self-fertilizing but cross fertilization should also be possible.

Life Span:

The organism should have a short but fast life cycle.

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Cross Fertilization:

Normally, the flowers of pea plant allow self-pollination. Cross pollination can also be done by transferring the pollen grains from the flower on one plant to the flower on another plant.

Mendel's Success:

Mendel's succeeded in his work not only because he selected the right organisms for his experiments but also because he analyzed the results by using the principles of statistics (ratiOS)

Q. No. 6 State and explain Mendel's Law of Segregation.

MENDEL'S LAW OF SEGREGATION

Introduction:

Gregor Mendel was a monk (priest) in Austria. He developed the fundamental principles of genetics.

Selection of plant:

Mendel selected pea plant (*Pisum sativum*) to carry out a large number of experiments.

Statement:

In each organism, the genes are present in pairs. During gamete formation, the genes (alleles) of each pair segregate from each other and each gamete receives one gene from the pair. When the gametes of male and female parents unite, the resulting offspring again gets the genes in pairs.

Selection of Phenotype:

Mendel studied the inheritance of seed shape first.

Monohybrid Cross:

He crossed two plants having one contrasting trait i.e. seed shape. A cross in which only one trait is studied at a time, is called as a monohybrid cross.

Cross Fertilization of True Breeding:

Mendel crossed a true-breeding round-seeded plant with a true-breeding wrinkled seeded plant.

P1 and F1 Generation:

The parental generation is denoted as P1 generation. The offspring of P1 generation are F1 generation (first filial).

Result:

All resulting seeds of the next generation were round.

Conclusion:

Mendel declared the trait "round Seeds" as dominant, while "wrinkled seeds" as recessive.

Self Fertilization of F1:

The following year, Mendel planted these seeds and allowed the new plants to self-fertilize. The cross in F1 generation produces F2 generation (second filial).

Results:

As a result, he got obtained:

Total seeds = 7324
Round seeds = 5474
Wrinkled seeds = 850

Ratio	Round	:	Wrinkled
	3	:	1

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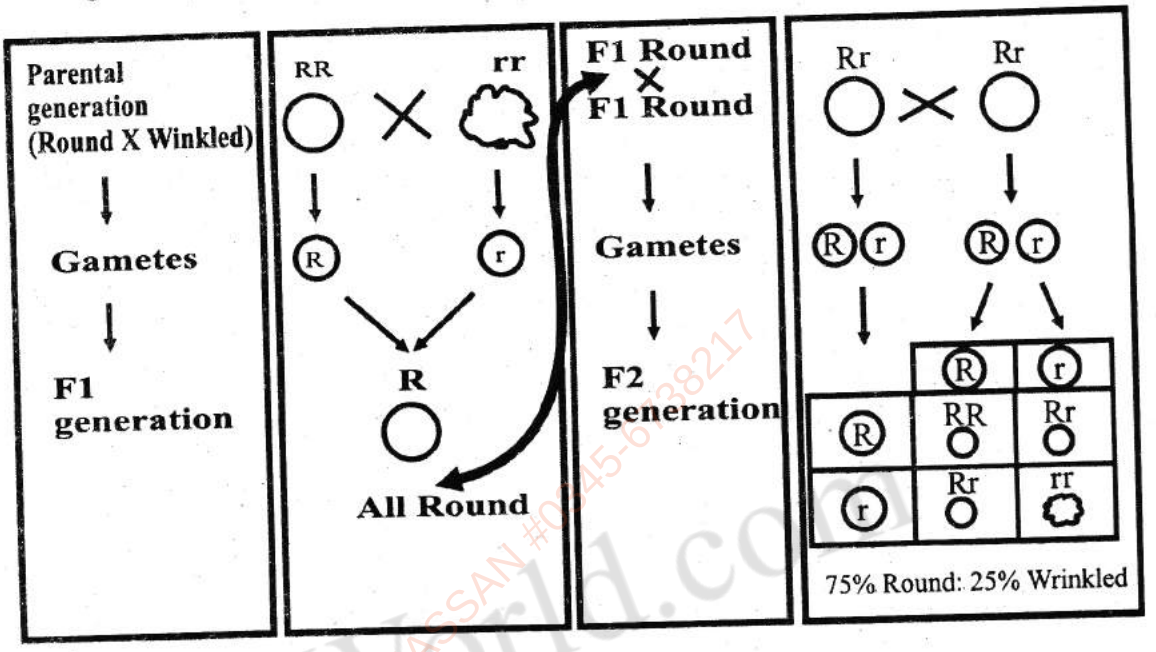
Experiments on Tall and Short Plants:

Similarly, when "true-breeding" tall plants were crossed with "true-breeding" short plants, all offspring of F1 were tall plants i.e. tallness was a dominant trait. When members of F1 generation were self-fertilized, Mendel got the ratio of tall to short plants in F2 as 3:1.

Conclusion:

Mendel concluded that the traits under study were controlled by discrete (separable) factors or genes.



Q. No. 7 State and explain Mendel's Law of Independent Assortment.

MENDEL'S LAW OF INDEPENDENT ASSORTMENT

Introduction:

Gregor Mendel was a monk (priest) in Austria. He developed the fundamental principles of genetics.

Selection of Plant:

Mendel selected pea plant (*Pisum sativum*) to carry out a large number of experiments.

Statement:

The alleles of a gene pair segregate (get separated and distributed to gametes) independently from the alleles of other gene pairs.

Explanation:

Mendel studied two contrasting traits at a time. Such crosses are called dihybrid crosses.

Selection of Phenotypes:

He performed experiments on two seed traits:

Seed shape:

The trait of round seeds, controlled by allele R, was parental dominant over wrinkled controlled by allele r) seeds.

Seed colour

Yellow seed colour controlled by Y was dominant over green controlled by y.

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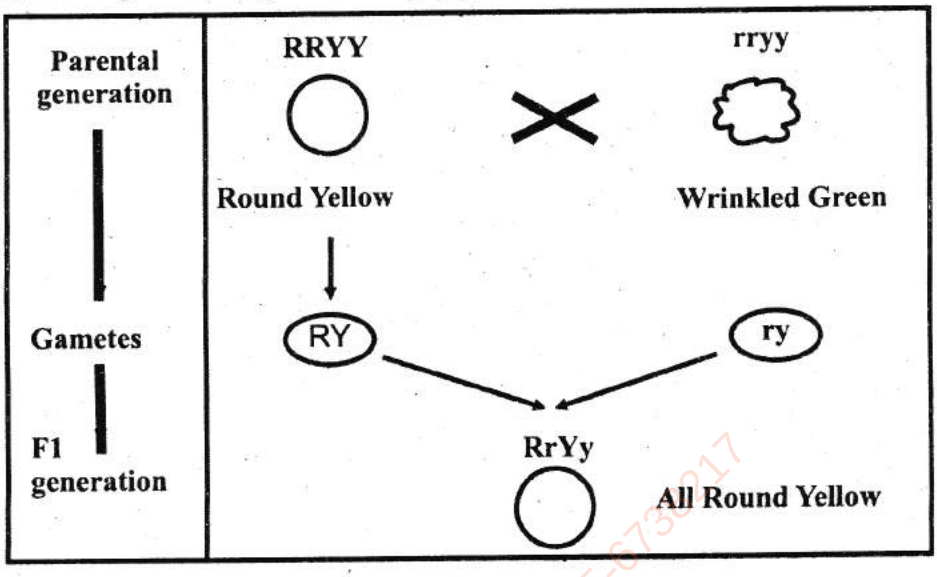
Cross Fertilization of True Breeding:

Mendel crossed a true- breeding plant-that had round yellow seeds (RRYY) with a true breeding plant having wrinkled green seeds (rryy).

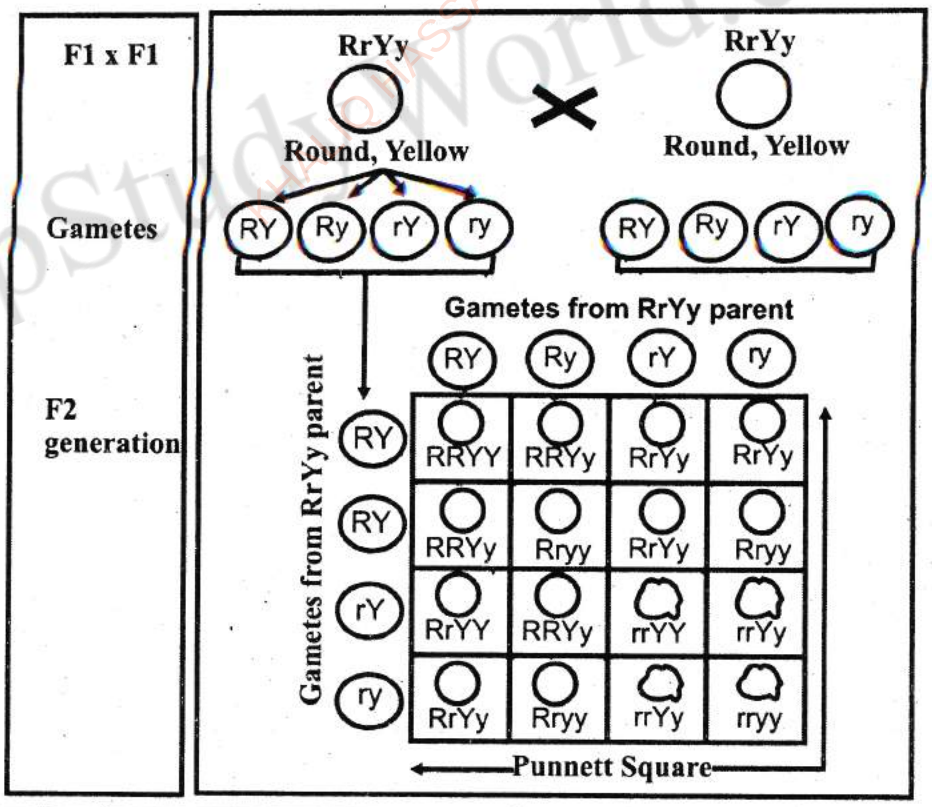


Results:

All seeds in F1 generation were round yellow.

Self Fertilization of F1:

When F1 seeds grew into plants, they were self-fertilized.



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Results:

This cross produced seeds with four phenotypes.

Round yellow seeds = 315
Round green seeds = 108
Wrinkled yellow seeds = 101
Wrinkled green seeds = 32

Phenotypic Ratio:

The ratio of these phenotypes was 9:3:3:1.

Conclusion:

Mendel concluded that the two traits i.e. seed shape and seed colour are not tied with each other. The segregation of 'R' and 'r' alleles happens independently of the segregation of 'Y' and 'y' alleles. From this, Mendel concluded that different traits are inherited independently of one another.

Q. No. 8 Write a note on Punnett Square.

PUNNETT SQUARE

Introduction:

Punnett square is named after R. C. Punnett who was an English mathematician.

Definition:

The diagram that is used to predict an outcome of a particular cross or breeding experiment is called Punnett square.

Determination of Gametes:

The gametes of both parents having all possible genetic set-ups are determined.

Usage of Checker Boarb:

A checker board is used to cross all the possible gametes of one parent with all the gametes of other parent.

Advantage:

A biologist can find all the possible genotypes of off springs.

Q. No. 9 Write a note on co-dominance.

CO-DOMINANCE

Introduction:

After the discovery of Mendel's work, scientists began experiments on the genetics of various organisms. These experiments proved that all the traits in organisms do not follow Mendel's laws. For example, it was found that there are many traits which are controlled by more than one pair of genes. Similarly for many traits there are more than two alleles in a gene pair.

Definition:

The situation where two different alleles of a gene pair express themselves completely, instead of showing a dominant-recessive relationship is called co-dominance.

Phenotype of Heterozygous:

As a result, the heterozygous organism shows a phenotype that is different from both homozygous parents.

Example:

Expression of Human Blood Group AB:

The ABO blood group system is controlled by the gene 'I'. This gene has three alleles:

- I^A
- I^B
- i

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Blood Group A:
The allele I^A produces antigen A in blood and the phenotype is blood group A.

Blood Group B:
The allele I^B produces antigen B in blood and the phenotype is blood group B.

Blood Group O:
The allele i does not produce any antigen and the phenotype is blood group O.

Blood Group AB:
The alleles I^A and I^B are dominant over i. When there is a heterozygous genotype of $I^A I^B$, each of the two alleles produces the respective antigen and neither of them dominates over the other.

Genotype	Antigen produced	Phenotype	Relationship between Alleles
$I^A I^A$ or $I^A i$	Antigen A	Blood Group A	Allele I^A is dominant over i
$I^B I^B$ or $I^B i$	Antigen B	Blood Group B	Allele I^B is dominant over i
ii	No Antigen	Blood Group O	Allele I is recessive
$I^A I^B$	Antigen A & Antigen B	Blood Group AB	Alleles I^A and I^B are co-dominant

Q. No. 10 Write a note on incomplete dominance.

INCOMPLETE DOMINANCE

Definition:

The situation where, in heterozygous genotypes, both the alleles express as a blend (mixture) and neither allele is dominant over the other is called incomplete dominance.

Intermediate Phenotype:

As a result of this blending, an intermediate phenotype is expressed.

Example:

In Four 0 Clock plants, there are three flower colours:

- Red
- Pink
- White

There is no specific gene responsible for producing pink flowers.

Explanation:

In four a clock plant, the trait of flower colour is controlled by two alleles 'R and r'.

- The true breeding red flower plants have RR alleles.
- The true breeding white flower plants have rr alleles.

Cross Fertilization of True Breeding:

A homozygous red flowered plant (RR) is crossed with homozygous white flowered plant (rr).

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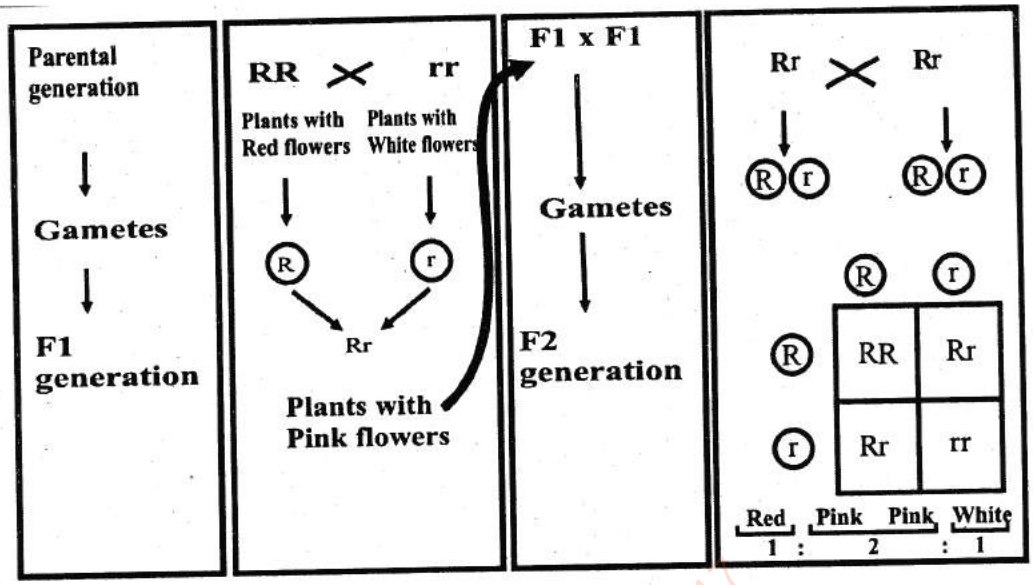

Result:

The heterozygous (Rr) plants of F1 generation produce pink flowers.

Conclusion:

Pink is a blend of red and white colours. This result clearly indicates that neither of the red flower allele (R) and white flower allele (r) is dominant.

Self Fertilization of F1:

When two heterozygous plants with pink flowers (Rr) are crossed, F2 generation shows phenotypes of red, pink and white flowers in the ratio 1:2:1.

Q. No. 11 Define variations. Discuss its sources.

VARIATIONS

Introduction:

Sexual reproduction produces variations in the next generation. No two individuals resulting from separate fertilizations are genetically identical.

Definition:

The differences among the individuals of the same species are called variations.

SOURCES OF VARIATIONS

The main sources of variations in sexually reproducing populations are as follow:

Genetic Recombinations:

The genetic recombination produced through crossing over occurring during meiosis results in gametes with variations.

Mutations:

Mutations which are the sudden changes in the structure of DNA are important source of variations. Mutations also happen during gametes formation through meiosis.

Random Fertilization of Gametes:

During fertilization, one of the millions of sperms combines with a single egg. The chance involved in this combination also act as the source of variations.

Gene Flow:

The movement of genes from one population to another is called as gene flow. It is also an important source of variations.

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Different Combinations of Chromosomes:

Variations are also caused by different combinations of chromosomes in gametes and then in zygote.

In the case of humans, the possible number of chromosomal combinations at fertilization is 70,368,744,177,664. In other words, a couple can produce more than 70 trillion genetically different children.

Q. No. 12 Differentiate between discontinuous and continuous variations.

DIFFERENCE BETWEEN DISCONTINUOUS AND CONTINUOUS VARIATIONS

Discontinuous Variations	Continuous Variations
Phenotypes: The individuals of a population either have distinct phenotypes, which can be easily distinguished from each other. ***Genes:*** Discontinuous variations are controlled by the alleles of a single gene pair. ***Environmental Effect:*** The environment has little effect on this type of variations. ***Examples:*** Blood Groups: In human population, an individual has one of the four distinct phenotypes of blood groups (A, B, AB or O) and cannot have in between. ***Other examples are:*** • Tongue rolling • Person with six fingers of hand or foot	***Phenotypes:*** In continuous variations, the phenotypes show a complete range of measurements from one extreme to the other. ***Genes:*** Continuous variations are controlled by many genes. ***Environmental Factors:*** Continuous variations are often affected by environmental factors. ***Examples:*** Height: In every human population, the individuals have a range of heights (from very small to tall). No population can show only two or three distinct heights. ***Other examples are:*** • Weight • Feet size • Intelligence

Q. No. !3 Discuss that variations lead to evolution. Explain role of Charles Darwin.

VARIATIONS LEAD TO EVOLUTION

Definition:

The change in the characteristics of a population or species of organisms over the course of generations is called organic evolution or biological evolution.

Explanation:

The evolutionary changes are always inheritable. The changes in an individual are not considered as evolution, because evolution refers to populations and not to individuals.

Processes of Organic Evolution:

Organic evolution includes two major processes:

- Alteration in genetic characteristics (traits) of a type of organism over time
- Creation of new types of organisms from a single type

Contribution of C. de Buffon:

French biologist C. de Buffon (1707-1788) was the first to hint at evolution.

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Contribution of Lamarck:

J. de Lamarck (1744-1829) was the first to propose a mechanism of evolution. Lamarck's ideas were soon rejected due to the vagueness of the mechanisms he proposed.

The study of evolution determines the ancestry and relationships among different kinds of organisms.

Theory of Special Creation:

This theory states that:

"All living things had been created in their current form only a few thousand years ago."

- This was an anti-evolution idea.

Scientific Research:

The scientific work in eighteenth century led to the idea that living things might change as well.

ROLE OF CHARLES DARWIN

Charles Darwin (1809-1882) proposed the mechanism of organic evolution in 1838. It was called as "The Theory of Natural Selection". Darwin proposed this theory after his 5-year voyage on the HMS (His Majesty's Ship) Beagle.

Publication:

He also published a book on "The Origin of Species by means of Natural Selection" in 1859.

Acceptance:

Darwin's theory of evolution was not widely accepted because of lack of sufficient evidence. Modern evolutionary theory began in the late 1920s and early 1930s. Some scientists proved that the theory of natural selection and Mendelian genetics are the same ideas just as Darwin had proposed.

Mechanism of Evolution

Variations:

Almost every population contains several variations for the characteristics of its members. In other words, there are morphological and physiological variations in all populations.

Natural Selection:

Natural selection is the process by which the better genetic variations become more common in successive generations of a population.

Adaptations:

Different populations face different environments and they have to adapt to different conditions.

Survival of the Fittest:

The central concept of natural selection is the evolutionary fitness of an organism. Fitness means an organism's ability to survive and reproduce. Organisms produce more offspring than can survive and these offspring vary in fitness. These conditions produce struggle for survival among the organisms of population. The organisms with favourable variations are able to reproduce and pass these variations to their next generations. On the other hand, the rate of the transmission of unfavourable to next generations is low.

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Selection of Variations:

The favourable variations are "selected for" their transmission to next generations, while the unfavourable variations are "selected against" their transmission to next generations.

Natural Selection in Rats:

A mouse population with variations in skin colour.

- Light coloured
- Medium coloured
- Dark coloured

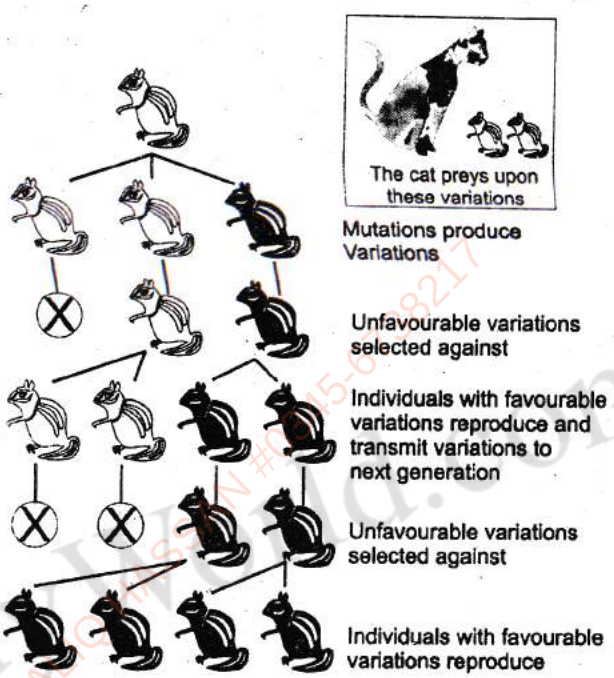

Figure: Natural Selection in Rats

Cat preys upon light and medium coloured mouse. In first generation, light coloured mouse is preyed upon by cat. Only medium and dark coloured mouse can make their next generations. In next generation, population again contains light, medium and dark coloured mouse. Cat preys upon the light and medium coloured mouse. Now only the dark coloured mouse make new generation. If this happens in many generations, the dark coloured (favourable variation) mouse in the population.

Outcome of Natural Selection:

As a result of natural selection, the allele that gives more fitness of characteristics (favourable variations) than other alleles becomes more common within population. So, the individuals with favourable variations become a major part of population while the individuals with harmful or unfavourable variations become rarer.

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NATURAL SELECTION IN MOTHS

In England, the moths had two variations:

- Dark coloured moths
- White coloured moths

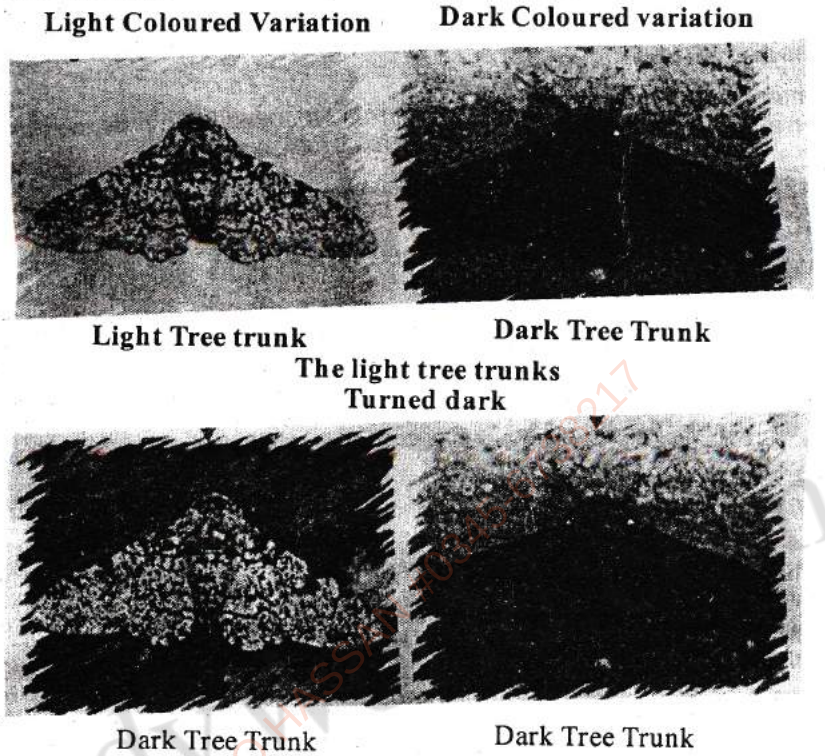

Figure: White and Dark Coloured Moths

The moths used to rest on the light coloured tree trunks on which white lichens had grown. In the 19th century when industries were established in England, the lichens on tree trunks died due to polluted air and the naked tree trunks turned dark. Now the white moth variation became harmful because a white moth resting on a dark tree trunk was easily visible to the predatory birds. The natural selection selected dark moths to reproduce. In this way dark coloured moth became more common and at last the white moths disappeared from population. In this case, the dark colour variation in moth may be considered an adaptation to environment.

Q. No. 14 Write a note on artificial selection.

ARTIFICIAL SELECTION

Introduction:

The term "artificial selection" was expressed by the Persian scientist Abu Rayhan Biruni in the 11th century. Charles Darwin also used this term in his work on natural selection.

Darwin's Observations:

Darwin noted that many domesticated animals and plants had special properties that were developed by:

- Intentional breeding among individuals with desirable characteristics
- Discouraging the breeding of individuals with less desirable characteristics

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Task:

In artificial selection, humans favour specific variations for selection while in natural selection the environment selects or rejects variations.

Definition:

The intentional breeding between individuals for certain traits or combination of traits is called artificial selection or selective breeding.

Revolution:

Selective breeding has revolutionized agricultural and livestock production throughout the world. Animals or having desirable characteristics are selected for breeding. In this way, many new generations with desirable characteristics are produced.

Breeds:

The bred animals are as breeds in artificial selection are called breeds.

Cultivars:

The bred plants in artificial selection are known as varieties or cultivars.

Examples:

Numerous breeds of the following animals have been produced by artificial selection:

- Sheep for wool
- Goat for meat
- Cow for milk
- Hen for eggs

Plant Varieties:

Similarly many plant varieties (cultivars) have been produced for better quantity and quality of:

- Cereals
- Fruits
- Vegetables

Plants Varieties of Wild Mustard:

- Kohlrabi
- Kale
- Cabbage
- Broccoli
- Cauliflower

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REVIEW QUESTION:

MULTIPLE CHOICE QUESTIONS

1. **An organism's expressed physical trait, such as seed colour or pod shape, is called its**
   (a) Genotype (b) Phenotype (c) Karyotype (d) Physical type
2. **An organism has two different alleles for a single trait. Its genotype is said to be:**
   (a) Homozygous (b) Heterozygous (c) Hemizygous (d) Homologous
3. **In the cross-pollination between a true-breeding yellow pod plant and true-breeding green pod plant, where green pod colour is dominant, the resulting offsprings (F1 generation) will be:**
   (a) ¼ green, ¾ yellow (b) All yellow
   (c) ¼ yellow, ¾ green (d) All green
4. **How many genetically different kinds of gametes an individual with genotype AAbb can produce?**
   (a) 1 (b) 2 (c) 4 (d) 8
5. **Which of the following statements regarding genes is FALSE?**
   (a) Genes are located on chromosomes
   (b) Genes consist of a long sequence of DNA
   (c) A gene contains information for the production of a protein
   (d) Each cell contains a single copy of every gene
6. **Mendel's primary contribution to our understanding of inheritance was:**
   (a) The idea that genes are found on chromosomes
   (b) Explanation of the patterns of inheritance
   (c) The discovery of alleles
   (d) Determining that informations contained in DNA are for protein synthesis
7. **A purple-flowered pea plant has the genotype PP, which of the following statement about this plant is FALSE?**
   (a) Its phenotype will be white flowers
   (b) It has a homozygous dominant genotype
   (c) When bred to a white-flowered plant, all offspring will be purple flowered
   (d) All the gametes produced will have the same flower colour allele
8. **Charles Darwin proposed that organisms produce many more off spring than can possible survive on the limited amount of resources available to them. According to Darwin, the offspring that are most likely to survive are those that:**
   (a) Are born first and grow fastest (b) Are largest and most aggressive
   (c) Have no natural predators (d) Are best adapted to the environment

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	b	2	b	3	d	4	A
5	d	6	b	7	a	8	d

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SHORT QUESTIONS

1. **Define genotype and phenotype.**
   Consult Long Question No. 4
2. **What do you mean by dominant and recessive alleles?**
   Consult Long Question No. 4
3. **What are the homozygous and heterozygous genotypes?**
   Consult Long Question No. 4
4. **Differentiate between natural and artificial selection.**
   Consult Long Question No. 13 and 14

UNDERSTANDING THE CONCEPT

1. **Describe the structure of chromatin.**
   Consult Long Question No. 1
2. **Describe Mendel's law of segregation.**
   Consult Long Question No. 6
3. **Explain how Model proved the law of independent assortment.**
   Consult Long Question No. 7
4. **How would you prove that variations lead to evolution?**
   Consult Long Question No. 13
5. **Explain the phenomenon of incomplete dominance with the help of example.**
   Consult Long Question No. 10
6. **What do you mean by co-dominance. Give an example.**
   Consult Long Question No. 9

THE TERMS TO KNOW

Allele:

The alternative form of a gene

Artificial selection:

Selective breeding; intentional breeding between individuals for certain traits or combination of traits

Breeds:

The animals which are bred through artificial selection

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Chromatin:

The chemical material that make the structure of the chromosome

Co-dominance:

The situation where two alleles of a gene pair express their traits independently instead of showing a dominant-recessive relationship

Cultivar:

The plants which are bred through artificial selection

Dihybrid cross:

A genetic cross in which two pairs of contrasting traits are studied

Dominant:

The trait that appears in the offsprings of a cross between two homozygous individuals showing contrasting form of the trait

Gene:

Unit of inheritance; consists of the length of DNA that contains specific instructions for the synthesis of a protein molecule

Genotype:

The specific combination of genes in an individual; may be homozygous or heterozygous

Heterozygous:

The genotype that has two different alleles of a trait

Histone:

The protein present in the structure of the chromosome

Homologous chromosomes:

A pair of chromosomes having the same size and shape and carrying alleles for the same trait

Homozygous:

Having two identical alleles of a trait

Incomplete dominance:

A type of inheritance in which neither of the pair of contrasting alleles is dominant over the other and the heterozygous individual is intermediate in phenotype

Inheritance:

The transmission of characteristics from parents to offsprings

Locus:

Plural Loci; the locations or positions of genes on chromosomes

Monohybrid cross:

A genetic cross in which only one pair of contrasting traits is studied

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Mutation:

Changes in chromosomes or DNA (gene); produces variations intraits

Natural selection:

The process in which organisms with favourable variations survive and produce more offsprings than less well-adapted organisms

Nucleosome:

The structure formed by the wrapping of DNA around histone proteins

Organic evolution:

Biological evolution; The modification of the characteristics in the species or populations of organisms during their descent, generation by generation

Phenotype:

The expression of the genotype in the form of trait

Recessive trait:

The trait which is masked in the offspring of a cross between two homozygous individuals showing contrasting forms of the trait

Trait:

The characteristics which are controlled and transmitted to next generation through genes

True-breeding:

The homozygous individual

Variations:

A characteristic in an individual that differs from the typical characteristics of other individuals of the same species

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CHAPTER 16

MAN AND HIS ENVIRONMENT

MULTIPLE CHOICE QUESTIONS

(1) A group of organisms which can interbreed freely in nature and produce fertile offsprings:
(a) Community (b) Population (c) Biosphere (d) Species

(2) An example of artificial ecosystem:
(a) River (b) Ocean (c) Rain forest (d) Aquarium

(3) Thickness of biosphere:
(a) 10 Km (b) 15 Km (c) 20 Km (d) 25 Km

(4) Which one is not a biotic component:
(a) Producer (b) Consumer (c) Decomposer (d) Air

(5) The consumers that feed on plants:
(a) Omnivores (b) Carnivores (c) Herbivores (d) Insectivores

(6) The first trophic level is made up of:
(a) Consumers (b) Producers (c) Herbivores (d) Carnivores

(7) Man is:
(a) Omnivore (b) Carnivore (c) Insectivore (d) Herbivore

(8) Which one is the primary source of energy?
(a) Moon (b) Mars (c) Sun (d) Mercury

(9) A network of food chains which are interconnected at various trophic levels:
(a) Community (b) Population (c) Species (d) Food web

(10) Who developed the concept of ecological pyramids?
(a) Charles Darwin (b) Charles Elton (c) Charles Brown (d) Charles Asker

(11) Which one cycle is a perfect cycle in nature?
(a) Carbon (b) Oxygen (c) Nitrogen (d) Water

(12) The green house effect and global warming are due to the increase in the concentration of:
(a) Sulphur (b) Carbon dioxide (c) Water vapours (d) Oxygen

(13) Ammonia is converted into nitrites by:
(a) *Streptococcus* (b) *Penicillium* (c) *Nitrobacter* (d) *Nitrosomonas*

(14) The biological process in which nitrates and nitrites are reduced to nitrogen gas:
(a) Ammonification (b) Nitrification (c) Denitrification (d) Assimilation

(15) Excessive denitrification reduces soil fertility and is stimulated by:
(a) Water logging (b) Lack of aeration (c) Organic matter (d) All of these

(16) Types of symbiosis:
(a) 1 (b) 2 (c) 3 (d) 4

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(17) **Plants show competition for:**
(a) Space (b) Light (c) Water (d) All of these

(18) **Which one is an ectoparasite?**
(a) Leech (b) Liver fluke (c) Tape worm (d) *Ascaris*

(19) **Parasitic plants have special roots called:**
(a) Rhizoids (b) Adventitious (c) Tap (d) Haustoria

(20) **Small plants found growing on the other larger plants:**
(a) Lichens (b) Parasites (c) Epiphytes (d) Barnacles

(21) **Large mammals that feed on honey:**
(a) Lions (b) Badgers (c) Deer (d) Zebras

(22) **Which are green house gases?**
(a) Carbon dioxide (b) Methane (c) Ozone (d) All of these

(23) **Average sea level rising in a year:**
(a) 0.6 cm (b) 0.7 cm (c) 0.8 cm (d) 0.9 cm

(24) **Since 1800, the amount of carbon dioxide in atmosphere has increase up to:**
(a) 20% (b) 25% (c) 30% (d) 35%

(25) **Clearing of forests by natural causes or humans:**
(a) Deforestation (b) Afforestation (c) Forestation (d) None of these

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	d	3	c	4	d	5	c
6	c	7	a	8	c	9	d	10	b
11	a	12	b	13	d	14	c	15	d
16	c	17	d	18	a	19	d	20	c
21	b	22	d	23	d	24	c	25	a

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SHORT QUESTIONS

Q. No. 1 Define environment

ENVIRONMENT

The sum of physical (abiotic) and biological (biotic) conditions which influence the organism is called the environment of that organism.

Q. No. 2 Define ecology.

ECOLOGY

The branch of Biology that deals with the study of the interrelationship between organisms and their environment is called ecology.

Q. No. 3 What are biogeochemical cycles?

BIOGEOCHEMICAL CYCLES

The cyclic pathways through which materials move from environment to organisms and back to environment are called biogeochemical cycles.

Q. No. 4 Why carbon cycle is called perfect cycle?

CARBON CYCLE AS PERFECT CYCLE

Carbon cycle is a perfect cycle in the sense that carbon is returned to atmosphere as soon as it is removed.

Q. No. 5 How the balance of carbon cycle has been upset by human activities?

UPSET OF CARBON CYCLE

The balance of carbon cycle has been upset by human activities such as:

- Deforestation
- Excessive burning of fossil fuels

Q. No. 6 What are the consequences of increasing concentration of carbon dioxide in atmosphere?

CONSEQUENCES OF INCREASED CONCENTRATION OF CARBON DIOXIDE

As a result, the amount of carbon dioxide in atmosphere is increasing, causing:

- The green house effect
- Global warming

Q. No. 7 How is denitrification stimulated?

STIMULATION OF DENITRIFICATION

Excessive denitrification reduces soil fertility and is stimulated by:

- Water logging
- Lack of aeration
- Accumulation of organic matter

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Q. No. 8 In what things plants show competition?

COMPETITION IN PLANTS

Plants also show competition for:

- Space
- Light
- Water
- Minerals

Q. No. 9 What is the relationship between survival of host and parasite?

SURVIVAL OF HOST AND PARASITE

Host can survive without parasite, but parasite cannot survive without host.

Q. No. 10 Why biogeochemical cycles are called as nutrient cycles?

BIOGEOCHEMICAL CYCLES AS NUTRIENT CYCLES

The movement of elements and inorganic compounds is essential for maintenance of life, they are also called 'nutrient cycles'. Carbon cycle is a perfect cycle in the sense that carbon is returned to atmosphere as soon as it is removed.

Q. No. 11 What is the difference between intraspecific and interspecific interactions?

Intraspecific Interactions	Interspecific Interactions
The interactions between the members of the same species are called intraspecific interactions.	The interactions between the members of different species are called interspecific interactions.

Q. No. 12 What is Intergovernmental Panel on Climate Change (IPCC)?

INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE (IPCC)

In 1990, the United Nations established Intergovernmental Panel on Climate Change (IPCC). It provides scientific advice to the world leaders on issues like the build-up of greenhouse gases and its prevention. According to IPCC, Earth's surface temperature has increased 0.2°C per decade in the past 30 years

Q. No. 13 How Maldives' survival is on threat?

THE MALDIVES' SURVIVAL

Scientists fear that the sea level is rising up to 0.9cm a year. Rise in sea level has worst effects on coastal countries. Most of the islands of the Maldives are less than 1 metre above sea level. It is estimated that within 100 years, the Maldives might become uninhabitable and the citizens would be forced to evacuate.

Q. No. 14 What is the relationship between honeyguide bird and badgers?

HONEYGUIDE BIRD AND BADGERS

The honeyguide bird feeds on wax and the larvae present in honeycombs. It flies around looking for honeycombs. but it is not strong enough to open the comb. Badgers are large mammals that feed on honey. When a honeyguide bird goes to find honeycombs, the badger follows it. When the bird finds a honeycomb, it calls the badger. Sometimes the bird has to stop and wait for the slow-moving badger. After reaching there, the badger opens the honeycomb and both of them eat their foods together. Traditionally, humans have also used these birds to find honeybee colonies.

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Q. No. 15 What are pollutants?

POLLUTANTS

The substances that actually cause pollution are called the pollutants.

Examples:

They may be:

- Industrial effluents
- Domestic wastes
- Medical wastes

Q. No. 16 What are two types of pollutants?

TYPES OF POLLUTANTS

Pollutants are of two types:

- Biodegradable
- Non-biodegradable

Q. No. 17 What is noise? What are its bad effects?

NOISE

The unwanted, unpleasant and annoying sounds are termed as noise. Noise is also considered as a form of pollution.

Immediate Effects:

Immediate effects of noise pollution are:

- Annoyance
- Aggression

Long Term Effects:

The long term effects of noise pollution are:

- Hearing loss
- Depression
- Hypertension

Q. No. 18 What will be the increase ion temperature in the next hundred years?

According to estimates, at the current rate of increase, the average global temperature will go up by 3°C to 8°C in the next 100 years.

Q. No. 19 In which countries the harmful effects of the UV rays are visible?

HARMFUL EFFECTS OF THE UV RAYS

The harmful effects of the UV rays are visible in the countries such as Australia and New Zealand where the rate of skin cancer is higher than the other regions of the world.

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LONG QUESTIONS

Q. No. 1 Describe levels of ecological organization.

LEVELS OF ECOLOGICAL ORGANIZATION

In ecology, the levels of organization range from organism to biosphere.

Organism:

An organism may be unicellular or multicellular.

Species:

The first level of ecological organization is species. A species may be defined as:
A group of organisms which can interbreed freely in nature to produce fertile offspring is called species.

Population:

A group of the organisms of the same species inhabiting a specific geographical area (habitat) at a particular time is called a population.

Community:

All the populations that live in a habitat and interact in various ways with one another are collectively called a community.

Ecosystem:

The self-sufficient unit of an environment that is formed as a result of interactions between its biotic community and the abiotic components is known as an ecosystem.

Explanation:

Living organisms cannot live isolated from the non-living part of their environment. The biotic and abiotic components of environment interact with each other to form an ecosystem.

Types of Ecosystem:

There are two types of ecosystems:

Natural Ecosystem:

The examples are:

- A pond
- A lake
- A forest

Artificial Ecosystem

Example:

- An aquarium

Biosphere:

All ecosystems of the world together form the biosphere. It includes all the ecosystems of the planet Earth. In other words, the biosphere consists of all organisms present on the Earth and all regions of the Earth where they live.

Range:

Biosphere ranges from the floor of oceans to the tops of the highest mountains.

Thickness:

It is about 20 kilometres thick. The biosphere makes a thin layer surrounding the planet Earth. If you consider the Earth as of the size of an apple, then the biosphere will be as thick as the apple's skin.

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Q. No. 2 Write a note on components of an ecosystem.

COMPONENTS OF ECOSYSTEM

Ecosystem:

The self-sufficient unit of an environment that is formed as a result of interactions between its biotic community and the abiotic components is known as an ecosystem.

An ecosystem comprises of two basic parts:

1. Abiotic components
2. Biotic components

1. Abiotic Components:

Definition:

The non-living factors present in ecosystem are called abiotic components.

Examples:

The important non-living factors are:

- Light
- Air
- Water
- Soil
- Basic elements and compounds

2. Biotic Components:

Definition:

All the living parts (organisms) of the ecosystem are called as biotic components.

Classification:

Biotic components are further classified as:

(i) Producers
(ii) Consumers
(iii) Decomposers

(i) Producers:

The producers are the autotrophs present in an ecosystem. These organisms are able to synthesize complex organic compounds (food) from inorganic raw materials. Producers form the basis of any ecosystem.

Examples:

Producers include:

- Plants
- Algae
- Photosynthetic bacteria

Terrestrial Ecosystem:

In terrestrial ecosystems, plants are the main producers.

Aquatic Ecosystem:

In aquatic ecosystems, the main producers are the floating photosynthetic organisms (mainly algae) called phytoplankton and shallow water rooted plants.

(ii) Consumers:

The consumers are heterotrophs. They cannot synthesize their food and so depend upon producers for food.

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Examples:

Consumers include:

- All animals
- Fungi
- Protozoans
- Many of the bacteria

Classification of Animals:

The animals are the major consumers of ecosystems. They are further classified as:

Herbivores (Primary Consumers):

The animals that feed on plants are called herbivores. The herbivores are the primary consumers. They feed directly on plants or products of plants. For example:

- Cattle
- Deer
- Rabbit
- Grasshopper

Primary Carnivores:

Primary carnivores (secondary consumers) feed on herbivores. For example:

- Fox
- Frog
- Predatory birds
- Many fishes
- Snakes

Secondary Carnivores:

Secondary carnivores (tertiary consumers) feed on primary carnivores. For example:

- Wolf
- Owl

Tertiary Carnivores:

Tertiary carnivores feed on secondary carnivores. For example:

- Lion
- Tiger

(iii) Decomposers:

Decomposers or reducers break down the complex organic compounds of dead matter (of plants and animals) into simple compounds. They secrete digestive enzymes into dead and decaying plant and animal remains to digest the organic material. After digestion, decomposers absorb the products for their own use. The remaining substances are added to environment.

Examples:

- Bacteria
- Fungi

Q. No. 3 Describe flow of energy in an ecosystem.

FLOW OF ENERGY

In an ecosystem, energy travels from one trophic level to the next.

Trophic Level:

Trophic level is the level at which an organism feeds in food chain. The first trophic level is made of producers; the second of primary consumers and so on.

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Unidirectional Flow:

The flow of energy in different trophic levels of ecosystem is unidirectional.

Primary Source:

The sun is the primary source of energy for all ecosystems.

Energy in Producers:

Producers get solar energy and transform it into chemical energy by the process of photosynthesis. They store their energy in their tissues and also transform it into mechanical and heat energy during their metabolic activities.

Energy in Herbivores:

The energy in producers' tissues flows to herbivores when producers are eaten. Herbivores transform it into mechanical and heat energy during their metabolic activities and store the rest in their tissues.

Energy in Carnivores:

Carnivores eat herbivores and get energy. They also use it for their body activities and store the rest in their tissues.

Energy in Decomposers:

After the death of producers and consumers, the energy stored in their tissues is used by decomposers.

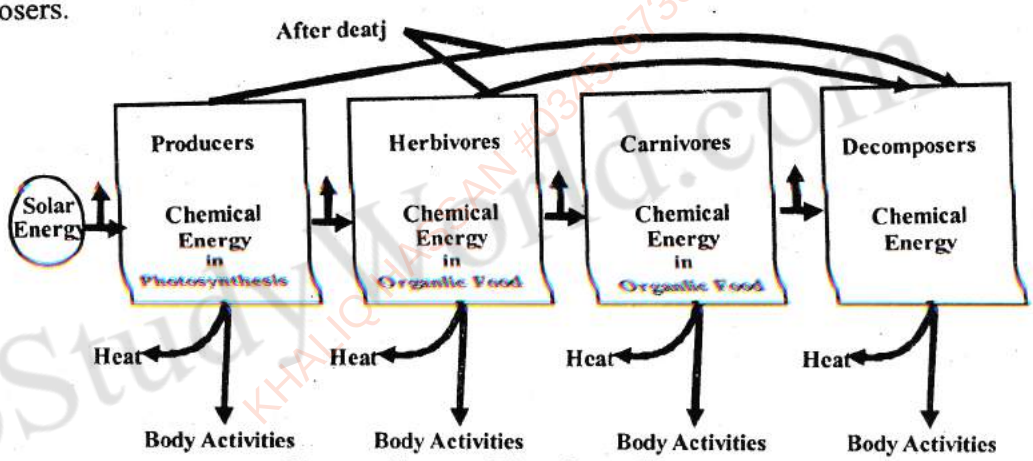

Figure: Energy Flow in an Ecosystem

Law of Thermodynamics:

The storage and expenditure of energy in an ecosystem is in accordance with the basic law of thermodynamics.

Statement:

Energy can neither be created nor destroyed but can be transformed from one form into another.

Energy within Ecosystem:

In an ecosystem there is:

Constant Flow:

Constant flow or transfer of energy from the Sun through producers to consumers and decomposer

Decrease in Energy:

A significant decrease in useful energy during transfer of energy at each trophic level

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Q. No. 4 Describe flow of materials in an ecosystem.

FLOW OF MATERIALS

The materials flow from one trophic level to the next by means of food chains and food webs.

Trophic Level:

Trophic level is the level at which an organism feeds in food chain. The first trophic level is made of producers; the second of primary consumers and so on.

FOOD CHAIN

Definition:

A series of organisms within an ecosystem, in which each organism feeds on the one before it and is fed by the one after it is called food chain.

Example:

Following is a food chain in an ecosystem:

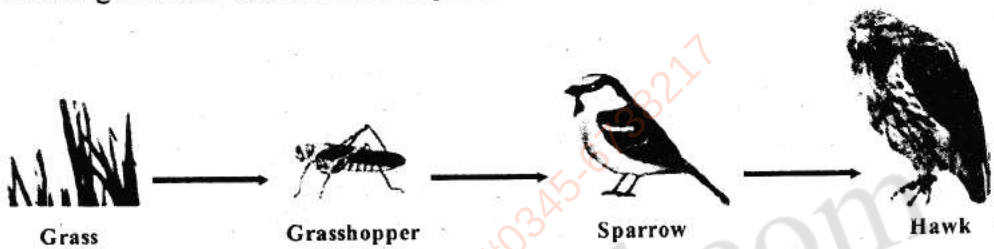

Figure: A Simple Food Chain

Explanation:

The base of food chain is always formed by a plant (producer). It is eaten by a primary consumer, which is preyed upon by a secondary consumer. The secondary consumer may be eaten by a tertiary consumer. A food chain, can therefore, be represented as,

Producer ⟶ Primary Consumer ⟶ Secondary Consumer ⟶ Tertiary Consumer

Advantage:

A food chain involves a nutritive interaction among the biotic components of an ecosystem.

Number of Trophic levels:

Usually there are 4 or 5 trophic levels. Shorter food chains provide greater available energy and vice-versa.

FOOD WEB

Definition:

A network of food chains which are interconnected at various trophic levels is called food web.

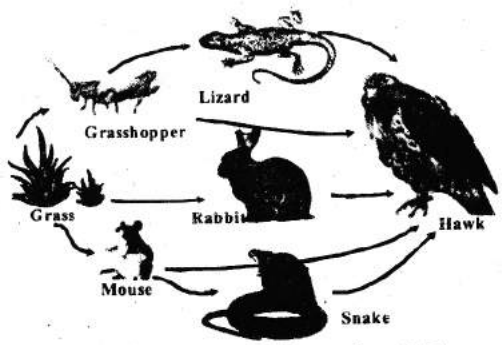

Figure: A Food Web in Grassland Ecosystem

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Explanation:

In nature, food chains are very complex, as one organism may be the food source of many other organisms. Thus, instead of a simple linear food chain, there is a web-like structure formed by these interlinked food chains. Such interconnected food chains collective make food web.

Q. No.5 Write a note on ecological pyramids.

ECOLOGICAL PYRAMIDS

Introduction:

In 1927, Charles Elton (an English ecologist) developed the concept of ecological pyramids. Organisms in Food Chain:

Observation:

He noted that the animals present at the beginning of food chain are abundant in number while the animals present at the end of food chain are fewer in number.

Definition:

A representation of the number of individuals or amount of biomass or energy present in various trophic levels of a food chain.

TYPES OF PYRAMIDS

Ecological pyramids are of three types. The two are:

1. Pyramid of Numbers
2. Pyramid of Biomass

1. Pyramid of Numbers:

Definition:

The graphic representation of the number of individuals per unit area at various trophic levels is called pyramid of numbers.

Explanation:

Usually, producers are present in large number, primary consumers are in lesser number, secondary consumers are fewer, and so on. So, the producers are of smallest size but maximum in number, while the tertiary consumers are larger in size but lesser in number.

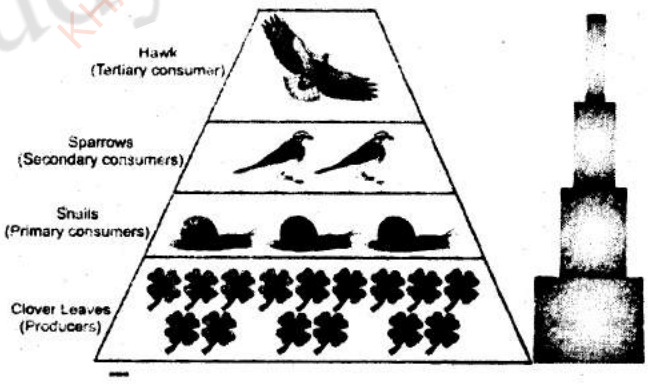

Figure: Pyramid of Numbers in an Ecosystem

2. Pyramid of Biomass:

Definition:

The graphic representation of biomass present per unit area at different trophic levels is called pyramid of biomass.

Biomass:

The total amount of living organic matter in an ecosystem at any time is called biomass.

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Explanation:

In a terrestrial ecosystem, the maximum biomass occurs in producers, and there is progressive decrease in biomass from lower to higher trophic levels.

Figure: Pyramid of Biomass in an Ecosystem

Q. No. 6 Write a note on carbon cycle.

CARBON CYCLE

Introduction:

Carbon atom is the principal building block of many kinds of biomolecules. Carbon is found as graphite and diamond in nature. It also occurs as carbon dioxide in atmosphere.

Sources:

Major source of carbon for the living world is carbon dioxide present in atmosphere and water. Carbonates of Earth's crust also give rise to carbon dioxide.

Fossil fuels also contain carbon like:

- Peat
- Coal
- Natural gas
- Petroleum

Photosynthesis:

The major process that brings carbon from atmosphere or water into living world is photosynthesis. Producers take in carbon dioxide from atmosphere and convert it into organic compounds. In this way, carbon becomes a part of the body of producers.

Carbon in Animals:

This carbon enters food chains and is passed to:

- Herbivores
- Carnivores
- Omnivores
- Decomposers

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Respiration:

Carbon dioxide is released back to environment by respiration of producers and consumers.

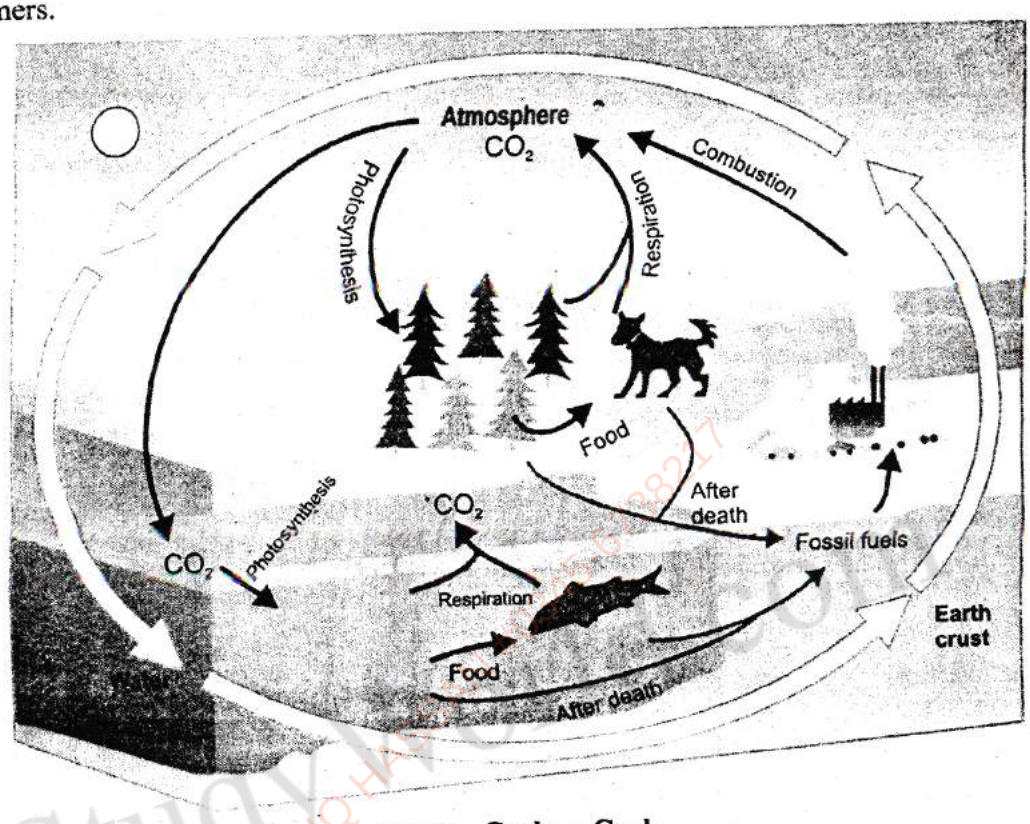

Figure: Carbon Cycle

Decomposition:

It is also released by the decomposition of organic wastes and dead bodies by decomposers.

Combustion:

Burning of wood and fossil fuels also adds large amount of carbon dioxide in atmosphere.

Perfect Cycle:

Carbon cycle is a perfect cycle in the sense that carbon is returned to atmosphere as soon as it is removed.

Upset of Carbon Cycle:

The balance of carbon cycle has been upset by human activities such as:

- Deforestation
- Excessive burning of fossil fuels

Consequences:

As a result, the amount of carbon dioxide in atmosphere is increasing, causing:

- The green house effect
- Global warming

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Q. No. 7 Write a note on carbon cycle.

NITROGEN CYCLE

Occurrence:

Nitrogen is an important component of many biomolecules, like:

- Proteins
- Nucleic acids (DNA and RNA)
- Atmosphere is the reservoir of free gaseous nitrogen

Living organisms cannot pickup this gaseous nitrogen directly from atmosphere except for nitrogen fixing bacteria. It has to be converted into nitrates to be utilised by plants.

Nitrogen cycling involves several stages:

1. Formation of Nitrates:

It is done by the following ways:

Nitrogen Fixation:

The conversion of nitrogen gas into nitrates is called nitrogen fixation. It occurs in the following ways.

Atmospheric Nitrogen Fixation:

Thunderstorms and lightning convert atmospheric gaseous nitrogen to oxides of nitrogen. These oxides dissolve in water and form nitrous acid and nitric acid. The acids in turn combine with other salts to produce 'nitrates'. It is called as atmospheric nitrogen fixation.

Biological Nitrogen Fixation:

Some bacteria also have the ability to transform gaseous nitrogen into nitrates. It is called biological nitrogen fixation. Some of these nitrogen fixing bacteria live as symbionts and many are free-living.

Industrial Nitrogen Fixation:

Nitrogen fixation is also done in industries. In industrial nitrogen fixation, hydrogen is combined with atmospheric nitrogen under high pressure and temperature. It produces ammonia which is further converted into ammonium nitrate.

2. Ammonification and Nitrification:

The breakdown of the proteins of dead organisms and nitrogenous wastes (urea, uric acid etc.) to ammonia is called ammonification. It is done by ammonifying bacteria.

Nitrification:

The process of conversion of ammonia into nitrites and nitrates is called nitrification. It is done by nitrifying bacteria.

Formation of Nitrites:

First, ammonia is converted into nitrites by bacteria like:

- *Nitrosomonas*

Formation of Nitrates:

The nitrites are then converted into nitrates by other bacteria like:

- *Nitrobacter*

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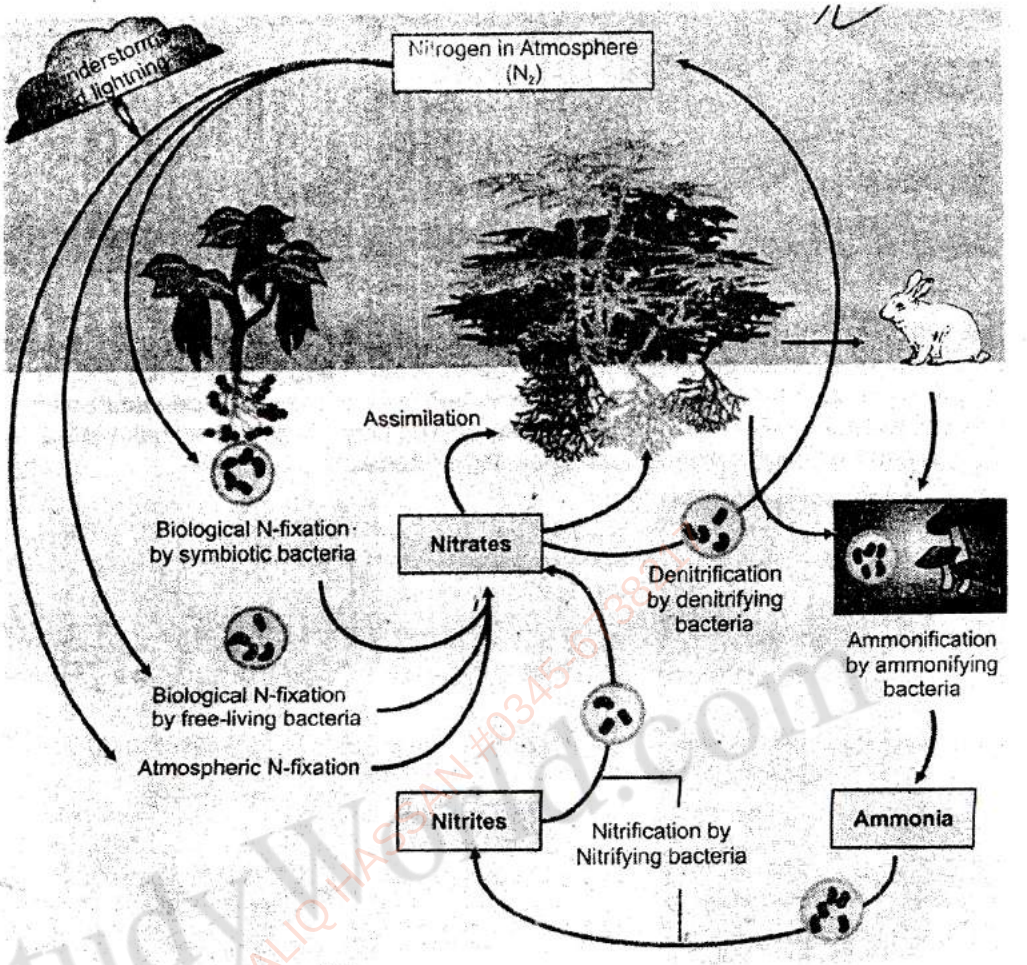

Figure: Nitrogen Cycle

3. Assimilation:

Definition:

The utilization of nitrates by organisms is called assimilation.

Explanation:

The nitrates formed by the processes of nitrification, are absorbed by plants and are utilized for making proteins etc. Animals take nitrogenous compounds from plants.

4. Denitrification:

The biological process in which nitrates and nitrites are reduced to nitrogen gas by denitrifying bacteria is called denitrification. By this process, nitrogen is returned to atmosphere.

Excessive denitrification reduces soil fertility and is stimulated by:

- Water logging
- Lack of aeration
- Accumulation of organic matter

Q. No. 8 Write a note on competition.

COMPETITION

In ecosystems, the natural resources e.g. nutrients, space etc. are usually in short supply. So there is a competition among the organisms of ecosystem for the utilization of resources.

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Types of Competition:

The competition may be intraspecific or interspecific.

Intraspecific Competition:

The competition among the individuals of the same species is called intraspecific competition.

Interspecific Competition:

The competition among the individuals of the different species is called interspecific competition.

Comparison:

Intraspecific competition is always stronger and more severe than the interspecific competition.

Advantage:

Competition helps in maintaining a balance between the available resources and the number of individuals of a species.

Competition in Plants:

Plants also show competition for:

- Space
- Light
- Water
- Minerals

Q. No. 9 Write a note on predation.

PREDATION

Introduction:

It is an interaction between two animals of different species or between a plant and an animal.

Definition:

The relationship between predator and prey is called predation.

Predator:

The organism that attacks, kills and feeds on other organism is called predator.

Prey:

The organism that is being hunted upon is called prey.

Examples:

All carnivore animals are predators. Some examples of predation are given below:

- Frog preys upon mosquito
- Fox preys upon rabbit

There are some examples where a predator is preyed upon by a second predator and then the second one is preyed upon by a third predator. For example, frog (predator 1) is preyed upon by a snake (predator 2) and the snake is preyed upon by an eagle (predator 3).

Predation in Plants:

Certain plants are carnivorous and live as predators. Such plants live in the areas where minerals and other nutrients are lacking. They feed on insects to fulfill their nitrogen requirements. These plants have mechanism to attract insects. For example, they secrete sweet nectar that attracts the insects searching for food. Their leaves are also modified to capture the prey.

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Examples:

- Pitcher plant
- Sundew
- Venus fly trap

Advantage:

Predation keeps the prey population under check, so as to maintain an ecological balance. Humans benefit from this interaction in the biological control of weeds and pests. In order to control pests in an area, their predators are released there.

Q. No. 10 Write a note on symbiosis.

SYMBIOSIS

Definition:

A relationship between members of different species, in which they live together for longer or shorter periods of time, is called symbiosis.

TYPES OF SYMBIOSIS

Symbiosis is of three types:

1. Parasitism
2. Mutualism
3. Commensalism

1. Parasitism:

Definition:

A type of symbiosis between members of different species, in which smaller partner (parasite) derives food and shelter from the body of larger partner (host) and, in turn, harms it is called parasitism.

Temporary Parasitism:

In temporary parasitism, the parasite spends most of its life cycle as independent free-living organisms. Only a part of its life cycle is spent as a parasite. Some common temporary parasites of humans are:

- Leech
- Bed bug
- Mosquito

Permanent Parasitism:

In permanent parasitism, the parasites spend their whole life cycle as parasites. Some examples of permanent parasites are:

- Disease causing bacteria
- All viruses

Classification of parasites:

Parasites may also be classified as ectoparasites and endoparasites.

Ectoparasites:

The parasites that live outside i.e. on the surface of host's body and get food from there are called ectoparasites

Examples:

- Mosquitoes
- Leeches
- Lice

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Endoparasites:

The parasites that live inside the body of host and get food and shelter are called endoparasites.

Examples:

- Bacteria
- Viruses
- Tapeworm
- *Ascaris*
- *Entamoeba*
- *Plasmodium*

Parasitic Plants:

Some plants are parasites on other plants. Parasitic plants grow special types of roots called haustoria into host body and suck the required nutrients from the vascular tissues of host.

Example:

- Cuscuta, also called dodder

Survival of Host and Parasite:

Host can survive without parasite, but parasite cannot survive without host.

2. Mutualism:

Definition:

The type of symbiotic interaction in which both partners of different species get benefit and neither is harmed is called mutualism.

Examples:

Termites and Zooflagelltes:

Termites eat wood but are not able to digest it. A protozoan lives in its intestine. It secretes 'cellulase' enzyme to digest the cellulose of wood. In return, the termite provides food and shelter to the protozoan.

Root Nodules:

The nitrogen fixer bacteria *Rhizobium* live in the root nodules of leguminous plants like pea, gram etc. The bacteria obtain food and shelter from plants while in return they fix gaseous nitrogen into nitrates for the plant which is required for their growth.

3. Commensalism:

Definition:

It is a type of symbiosis in which one partner is benefited while the other is neither benefited nor harmed.

Examples:

Epiphytes:

Epiphytes are small plants found growing on other larger plants for space only. They absorb water and minerals from atmosphere and prepare their own food. The larger plants are neither benefited nor harmed in any way.

Sucker Fish:

Sucker fish attaches to the surface of sharks by its sucker. In this way, the shark provides easy transport to the sucker fish to new feeding grounds.

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Q. No. 11 Write a note on global warming.

GLOBAL WARMING

Definition:

The increase in the temperature of the earth due to the accumulation of green house gases is called global warming.

Explanation:

The addition of greenhouse gases in atmosphere increases the temperature of the Earth. These green houses gases are:

- Carbon dioxide
- Methane
- Ozone

Reflecting Back of Solar Radiations:

These gases remain in the lowest part of Earth's atmosphere and do not allow solar radiations to reflect back into space. As a result, heat remains within the Earth's atmosphere and increases its temperature.

Effects:

Due to global warming:

- Polar ice-caps and glaciers are melting faster than the time taken for new ice layers to form.
- Sea water is also expanding causing sea levels to rise.
- Due to melting glaciers, rivers overflow and cause floods.

Q. No. 12 Write a note on green house effect.

GREEN HOUSE EFFECT

The term 'Greenhouse Effect' refers to the phenomenon in which certain gases called greenhouse gases trap heat in the atmosphere.

Escape of Heat:

These gases act like the glass in a greenhouse, which does not allow the inner heat to escape. When sunlight reaches the surface of the Earth, much of its energy is transformed into heat energy. The Earth surface reflects this heat energy towards space as infrared radiation. The greenhouse gases trap infrared radiation and send it back to Earth.

Green House Gases:

Some important green house gases are:

- Carbon dioxide
- Methane
- Nitrous oxide

Increase in Concentration:

Since 1800, there is remarkable increase in the amount of the following gases in atmosphere:

- Carbon dioxide 30 %
- Methane Doubled
- Nitrous oxide 8%

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Q. No. 13 Write a note on acid rain.

ACID RAIN

Definition:

The rain consisting of sulphuric and nitric acids is called acid rain.

Formation of Acids:

When rain falls through polluted air, it comes across chemicals such as oxides of sulphur and nitrogen. These chemicals interact with water vapours in the presence of sunlight to form sulphuric acid and nitric acid. These acids remain as vapours at high temperatures.

pH Level:

As temperature falls, the acids begin to condense into liquid form and mix with rain or snow, on the way down to the Earth. This makes rain acidic with pH range of 3 to 6.

SIGNIFICANT ILL EFFECTS OF ACID RAIN

Some of the significant ill effects of acid rain are:

Destruction of Nutrients:

Acid rain destroys the necessary nutrients present in the waters of rivers and lakes. Acid rain washes nutrients out of soil.

Lowering pH Level:

It also lowers the pH of water. Most of the aquatic animals cannot survive at this pH.

Damage to Trees:

It damages the bark and leaves of trees and harms root hairs. Leaf pigments like chlorophyll are also destroyed.

Corrosion:

Metallic surfaces exposed to acid rain are easily corroded.

Loss of Strength:

The following products lose their material strength or disintegrate easily due to acid rains:

- Fabrics
- Paper
- Leather

Damage to Building Materials:

Building materials are weakened with acid rains because of the formation of soluble compounds such as:

- Limestone
- Marble
- Dolomite
- Mortar
- Slate

Damage to Historical Monuments:

Acid rain is dangerous for historical monuments. The building of famous Taj Mahal has been corroded at many places, due to acid rains.

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Q. No. 14 Write a note on deforestation.

DEFORESTATION

Definition:

The clearing of forests by natural causes or humans is called deforestation.

Purposes:

Large areas of forests have been cleaned for:

- Agriculture
- Factories
- Roads
- Rail tracks
- Mining

Wood:

Humans cut trees for getting wood (lumber), which is then used for making structures and for heat production.

Animals:

Humans prey upon forest animals which are the predators of many insect pests. In this way, insect pests destroy forests by eating the shoots and spreading diseases.

Effects:

The effects of deforestation include:

- Floods
- Droughts
- Landslides
- Soil erosions
- Global warming
- Loss of habitat of many species

Q. No. 15 Write a note on overpopulation.

OVERPOPULATION

Population:

A group of the organisms of the same species inhabiting a specific geographical area (habitat) at a particular time is called a population.

World Population:

Past:

When the industrial revolution started some 250 years ago, the world population was 600 million.

Present:

Now the world population is almost ten times at 6 billion.

Future:

The population will grow to 8 billion by 2025.

Reasons:

The following things have contributed in population growth:

- Better health facilities
- Lowered mortality rates

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Q. No. 16 Write a note on urbanization.

URBANIZATION

Definition:

The growing of cities is called urbanization.

Reasons:

People move from rural areas to cities in search of:

- Better jobs
- Education opportunities
- Higher standards of living

Problems for Government:

If there is rapid urban growth, the governments find it difficult to provide even the basic facilities like:

- Health
- Education
- Shelter
- Water
- Electricity

Urban Poverty:

Most of the migrants in cities do not find good jobs and become the part of urban poor.

Overcrowding:

There is overcrowding in schools, hospitals etc. The slum areas increase in number and people living there are at greater risk of diseases.

Global Problem:

Urbanization is a global problem and cannot be stopped but it can be managed.

Urbanization in Pakistan:

The current level of urbanization in Pakistan is about 32% which is not high by global standards.

Planned Urbanization:

A planned urbanization can solve many problems.

- The cities should have thick green belts in their surroundings to control pollution.
- The open spaces in cities should be reserved through zoning and land plans.
- The urban spread-out should also be controlled.
- Utilization of public transport instead of individual transports also proves effective way to manage urbanization.

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Q. No. 17 Write a note on air pollution.

Definition:

Any undesirable change in the physical, chemical or biological characteristics of air, water and land that may harmfully affect living organisms and natural resources is called pollution.

AIR POLLUTION

Major Issue:

Air pollution is one of the major environmental issues of today.

Definition:

The change of composition of air by the addition of harmful substances is called air pollution.

Harmful Substances:

The harmful substances causing air pollution may be:

- Industrial gases
- Automobile gases
- Particulate matter

Sources:

All sources of air pollution are related to human activities.

Combustion:

Burning of coal produces a lot of smoke and dust whereas burning of petroleum produces sulphur dioxide.

Air Pollutants:

In addition to these, air pollutants include:

- Carbon monoxide
- Carbon dioxide
- Nitrogen oxides
- Hydrocarbons
- Particulate matter
- Traces of metals

Industrial Air Pollution:

Different industries produce air pollution in the following way:

Fertilizer Industries:

Fertilizer industries release:

- Oxides of sulphur and nitrogen
- Hydrocarbons
- Particulate matter
- Fluorine

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Thermal Industries:

Thermal industries are coal based and their pollutants are:

- Fly ash
- Soot
- Sulphur dioxide

Textile Industries:

Textile industries release:

- Cotton dust
- Nitrogen oxides
- Chlorine
- Smoke
- Sulphur dioxide

Steel Industries:

Steel industries release:

- Carbon monoxide
- Carbon dioxide
- Sulphur dioxide
- Phenol
- Fluorine
- Cyanide
- Particulate matter

EFFECTS OF AIR POLLUTION

Global Warming:

Global warming is one of the consequences of air pollution.

Smog Formation:

When pollutants like hydrocarbons and nitrogen oxides combine in the presence of sunlight, smog is formed. This is a mixture of gases. It forms a yellowish brown haze especially during winter.

Effects:

The bad effects of smog are:

- It hampers visibility during winter.
- It causes many respiratory disorders.
- It causes allergies as it contains polluting gases.

Acid Rains:

The air pollutants like sulphur dioxide and nitrogen oxides react with water in the atmosphere producing acid rains.

Ozone depletion:

The upper layer (stratosphere) of the atmosphere has ozone (O_3) which absorbs ultraviolet (UV) rays present in the sun's radiation. However, the air pollutants like chlorofluorocarbons (CFCs) destroy the ozone molecules and so break the ozone layer. Ozone holes are created which permit UV rays to reach the Earth's surface. The UV rays increase the temperature and also cause skin cancers.

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CONTROL OF AIR POLLUTION

For effective control of air pollution, it is important to create public awareness about the ill-effects of air pollution. Air pollution can be controlled by the following ways:

Afforestation:

The establishment of new forests by planting on non-forest areas is called afforestation. Forests are effective means to control air pollution because plants can filter and absorb air pollutants.

Modification of Industrial Effluents:

The air pollutants coming from industries should be passed through filters and other devices, so that the particulate matter is removed before the waste gases are released out.

The smoke producing units should have long chimneys to take the polluting gases far above and then disperse over a larger area.

Industries should also invest for solar cookers or for producing bio gas.

Environment Friendly Fuels:

Lead-free fuels should be used in automobiles. Similarly, sulphur-free fuel should be used in coal-based industry to reduce pollution by sulphur dioxide.

Q. No. 18 Write a note on water pollution.

WATER POLLUTION

Definition:

The change in the composition of water by the addition of harmful substances is called water pollution.

SOURCES OF WATER POLLUTION

Following are the important sources of water pollution:

Sewage:

Sewage is one of the major pollutants of water. It contains organic matter and the excreta of human and other animals. Organic matter encourages the growth of micro-organisms which spread diseases.

Industrial Wastes:

The wastes of industries (acids, alkalis, dyes and other chemicals) are disposed in nearby water bodies. These wastes change the pH of water and are harmful or even fatal to aquatic organisms.

Hot Water:

Certain industries release a lot of hot water from their cooling plants. It results in heating up of water bodies and kills aquatic life.

Fertilizers and Pesticides:

Fertilizers and pesticides enter into water bodies with the rain water flow and the ground water by seepage. These chemicals remain in water for a long time and can enter food chains. They cause a number of diseases in animals.

Oil Leakage:

Oil tankers and offshore petroleum refineries cause oil leakage into water. Oil floats on the water surface and prevents atmospheric oxygen from mixing in water. So, aquatic animals begin to die due to oxygen shortage.

Heavy Metals:

Some heavy metals e.g. lead, mercury, arsenic and cadmium also make the water polluted. Such metals can be present in the water, released from industrial and urban areas.

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EFFECTS OF WATER POLLUTION

Water pollution severely affects the health of people.

The following are major effects of water pollution.

Eutrophication:

The enrichment of water with inorganic nutrients like nitrates and phosphates is called eutrophication.

Algal Blooms:

The sewage and fertilizers contain large amount of inorganic material (nutrients). When sewage and fertilizers reach water bodies, the nutrients present in them promote algal blooms (excessive growth) there. Rich algal growth leads to increase in the number of the decomposers. Decomposers use the oxygen present in water and it results in the depletion of oxygen. Algal bloom also reduces the light reaching the lower layers in water.

Food Chain Contamination:

The non-biodegradable water pollutants may stay in water for long times. From water, they enter into small organisms, which are fed upon by fish. The fish in turn are fed upon by land animals including human.

Epidemics:

Organic pollutants in water facilitate the growth of germs. Such polluted water causes epidemics like cholera, gastroenteritis etc.

Effect of Heavy Metals:

If water with such heavy metals is given to plants, the metals enter the vegetables that grow on these plants. Such contaminated vegetables are harmful for human health. Following are the adverse effects of heavy metals:

- Reduce growth and development
- Cause cancer
- Damage to nervous system

Effect of Mercury and Lead:

Mercury and lead can cause:

- Joint diseases such as rheumatoid arthritis
- Diseases of kidneys
- Diseases of circulatory system and nervous system

CONTROL OF WATER POLLUTION

Public Awareness:

Public should be made aware of the dangers of water pollution.

Sewage Treatment:

Before releasing the sewage into water bodies, it must be purified through sewage treatment techniques.

Industrial Waste Treatment:

Industrial wastes should also be treated before they are released into water bodies.

Q. No. 19 Write a note on land pollution.

LAND POLLUTION

Definition:

The change in the composition of soil by the addition of harmful substances is called land pollution.

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Importance of Land:

Land (soil) is an important resource as it is the basis for the growth of producers. In the recent times, soil has been subjected to pollution.

SOURCES OF LAND POLLUTION

Following are the main sources of land pollution:

Pesticides:

The pesticides used in agriculture have chemicals that stay in soil for long times.

Acid Rains:

The acid rains change the pH of soil making it unsuitable for cultivation.

Garbage:

The household and other city garbage lies scattered in soil in the absence of a proper disposal system.

Polythene Materials:

Materials like polythene block the passage of water into soil and so decrease the water-holding capacity of soil.

Industrial Wastes:

Many industries produce harmful chemicals which are disposed of without being treated.

Nuclear Wastes:

Improper disposal of nuclear wastes also causes radioactive substances to remain in soil for a long time.

Improper Sewerage System:

Open latrines in villages and some parts of cities are also the source of land pollution.

CONTROL OF LAND POLLUTION

Disposal of Wastes:

There should be suitable and safe disposal of wastes including nuclear wastes.

Recycling:

Non-biodegradable materials like plastic, glass, metals etc. should be recovered and recycled.

Less use of Inorganic Pesticides:

Inorganic pesticides should be replaced by organic pesticides.

Q. No. 20 Write a short note on Kasur tanneries.

KASUR TANNERIES

Tannery:

The industry where raw skin is treated to make leather is called tannery.

Number:

There are more than 200 tanneries operating in Kasur city.

Quantity of Waste:

The industry discharges 9000 cubic metres of waste water daily into the nearby water bodies. This water contains heavy metals and becomes a part of the underground water.

Adverse Effects:

In 2003, a survey showed that two-thirds of residents and 72 percent of tannery workers suffered from:

- Cancer
- Infections of the kidney
- Loss of eye sight

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Cause:

Tests showed that the drinking water was contaminated with:

- Lead
- Mercury
- Chromium

Step Taken:

The Pakistan government and the United Nations Development Programme (UNDP) launched the Kasur Tannery Pollution Control Project. The project has established an effluent treatment plant, chromium plant and a solid waste disposal site.

Q. No. 21 Write a note on conservation of nature and explain 3R Principle.

CONSERVATION OF NATURE

Meaning:

Conservation of nature means the conservation of natural resources.

Need of Conservation:

Everything that we use or consume e.g. food, petrol etc. is obtained from natural resources. The renewable natural resources e.g. air are reproduced easily but the' non-renewable resources (e.g. minerals and fossil fuels) are not replenished once they get depleted. We have to conserve the non-renewable resources because their reserves are limited' and humans are heavily dependent on them for daily needs. The renewable resources too have to be judiciously used.

3R PRINCIPLE

To ensure sustainable use of resources in our environment, we should act upon the principle of the 3R' i.e. Reduce, Reuse, and Recycle.'

The R1: Reduce:

- We should use the natural resources less and should not waste them.
- We should use this principle at different places, in our daily lives.
- We should not waste water, electricity, fuel etc.
- We should turn off the tap when not in use.
- We should bathe with a bucket instead of shower.
- The lights and fans should be off, when we are not in room.
- We should take public transport (like buses) or walk short distances instead of using motor fuel.
- We should not waste food and should give unused food to poor people.

The R2: Reuse:

We should use things again and again. We should not throw away materials such as glass containers, plastic bags, paper, cloth etc. These should be reused at domestic levels rather than being thrown. It also reduces solid waste pollution.

The R3: Recycle:

Materials such as paper, plastic, glass etc. can be recycled. This decreases the volume of refuse and helps in the conservation of natural resources. A recycling of one tonne of paper can save 17 trees.

Addition of R4: Reforest:

We can add the R4 i.e. Reforest. Trees should be planted during the rains. Trees make our environment more cool, shady and green.

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Q. No. 22 Describe the plans for the conservation of nature.

PLANS FOR THE CONSERVATION OF NATURE

The following are the projects and plans of our government for the conservation of resources.

National Conservation Strategy:

In 1992, Pakistan developed the National Conservation Strategy. The main objectives of the strategy are conservation of natural resources and improved efficiency in the use of resources. It also covers the policies for promoting efficiency and conservation of energy resources.

National Drinking Water and Sanitation Policy:

The Federal Ministry of Environment has launched the National Drinking Water and Sanitation Policy. It focuses on the provision of clean drinking water to entire population and the conservation of water resources. Water purification plants are being installed all over the country.

Mass Awareness for Water Conservation and Management:

In 2006, the UNDP launched the project "Mass Awareness for Water Conservation and Management". The objective of the project was to launch a comprehensive awareness campaign for the conservation and management of water resources in Pakistan.

Role of SCOPE:

The organization SCOPE (Society for Conservation and Protection of Environment) works with government for mass awareness and research for the conservation of natural resources in Pakistan.

Role of World Wide Fund for Nature:

The WWF (old name is World Wildlife Fund but now it is called World Wide Fund for Nature) is working on many projects related to the conservation of nature.

The following are some important programmes of WWF-Pakistan in collaboration with the government of Pakistan:

- Improving sub-watershed management and environmental awareness around Ayubia National Park
- Plantation of the trees of Jatropha and Mangroves at District Thatta, Sindh
- District-wise forest cover assessment of Pakistan
- Saving Wetlands Sky High Programme (for the conservation and management high altitude wetlands)
- Indus Basin Water Security Project (to protect the water-flow needed for the maintenance of river ecosystem and for the benefit of nearby areas)
- Regional Climate Risk Reduction in Himalayas

Q No. 23 Write a note on Dengue Fever.

DENGUE FEVER

Causative Organism:

Dengue fever is a viral infection transmitted through a mosquito *Aedes aegypti*.

Major Health Problem:

It has become a major health problem in tropical and sub-tropical countries, including Pakistan.

Viral Attack and Immunity:

There are four types of dengue virus. Recovery from infection by one provides lifelong immunity against that virus but provides no protection against infection by the other three viruses.

Incidence:

According to the World Health Organization, there are 50 million dengue infections worldwide every year. Now, there are 2.5 billion people at risk from dengue.

Spread of Disease:

The female *Aedes* mosquito gets the virus when it bites an infected person. When an infected mosquito bites another person, viruses enter his/her blood and attack white blood cells. Inside WBCs, viruses reproduce and destroy them.

Symptoms:

In severe cases, the virus affects liver and bone marrow. As a result there is a decrease in the production of blood platelets and patient suffers from bleeding.

Other symptoms of dengue include:

- High fever
- Severe headache
- Pain behind the eyes
- Muscle and joint pains
- Rash

Complications:

Sometimes, dengue fever converts into:

Dengue Haemorrhagic Fever (DHF):

DHF results in:

- Bleeding
- Low levels of blood platelets
- Blood plasma leakage

Dengue Shock Syndrome (DSS):

In DSS the blood pressure falls dangerously low.

Prevention:

There is no vaccine for dengue fever.

Treatment:

There is no treatment of dengue fever.

Control:

At present, the only method of controlling dengue virus transmission is to check the spread of *Aedes* mosquitoes.

Breeding of Aedes aegypti:

Aedes aegypti breeds primarily in the:

- Containers used for water storage
- Discarded plastic containers
- Used automobile tyres
- Items that collect rain water

Control over Mosquitoes:

The mosquitoes can be controlled through:

- Proper solid waste disposal
- Improved water storage practices

Use of Predators:

Small fish and crustaceans have also been used for killing the larvae of the mosquito.

Use of Insecticides:

Insecticide sprays have not proved efficient in killing the mosquitoes, because spray does not penetrate all habitats of adult mosquitoes.

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REVIEW QUESTION:

MULTIPLE CHOICE QUESTIONS

1. **Which of the following is the abiotic component of the ecosystem?**
   (a) Producers (b) Herbivores
   (c) Carnivores (d) Oxygen
2. **When we eat onions, our trophic level is;**
   (a) Primary consumer (b) Secondary consumer
   (c) Decomposer (d) Producer
3. **Identify the correctly matched pair:**
   (a) Rainfall - biotic factors in ecosystem
   (b)Global warming - formation of fossil fuels
   (c) Renewable natural resource - air
   (d) Corn - secondary consumer
4. **In the food chain tree → caterpillar → robin → hawk → coyote, which is the secondary consumer?**
   (a) Caterpillar (b) Robin
   (c) Hawk (d) Coyote
5. **In ecosystems, the flow of ------ is one way, while ------ is/are constantly recycled.**
   (a) Minerals, energy (b) Energy, minerals
   (c) Oxygen, energy (d) Glucose, water
6. **In the food chain "grass → rabbit → fox → bear → mushroom", how many types of decomposers are present?**
   (a) 1 (b) 2
   (c) 3 (d) 4
7. **Organisms in the ecosystem that are responsible for the recycling of plant and animal wastes are:**
   (a) Producers (d) Consumers
   (c) Decomposers (d) Competitors
8. **Which form of Nitrogen is taken by the producers of the ecosystem?**
   (a) Nitrogen gas (b) Ammonia
   (c) Nitrites (d) Nitrates

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	a	3	c	4	b
5	b	6	a	7	c	8	d

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SHORT QUESTIONS

1. **What are the different levels of ecological organization?**
   Consult Long Question No. 1
2. **Define ecosystem and its components.**
   Consult Long Question No. 2
3. **How the flow of energy is different from that of materials?**
   Consult Long Question No. 3
4. **Define food chain and food web.**
   Consult Long Question No. 4
5. **What do you mean by the concept of 3Rs with reference to the conservation of natural resources?**
   Consult Long Question No. 21

UNDERSTANDING THE CONCEPT

1. **Explain what do you mean by the pyramids of number and biomass?**
   Consult Long Question No. 5
2. **Write a note on Carbon cycle.**
   Consult Long Question No. 6
3. **W are the different stages of Nitrogen cycle?**
   Consult Long Question No. 7
4. **Write notes on competition, predation and symbiosis.**
   Consult Long Question No. 8, 9 and 10
5. **Explain how human activities have contributed to the loss of balance in nature.**
   Consult Long Question No. 14 to 19
6. **Write note on the causes and effects of the air and water pollutions.**
   Consult Long Question No. 17 and 18

THE TERMS TO KNOW

Abiotic:
The non-living components of the environment like water, sunlight, soil, heat etc.

Acid Rain:
The rain containing sulphuric acid and nitric acid; with pH range of 3 to 6

Ammonification:
The decomposition of protein of dead plants and animals, and nitrogenous wastes to ammonia by ammonifying bacteria.

Atmospheric nitrogen fixation:
The conversion of atmospheric gaseous nitrogen to nitrates by thunderstorms and lighting.

Biogeochemical Cycle:
The cyclic pathway through which chemical elements move from environment to organisms and back to the environment.

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Biosphere:
The last level of ecological organization; all the ecosystems of the world together form the biosphere.

Biotic:
The living components of the environment; include producers, consumers and decomposers.

Carbon Cycle:
The biogeochemical cycle in which carbon flows between organisms and the environment.

Carnivore:
The consumers which eat only animal flesh

Commensalism:
A type of symbiosis in which one of the partners gests benefit while the other is neither benefited nor harmed.

Consumer:
The part of the biotic components of the ecosystem that consists of animals.

Decomposer:
An organism which decomposes the dead bodies and dead matter.

Deforestation:
Clearing of forests by natural causes or by humans.

Denitrification:
The conversion of nitrities and nitrates into nitrogen gas.

Ecological pyramid:
A representation of the number of individuals or amount of biomass or energy present in various trophic levels of a food chain.

Environment:
The sum total of physical (abiotic) and biotic conditions which influence the organism.

Eutrophication:
The enrichment of water with inorganic nutrients; the nutrients promote the growth of age and it leads it increase in the number of the decomposers and depletion of oxygen.

Food chain:
The series of organisms in an ecosystem, in which an ecosystem, in which an organism eats the preceding one and is eaten by the next one.

Food Web:
A network of interconnected food chains; has a number of feeding connections amongst different organisms of a community.

Global Warming:
Increase in the temperature of the Earth; due to the addition of greenhouse gases in atmosphere, which do not allow solar radiations to reflect back into the space.

Interspecific interactions:
Interactions between the members of the different species.

Intraspecific interactions:
Interactions between the members of the same species.

Mutualism:
The symbiotic association in which both the partners get benefits and neither is harmed.

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Natural resources:

The resources on Earth, which provide everything the humans use or consume

Nitrification:

The oxidation of ammonia to nitrites and nitrates by the nitrifying bacteria.

Nitrogen Cycle:

The flow of nitrogen between environment and the organisms.

Nitrogen fixation:

Conversion of nitrogen into nitrates.

Non-renewable resource:

A resource that is formed over very long periods; the rate of formation is extremely slow so cannot be replaced; e.g. minerals and fossil fuels.

Overpopulation:

Increase in population beyond the carrying capacity of an area or environment.

Ozone:

The O_3 gas; also present in the upper layer of the atmosphere where it absorbs the ultraviolet rays present in the sun's radiation.

Parasitism:

A type of interspecific interaction in which smaller partner (parasite) derives food and shelter from the body of larger partner (host) and harms the host.

Phytoplankton:

Photosynthetic organisms that float on the surface of water.

Pollutant:

The substance that causes pollution.

Pollution:

Undersirable change in the physical, chemical or biological characteristics of air, water and land that may harmfully affect living organisms and other resources.

Predation:

An interaction between animals of two species or an pant and an animal, in which the predator attacks, kills and feeds on the smaller animal called prey.

Producer:

An organism that produces organic compounds from inorganic compound; an autotroph

Pyramid of biomass:

The graphic representation of biomass present per unit area at different torophic levels in an ecosystem.

Pyramid of numbers:

The graphic representation of the number of individuals per unit area at various trophic levels in an ecosystem.

Renewable resources:

The resources which are repleni9shed or reproduced easily e.g. sunlight, air, wind etc.

Symbiosis:

Long or short term relationship between members of different species; three forms are parasitism, commensalism and mutualism.

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CHAPTER 17

BIOTECHNOLOGY

MULTIPLE CHOICE QUESTIONS

(1) **The use of living organisms in process for the manufacture of useful products:**
(a) Parasitology (b) Biotechnology (c) Pharmacology (d) Drug abuse

(2) **When the work on genetic engineering started?**
(a) 1941 (b) 1942 (c) 1943 (d) 1944

(3) **When was human genome project launched?**
(a) 1990 (b) 1991 (c) 1992 (d) 1993

(4) **What was the name of sheep produced by Ian Wilmut from the body cell of an adult sheep?**
(a) Dolly (b) Nancy (c) Ethal (d) Dhani

(5) **Ian Wilmut belonged to**
(a) Newzealand (b) Australia (c) Scotland (d) America

(6) **An antiviral protein:**
(a) Insulin (b) Thymosin (c) HGH (d) Interferon

(7) **Insulin is a:**
(a) Hormone (b) Antiseptic (c) Antifungal (d) Sedative

(8) **How much yeast is required to produce 250 tons of protein in one day?**
(a) 20 kg (b) 30 kg (c) 40 kg (d) 50 kg

(9) **Who convinced the scientific community that all fermentations are the result of microbial activity?**
(a) Alexander Fleming (b) Joseph Lister (c) Louis Pasteur (d) Edward Jenner

(10) **In glycolysis, the glucose molecule is broken down into two molecules of:**
(a) Acetic acid (b) Lactic acid (c) Pyruvic acid (d) Formic acid

(11) **When milk protein is coagulated, what is formed?**
(a) Yogurt (b) Cheese (c) Alcohol (d) Wine

(12) **The enzymes that are used to cut the identified gene from the DNA of donor organism:**
(a) Restriction endonucleases (b) Ligases
(c) Lipases (d) Amylases

(13) **Diabetes is cured by:**
(a) Human Growth Hormone (b) Insulin
(c) Glucagon (d) Parathormone

(14) **A pain killer produced by brain:**
(a) Thymosin (b) Beta-endorphin
(c) Insulin (d) Human Growth Hormone

(15) **How many sheep brains are required to produce 1 mg human growth hormone:**
(a) 10,000 (b) 1,000 (c) 100,000 (d) 1,000,000

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	b	2	d	3	a	4	a	5	c
6	d	7	a	8	d	9	c	10	c
11	b	12	a	13	b	14	b	15	c

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SHORT QUESTIONS

Q. No. 1 Till how long the humans are using biotechnology?

USE OF BIOTECHNOLOGY

Humans have been making use of biotechnology since they discovered farming. This use extended from the planting of seeds to the control of plant growth and crop production.

Examples:

- Animal breeding is also a form of biotechnology.
- Cross-pollination of plants and cross-breeding of animals were major techniques in biotechnology.

Q. No. 2 Why Cross-pollination and cross-breeding techniques were used?

USAGE OF TECHNIQUES

Cross-pollination of plants and cross-breeding of animals were major techniques in biotechnology used to:

- Enhance product quality
- Meet specific requirements

Q. No. 3 What is old biotechnology?

OLD BIOTECHNOLOGY

Although the term biotechnology is new, the discipline itself is very old. Fermentation and other such processes, which are based on the natural capabilities of organisms, are commonly considered as old biotechnology.

Q. No. 4 Define biotechnology.

BIOTECHNOLOGY

The use of living organisms in processes for the manufacture of useful products or for services is called biotechnology.

Q. No. 5 When human being started use of microorganisms?

USE OF MICROORGANISMS

Human began using microorganisms as early as 4000 BC for making:

- Wine
- Vinegar
- Cheese
- Yogurt

Some of these processes have become a part of every home that we may even hesitate to refer them as biotechnology.

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Q. No. 6 What do you know about Dolly Sheep?

DOLLY SHEEP

In Scotland, in 1997, an embryologist Ian Wilmut produced a sheep (Dolly) from the body cell of an adult sheep.

Q. No. 7 What is the difference between cellular respiration and fermentation?

DIFFERENCE BETWEEN CELLULAR RESPIRATION AND FERMENTATION

Cellular Respiration	Fermentation
In cellular respiration, glucose molecule goes through oxidation-reduction reactions to release energy in the form of ATP.	The process in which there is incomplete oxidation-reduction of glucose is called fermentation.

Q. No. 8 What are the fears related to biotechnology?

FEARS RELATED TO BIOTECHNOLOGY

Fears are also being expressed about the advances in biotechnology in terms of release of harmful organisms developed through recombinant DNA technology.

Q. No. 9 What is the role of Louis Pasteur in fermentation?

ROLE OF LOUIS PASTEUR IN FERMENTATION

In 1857, Pasteur convinced the scientific community that all fermentations are the results of microbial activity. He showed that fermentation is always accompanied by the development of microorganisms. There are many kinds of fermentation and each kind is a characteristic of particular microbial group.

Q. No. 10 How are the fermentations classified?

CLASSIFICATION OF FERMENTATIONS

Fermentations are classified in terms of the products formed.

Q. No. 11 What are the steps of carbohydrate fermentation?

STEPS OF CARBOHYDRATE FERMENTATION

The initial steps of carbohydrate fermentation are identical to those of respiration. The process begins with glycolysis, in which the glucose, molecule is broken into two molecules of pyruvic acid. Different microorganisms proceed the further reactions in different ways. It results in the formation of various products from pyruvic acid.

Q. No. 12 Before genetic engineering, how many sheep brains were required to produce 5 mg of human growth hormone?

REQUIREMENT OF SHEEP BRAINS

Before genetic engineering, 500,000 sheep brains were required to produce 5 mg of human growth hormone.

Q. No. 13 Why single cell proteins are called so?

NAMING OF SINGLE CELL PROTEIN

The single cell proteins are called so because the microorganisms used as producers are unicellular or filamentous individuals.

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LONG QUESTIONS

Q. No. 1 Write a note on genetic engineering.

GENETIC ENGINEERING

Introduction:

It deals with the following aspects of genetic material (DNA):

- Artificial synthesis
- Modification
- Removal
- Addition
- Repair

Modern Biotechnology:

It is considered as modern biotechnology.

Need:

It is done to alter the characteristics of organisms.

Work Done on DNA:

The work on genetic engineering started in 1944 when it was proved that DNA carries the genetic information. Scientists isolated the enzymes of DNA synthesis and then prepared DNA outside cells. In 1970s, they were able to cut and paste the DNA of organisms.

Preparation of Insulin:

In 1978, scientists prepared human insulin by inserting the insulin gene in bacteria.

Preparation of Human Growth Hormone:

Human growth hormone was also synthesized in bacteria.

Human Genome Project:

In 1990, the Human Genome Project was launched to map all the genes in human cell. The complete map of human genome was published in 2002. '

Q. No. 2 Explain the scope and importance of biotechnology.

BIOTECHNOLOGY

The use of living organisms in processes for the manufacture of useful products or for services is called biotechnology.

SCOPE AND IMPORTANCE OF BIOTECHNOLOGY

In recent years, biotechnology is growing as a separate science. It has attracted the attention of many intellectuals from diverse fields like:

- Agriculture
- Medicine
- Microbiology
- Organic chemistry

The scope for biotechnology is so wide that it is difficult to recognize the limits. The following are some areas of the application of biotechnology:

Biotechnology in the Field of Medicine:

In the field of medicine, biotechnologists have synthesized:

- Insulin
- Interferon, an antiviral protein
- Vaccines
- Antibodies
- Human Growth Hormone

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Enzymes:

Various enzymes are being synthesized for medicinal as well as industrial use.

Gene Therapy:

Gene therapy (treatment through genes) has become important in recent years.

Forensic Medicine:

Biotechnology also proved much beneficial in forensic medicine.

Identification of Criminals:

The study of DNA helps in the identification of criminals.

Biotechnology in the Field of Food and Agriculture:

Food Industry:

The following products are being produced by using microorganisms:

- Fermented foods (e.g. pickles, yogurt)
- Malted foods (e.g. powdered milk: a mixture of barley, wheat flour and whole milk)
- Various vitamins
- Dairy products

Beverage Industry:

Wine and beer are produced in beverage industry.

Agriculture:

Biotechnology has also revolutionized research activities in the area of agriculture.

Transgenic Organisms:

The organisms with modified genetic set-up are called transgenic organisms.

Transgenic Plants:

Transgenic Plants: plants are being developed in which desirable characteristics are present:

- More yields
- Resistance against diseases
- Resistant against insects
- Resistant against herbicides

Transgenic Animals:

- Transgenic goats, chickens, cows give more food and milk.
- Many animals like mice, goats, cows etc. have been made transgenic to get medicines through their milk, blood or urine.

Biotechnology and Environment:

Biotechnology is also being used for dealing with environmental issues like:

- Pollution control
- Development of renewable sources for energy
- Restoration of degraded lands
- Biodiversity conservation

Treatment of Sewage Water:

Bacterial enzymes are used to treat sewage water to purify.

Use of Microbes:

Microbes are being developed to be used as biopesticides, biofertilizers, biosensors etc.

Other Purposes:

Such transgenic microorganisms are also used for:

- Recovery of metals
- Cleaning of spilled oils

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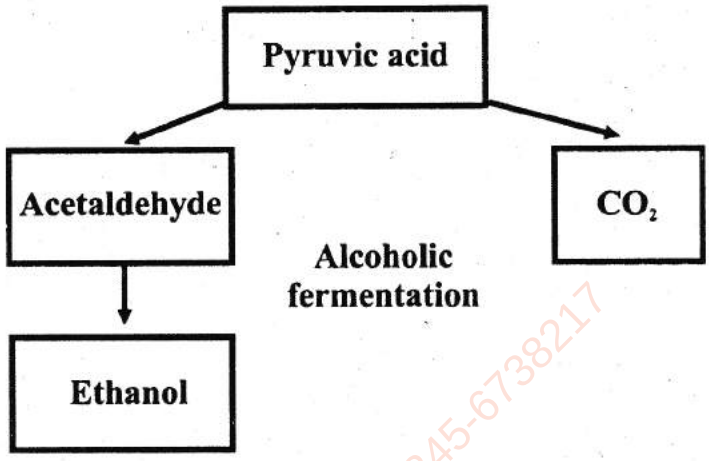
Usage:

This process is quite important and is used to produce:

- Bread
- Beer
- Wine
- Distilled spirits



2. Lactic Acid Fermentation (by bacteria):

Mechanism:

In this process, pyruvic acid is reduced to lactic acid.

Organisms Involved:

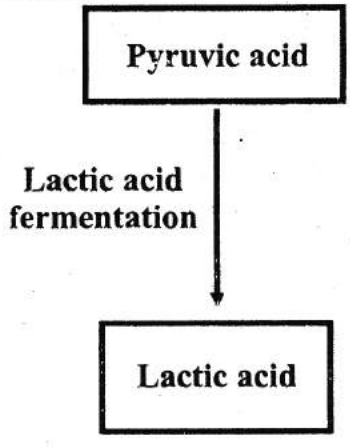
It is carried out by many bacterial species like:

- *Streptococcus*
- *Lactobacillus*

Usage:

It is quite important in dairy industry where it is used for:

- Souring milk
- Production of various types of cheese



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Q. No. 4 Write a note on applications of fermentation.

APPLICATIONS OF FERMENTATION

In fermentation, maximum growth of an organism is obtained for the production of desired products of commercial value. Traditionally, only food and beverage products were produced by using fermentation. Now many other products e.g. industrial chemicals are also being produced.

This can be categorized into two groups:

1. Fermented Foods
2. Industrial Products

1. Fermented Foods:

Fermentation often makes the food more nutritious, more digestible and tastier. It also tends to preserve the food, lowering the need for refrigeration. The following groups are included in the fermented foods.

(i) Cereal Products:

Bread is the commonest type of fermented cereal product. Wheat dough is fermented by *S. cerevisiae* along with some lactic acid bacteria.

(ii) Dairy Products:

Cheese and yogurt are important fermentation products.

Cheese Formation:

Cheese is formed when a milk protein is coagulated. This happens when the acid produced by lactic acid bacteria reacts with milk protein.

Yogurt Formation:

Yogurt is made from milk by different lactic acid bacteria.

(iii) Fruit and Vegetable Products:

Fermentation is usually used, along with salt and acid, to preserve:

- Pickle
- Fruits
- Vegetables

(iv) Beverage Products:

Beer is produced from cereal grains which have been malted, dried and ground into find powder. Fermentation of the powder is done by yeast. This process breaks the glucose present in powder into pyruvic acid and then into ethanol. Grapes can be directly fermented by yeasts to wine.

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Q. No. 3 Define fermentation. What is carbohydrate fermentation? Explain its types.

FERMENTATION

Definition:

The process in which there is incomplete oxidation-reduction of glucose is called fermentation.

History:

Fermentation has been in the knowledge of man since centuries, but it was believed that it is purely a chemical process.

Carbohydrate Fermentation:

The initial steps of carbohydrate fermentation are identical to those of respiration. The process begins with glycolysis, in which the glucose, molecule is broken into two molecules of pyruvic acid. Different microorganisms proceed the further reactions in different ways. It results in the formation of various products from pyruvic acid.

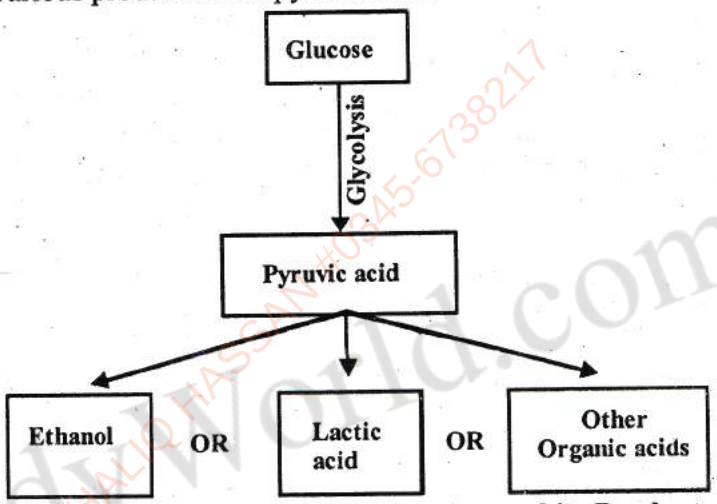

Figure: Carbohydrate Fermentation and its Products

TYPES OF CARBOHYDRATE FERMENTATION

The two basic types of carbohydrate fermentation are as follow:

1. Alcoholic Fermentation (by yeast)
2. Lactic Acid Fermentation (by bacteria)

1. Alcoholic Fermentation (by yeast):

Organism Involved:

This fermentation is carried out by many types of yeast such as *Saccharomyces cerevisiae.*

Mechanism:

In this process, carbon dioxide is removed from pyruvic acid. The product i.e. acetaldehyde is then reduced to ethanol. The carbon dioxide produced during this fermentation causes the rise of the bread.

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2. Industrial Products:

The following are the important industrial products produced through the process of fermentation.

Products	Microorganisms Used	Some Uses
Formic acid	*Aspergillus*	• Textile dyeing • Leather treatment • Eelectroplating • Rrubber manufacture
Ethanol	*Saccharomyces*	• Used as solvent • Production of vinegar • Production of beverages.
Glycerol	*Saccharomyces*	• Used as solvent • Production of plastics • Production of cosmetics • Production of soaps • Used in printing • Used as sweetener
Acrylic Acid	*Bacillus*	• Used in the production of plastics

Q. No. 5 What is a fermenter? Describe types of fermentation. What are the advantages of using fermenters?

FERMENTER

Fermenter is a device that provides optimum environment to microorganisms to grow into a biomass, so that they can interact with a substrate, forming the product.

Heart of Industry:

In fact, the fermenter constitutes the heart of any industrial fermentation process.

TYPES OF FERMENTATION:

Fermentation is carried out in fermenters, in the following two ways.

1. Batch Fermentation
2. Continuous Fermentation

1. Batch Fermentation:

In this process, the tank of fermenter is filled with the raw materials to be fermented. The temperature and pH for microbial fermentation is properly adjusted, and nutritive supplements are added. All the material is steam sterilized, the pure culture of microorganisms is added to fermenter from a separate vessel. Fermentation proceeds and after the proper time the contents of fermenter are taken out. Fermenter is cleaned and the process is repeated. Thus, fermentation is a discontinuous process divided into batches.

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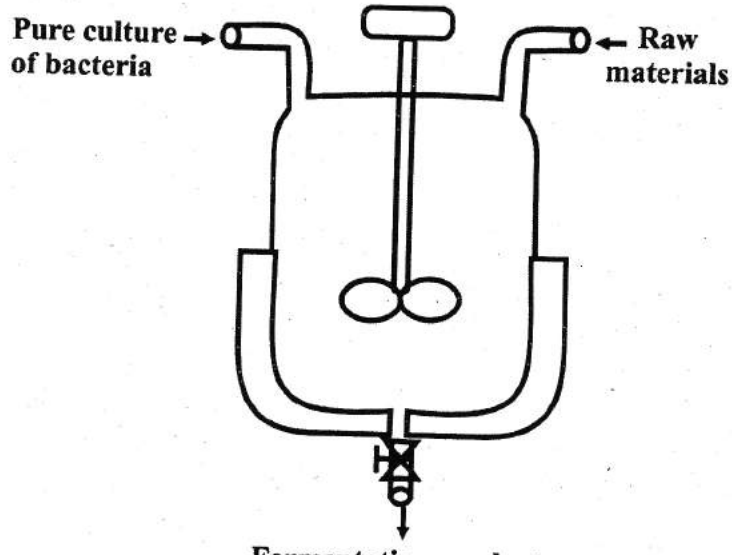

Figure: A Batch Fermenter

2. Continuous Fermentation:

In this process, the substrate is added to fermenter continuously at a fixed rate. This maintains the microorganisms in growth phase. Fermentation products are taken out continuously.

The design and arrangements for continuous fermentation are more complex.

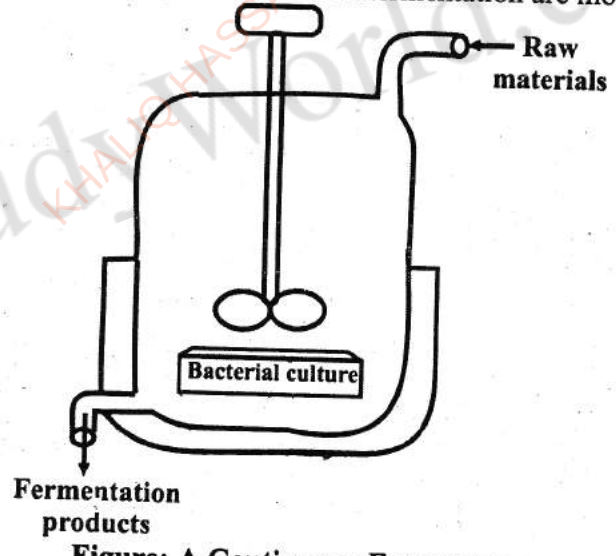

Figure: A Continuous Fermenter

ADVANTAGES OF USING FERMENTERS

Controlled Environment:

For each biotechnological process, the environment provided to the organisms must be monitored and controlled. Such a controlled environment is provided by fermenters.

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Controlling Different Factors:

A fermenter optimizes the growth of the organisms by controlling many factors like:

- Nutrients
- Oxygen
- Growth inhibitors
- pH
- Temperature

Capacity:

A fermenter may hold several thousand liters of the growth medium. So, fermenters allow the production of materials in bulk quantities. Massive amounts of the following products are being produced in fermenters:

- Medicines
- Insulin
- Human growth hormone
- Proteins

Inexpensive:

This production proves much inexpensive.

Q. No. 6 What is genetic engineering? What are its objectives?

GENETIC ENGINEERING

Introduction:

Genetic engineering or recombinant DNA technology involves the artificial synthesis, modification, removal, addition and repair of the genetic material (DNA).

Development:

Genetic engineering developed in the mid-1970s when it became possible to cut DNA and to transfer particular pieces of DNA from one type of organism into another. As a result, the characteristics of the host organism could be changed. If host organism is a microorganism, such as a bacterium, the transferred DNA is multiplied many times as the microorganism multiplies. Consequently, it is possible to obtain millions of copies of a specific DNA inside a bacterial cell.

OBJECTIVES OF GENETIC ENGINEERING

The important objectives of genetic engineering are as follows:

Gene Therapy:

Isolation of a particular gene or part of a gene for various purposes such as gene therapy

RNA and Proteins:

Production of particular RNA and protein molecules

Production Improvement:

Improvement in the production of enzymes, drugs and commercially important organic chemicals

Plant Varieties:

Production of varieties of plants having particular desirable characteristics

Treatment Purposes:

Treatment of genetic defects in higher organisms

Q. No. 7 Describe the process of genetic engineering.

GENETIC ENGINEERING

The following basic steps are involved in genetic engineering:

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Isolation of the Gene of Interest:

In the first step, the genetic engineer identifies the gene of interest in a donor organism. Special enzymes, called restriction endonucleases, are used to cut the identified gene from the total DNA of donor organism.

Insertion of the Gene into a Vector:

A vector is selected for the transfer of the isolated gene of interest to the host cell. The vector may be a plasmid (the extra-chromosomal DNA present in many bacteria) or a bacteriophage. The gene of interest is attached with the vector DNA by using endonuclease (breaking enzymes) and ligase {Joining enzymes).

Formation of Recombinant DNA:

The vector DNA and the attached gene of interest are collectively called recombinant DNA.

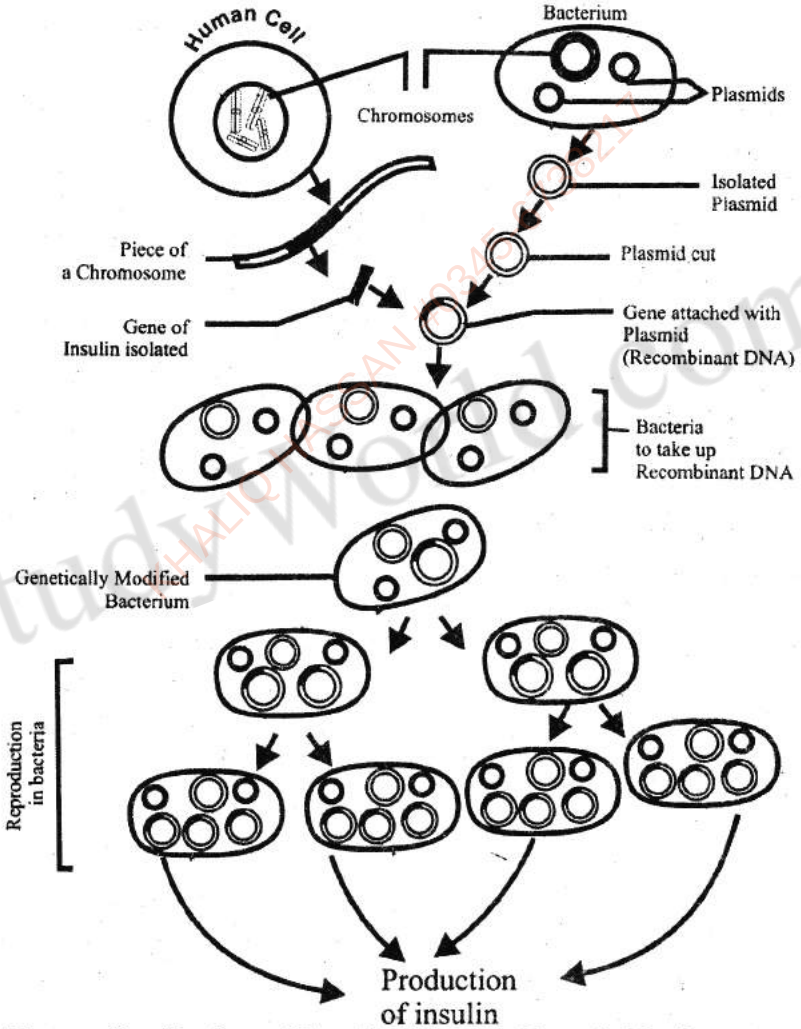

Figure: Production of Insulin through Genetic Engineering

Transfer of Recombinant DNA into Host Organism:

Recombinant DNA is transferred to the target host. In this way, host organism is transformed into a genetically modified organism (GMO).

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Growth of the GMO:

The GMO are provided suitable culture medium for growth to give as much copies of the gene of interest as needed.

Expression of the Gene:

The GMO contains the gene of interest and manufactures the desired product, which is isolated from culture medium.

Q. No. 8 Describe the achievements of genetic engineering.

ACHIEVEMENTS OF GENETIC ENGINEERING

Various achievements of genetic engineering are as follows.

Preparation of Insulin:

Human insulin gene was transferred into bacteria. The genetically modified bacteria became able to synthesize insulin. Diabetics are now receiving this insulin. The steps of genetic engineering for the production of insulin are shown in the following figure:

Preparation of Human Growth Hormone:

In 1977 an *E. coli* bacterium was created that was capable of synthesizing the human growth hormone.

Preparation of Thymosin:

The hormone thymosin which may prove effective against brain and lung cancer has been produced by genetically modified microorganisms.

Preparation of Beta-endorphin:

Beta-endorphin, a pain killer produced by the brain, has also been produced by genetic engineering techniques.

Preparation of Vaccines:

Genetic engineers produced a safe vaccine against the foot and mouth disease (a viral disease in cattle, goats and deer). Similarly many vaccines have been produced against human diseases such as hepatitis B.

Preparation of Interferons:

Interferons are anti-viral proteins produced by cells infected with viruses. In 1980 interferon was produced in the genetically modified microorganisms, for the first time.

Preparation of Urokinase:

The enzyme urokinase, which is used to dissolve blood clots, has been produced by genetically modified microorganisms.

Eliminating Inherited Diseases:

It has become possible to modify the genes in the human egg cell. This can lead to the elimination of inherited diseases like haemophilia.

Cure of Blood Diseases:

Genetic engineering techniques can also be used to cure blood diseases like thalassemia and sickle-cell anaemia, which result from defects in single genes. Normal genes could be transferred into the bone marrow.

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Algae:

Algae grown in ponds produce 20 tons (dry weight) of protein per acre/year. This yield of protein is 10-15 times higher than soybeans and 20-50 times higher than corn.

High Vitamin Contents:

When single-cell proteins are produced by using yeasts, the products also contain high vitamin content.

Usage of Wastes:

In the production of single- cell proteins, industrial wastes are used as raw materials for microorganisms. It helps in controlling pollution.

Availability of all Amino Acids:

The use of single-cell proteins has good prospects in future because they contain all essential amino acids.

Seasonal Variations:

The production of single-cell proteins is independent of seasonal variations.

Limited Land Area:

SCP is gaining popularity day by day because it requires limited land area for production.

MULTIPLE CHOICE QUESTIONS

1. **Find the correct match for the fermentation product and the organism involved.**
   (a) Formic acid – *Saccharomyces*
   (b) Ethanol - *Saccharomyces*
   (c) Ethanol - *Aspergillus*
   (d) Glycerol - *Aspergillus*
2. **Which one is NOT an objective of genetic engineering?**
   (a) Production of cheese and yogurt by lactic acid bacteria
   (b) Isolation of a particular gene or part of a gene
   (c) Production of RNA and protein molecules
   (d) Correction of genetic defects in higher organisms
3. **Which of these is an anti-viral protein?**
   (a) Urokinase (b) Thymosin
   (c) Insulin (d) Interferon
4. **The first step in genetic engineering is;**
   (a) Growth of the genetically modified organism
   (b) Transfer of the Recombinant DNA into the host organism
   (c) Isolation of the gene of interest
   (d) Insertion of a gene into a vector

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	b	2	a	3	d	4	c

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SHORT QUESTIONS

1. **How would you define fermentation with reference to biotechnology?**

Consult Long Question No. 3

2. **Name any two industrial products made by fermentation. Also describe their uses in the industry.**

Consult Long Question No. 4

3. **What are the products of the two types of carbohydrate fermentation?**

Consult Long Question No. 3

4. **Give an example how biotechnology is helping for better environment.**

Consult Long Question No. 2

5. **In biotechnology, what is meant by Genetically Modified Organism (GMO)? How is it made?**

Consult Long Question No. 7

UNDERSTANDING THE CONCEPT

1. **Define biotechnology and describe its importance.**

Consult Long Question No. 2

2. **What is a fermenter? What are the two types of fermentation carried out in fermenters?**

Consult Long Question No. 5

3. **Describe the achievements of genetic engineering in medicine, agriculture and environment.**

Consult Long Question No. 8

4. **What basic steps a genetic engineer adopts during the manipulation of genes?**

Consult Long Question No. 7

5. **What are single cell proteins? Describe their importance.**

Consult Long Question No. 9

THE TERMS TO KNOW

Batch fermentation:

The discontinuous fermentation process, divided into batches

Biotechnology:

The use of living organisms in systems or processes for the manufacture of useful products or for services for mankind

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Nitrogen Fixation:

Genetic engineers have developed plants that can fix nitrogen directly from the atmosphere. Such plants need less fertilizers.

Q. No. 9 Write a note on single cell proteins.

SINGLE-CELL PROTEIN

Introduction:

This technique was introduced by Prof. Scrimshow of Massachusetts Institute of Technology.

Naming:

It is known as single cell protein because the microorganisms used as producers are unicellular or filamentous individuals.

Explanation:

Single-Cell Protein (SCP) refers to the protein content extracted from pure or mixed cultures of algae, yeasts, fungi or bacteria.

Fermenters:

For the production of single-cell proteins, the microorganisms are grown in fermenters.

Substrates:

These microorganisms utilize a variety of substrates like:

- Agricultural wastes
- Industrial wastes
- Natural gas like methane

Growth of Microorganisms:

Microorganisms grow very vigorously and produce a high yield of protein. The protein content produced by microorganisms is also known as novel protein or minifood.

Need:

Due to over-population, the world is facing the problem of food shortage. In future, the conventional agricultural methods might not be able to provide a sufficient supply of food (especially proteins). For a better management of food shortage problems (in humans and domestic animals), the use of microbes as the producers of single-cell proteins has been successful on experimental basis.

Belief of Scientists:

Scientist and food technologists believe that single-cell proteins will substitute the other protein-rich foods in human and animal feeds.

Significance:

All scientists recognize the significance of the production of single-cell proteins. The microorganisms grow very vigorously and produce a high yield.

Yeast:

It has been calculated that 50 kilogram of yeast produces about 250 tons of protein within 24 hours.

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Continuous fermentation:

The fermentation in which substrate is added to the fermenter continuously, at a fixed rate

Fermentation:

The process in which there is an incomplete oxidation-reduction of the organic substrate

Fermenter:

A device that provides optimum environment in which organisms can grow to produce biomass and to form the product

Genetically Modified Organism:

The organism in which DNA (gene) from some other organism has been transferred

Recombinant DNA:

The vector DNA and the attached gene of interest

Restriction endonucleases:

Enzymes used to cut the gene from the total DNA of the organism

Single-Cell Protein:

The protein contents extracted from pure or mixed cultures of algae, yeast, fungi or bacteria; the micro-organisms are grown in fermenters where they produce a high yield of protein

Transgenic:

Organisms with modified genome (genetic make-up)

Vector:

The DNA (plasmid) or bacteriophage that transfers the isolated gene of interest to the host cells

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CHAPTER 18

PHARMACOLOGY

MULTIPLE CHOICE QUESTIONS

(1) The drugs obtained from plants and fungi:

(a) Antibiotics (b) Cardiotonics (c) Certain analgesics (d) All of these

(2) Digitalis is obtained from

(a) *Pencillium* (b) *Notatum* (c) Foxglove (d) *Spirogyra*

(3) Morphine is used to relieve:

(a) Fever (b) Pain (c) Inflammation (d) Swelling

(4) Powder form of which is applied on wounds to stop bleeding and prevent infection?

(a) Sodium nitrate (b) Silver nitrate (c) Potassium sulphate (d) Copper sulphate

(5) Which one is obtained from bacteria?

(a) Penicillin (b) Digitalis (c) Morphine (d) Streptomycin

(6) Drugs are classified on the basis of:

(a) Chemical properties (b) Modes of action

(c) Both a and b (d) None of these

(7) Which one is an antibiotic?

(a) Tetracycline (b) Aspirin (c) Diazepam (d) Morphine

(8) Which medicines inhibit or kill bacteria within or on the body?

(a) Antiseptics (b) Antibiotics (c) Disinfectants (d) Sedatives

(9) Which destroy microorganisms found on non-living objects?

(a) Antiseptics (b) Antibiotics (c) Disinfectants (d) Sedatives

(10) Which antibiotic was discovered by Sir Alexander Fleming?

(a) Morphine (b) Penicillin (c) Cephalosporin (d) Tetracycline

(11) Vaccines have been developed for?

(a) Small pox (b) Whooping cough (c) Both a and b (d) None of these

(12) Date of birth of Alexander Fleming:

(a) 1880 (b) 1881 (c) 1882 (d) 1883

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(13) **Date of death of Joseph Lister:**

(a) 1910 (b) 1911 (c) 1912 (d) 1913

(14) **Which medicines reduce pain?**

(a) Antiseptics (b) Antibiotics (c) Analgesics (d) Sedatives

(15) **Which belong to narcotics?**

(a) Morphine (b) Codeine (c) Heroin (d) All of these

(16) **In which century pharmacology developed?**

(a) 17^{th} (b) 18^{th} (c) 19^{th} (d) 20^{th}

(17) **Until 1890, the subject of pharmacology was known as:**

(a) Materia Medica (b) Biomedical science (c) Both a & b (d) None of these

(18) **Which one is a non-prescription drug?**

(a) Barbiturates (b) Tranquilizers (c) Antibiotics (d) Aspirin

(19) **Antitoxins are obtained from which source?**

(a) Plant (b) Fungi (c) Animal (d) Bacterium

(20) **A liquid that helps prevent infection when applied to cuts:**

(a) Tincture iodine (b) Glycerine (c) Cardiotonic (d) Expectorant

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ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	d	2	c	3	b	4	b	5	d
6	c	7	a	8	b	9	c	10	b
11	c	12	b	13	c	14	c	15	d
16	c	17	a	18	d	19	c	20	a

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SHORT QUESTIONS

Q. No. 1 What is pharmacology?

PHARMACOLOGY

The study of drug composition, properties, medical applications sources of drugs is called pharmacology.

Q. No. 2 What do you know about clinical pharmacology?

CLINICAL PHARMACOLOGY

Clinical pharmacology was present in Middle Ages. Early pharmacologists focused on natural substances, mainly plant extracts. Pharmacology developed in the 19th century as a biomedical science.

Q. No. 3 What is a drug?

DRUG

Any substance that, when absorbed into the body of a living organism, alters normal body functions is known as a drug.

Q. No. 4 What is a pharmaceutical drug?

PHARMACEUTICAL DRUG

Any chemical substance that is used in the diagnosis, cure, treatment or prevention of a disease is called a pharmaceutical or medicinal drug.

Q. No. 5 What is an addictive drug?

ADDICTIVE DRUG

The drugs that make person dependent on them or addicted are called addictive drugs.

Effect:

By using addictive drug, the person's body becomes familiar to it and the user can not function well with out it.

Q. No. 6 What are prescription drugs?

PRESCRIPTION DRUGS

The drugs that are sold only on physician' prescription are called prescription drugs.

Examples:

These include

- Barbiturates
- Tranquilizers
- Antibiotics

Q. No. 7 What are non-prescription drugs?

NON-PRESCRIPTION DRUGS

The drugs that can be sold over the counter with out physicians' prescription because these are considered safe enough are called non-prescription drugs.

Examples:

These include:

- Aspirin
- Some cough medicines

Q. No. 8 What do know about Materia Medica?

MATERIA MEDICA

Until 1890, the subject of pharmacology was known as Materia Medica.

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Q. No. 9 How terramycin was developed?

DEVELOPMENT OF TERRAMYCIN

Researchers of a pharmaceutical company spent two years testing soil from all parts of the world to find new antibiotics. The project resulted in the development of one antibiotic, Terramycin, which is used to treat many infections.

Q. No. 10 What is the contribution of Sir Alexander Fleming?

CONTRIBUTION OF SIR ALEXANDER FLEMING

Period:

1881 – 1955

Place:

He was a Scottish Biologist.

Contribution:

He discovered the antibiotic penicillin from the fungus *Penicilliun notatum*.

Noble Prize:

He was awarded Noble Prize in 1945.

Q. No. 11 What is the contribution of Joseph Lister?

CONTRIBUTION OF JOSEPH LISTER

Period:

1827 – 1912

Place:

He was an English Surgeon.

Contribution:

He promoted the idea of sterile surgery for the first time. He introduced carbolic acid to sterilize surgical instruments and to clean wounds.

Q. No. 12 Why heroin is prescribed?

HEROIN PRESCRIPTION

In many western countries, heroin is prescribed as a strong analgesic under the name diamorphine. Its use includes treatment for acute pain, such as:

- Severe physical trauma
- Myocardial infarction
- Post-surgical pain

Q. No. 13 What are hallucinations?

HALLUCINATIONS

The perceptions that have no basis in reality, but that appear entirely realistic are called hallucinations.

Q. No. 14 What is the difference between broad-spectrum and narrow-spectrum antibiotics?

BROAD-SPECTRUM ANTIBIOTICS

The antibiotics that can be used to treat a wide range of infections are known as broad spectrum antibiotics.

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NARROW-SPECTRUM ANTIBIOTICS

The antibiotics that are only effective against a few types of bacteria are known as narrow spectrum antibiotics.

Q. No. 15 What is the most common method of administering vaccine?

MOST COMMON METHOD OF ADMINISTERING VACCINE

The most common method of administering vaccines is by injection, but some vaccines are given by mouth or nasal spray.

Q. No. 16 When children are required to be vaccinated?

VACCINATION TO CHILDREN

The children are required to be vaccinated before attending school.

Q. No. 17 What are the outcomes of vaccination to children?

OUTCOMES OF VACCINATION TO CHILDREN

The vaccination of children has resulted in marked decrease of many once common diseases including:

- Whooping cough
- Polio
- Smallpox

Q. No. 18 Why booster shots of vaccines are required?

REQUIREMENT OF BOOSTER SHOTS OF VACCINES

Some vaccines do not provide life time immunity. Foe example, tetanus vaccines are only effective for a limited period of time. In such cases, booster shots are necessary to maintain continuous protection.

Q. No. 19 What is the side effect of expired drugs?

SIDE EFFECT OF EXPIRED DRUGS

The expired drugs can cause damage to kidneys.

Q. No. 20 On what basis the drugs are classified?

BASIS OF CLASSIFICATION OF DRUGS

The drugs are classified on the basis of:

- Chemical properties
- Modes of action

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LONG QUESTIONS

Q. No. 1 What are medicinal drugs? Explain the sources of medicinal drugs.

MEDICINAL DRUGS

Definition:

The chemical substances that are used in the diagnosis, cure, treatment or prevention of a disease are called medicinal drugs.

Advantage:

Various diseases have been made easier to treat in recent years by the production of medicinal drugs.

SOURCES OF DRUGS

Drugs are obtained from the following sources:

1. Synthetic drugs
2. Drugs from plants
3. Drugs from fungi
4. Drugs from animals
5. Drugs from minerals
6. Drugs from bacteria

1. Synthetic drugs:

Such drugs do not occur naturally but are synthesized in laboratory. Pharmaceutical companies produce these drugs.

Example:

- Aspirin

2. Drugs from Plants:

Many important drugs are obtained from plants. These medicines include:

- Antibiotics
- Cardiotonics
- Certain analgesics

Examples:

Digitalis:

Function:

It is a cardiotonic which is used to stimulate the heart.

Source:

It is made from the leaves of purple flowered plant, foxglove.

Morphine:

Function:

It is a pain reliever.

Source:

It is obtained from opium, which comes from the juice of poppy plant.

Addictive Drugs:

Many addictive illegal drugs are also obtained from plants e. g.

- Marijuana

3. Drugs from Fungi:

Some medicines are obtained from fungi.

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Example:
The antibiotic penicillin is obtained from a fungus, *Penicillium notatum*.

4. **Drugs from Animals:**
Drugs obtained from animals are usually their glandular products.
Examples: The following are obtained from animal sources:
- Fish liver oil
- Musk
- Bee's wax
- Certain hormones
- Antitoxins

5. **Drugs from Minerals:**
Several common drugs are produced from minerals:

Examples:
Iodine:
The mineral iodine is used in making tincture of iodine.
Function:
It is a liquid that helps prevent infection when applied to cuts and bruises.
Silver Nitrate:
The powder form of silver nitrate is applied on wounds to stop bleeding and prevent infections.

6. **Drugs from Bacteria:**
Many antibiotics are obtained from bacteria:
Example:
- Streptomycin

Q. No. 2 Describe the principal usage of important medicinal drugs.

PRINCIPAL USAGE OF IMPORTANT MEDICINAL DRUGS

The drugs are classified on the basis of:
- Chemical properties
- Modes of action

The principal usage of important medicinal drugs is as follow:

1. **Analgesics:**
These are the pain killers. These reduce pain.
Examples:
- Aspirin
- Paracetamol

2. **Antibiotics:**
These inhibit or kill bacteria with in or on the body and treat bacterial infections.
Examples:
- Tetracycline
- Cephalosporins

3. **Sedatives:**
These induce sedation by reducing irritability or excitement.
Example:
- Diazepam

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4. **Vaccines:**
These are used to develop immunity against viral and bacterial infections.
Examples:
- Smallpox
- Whooping cough
- Hepatitis B

5. **Antiseptics:**
These reduce the possibility of infections on skin.
Example:
- Tincture of iodine

6. **Disinfectants:**
These destroy microorganisms found on non-living objects.

Q. No. 3 What precautions should be taken before taking medicines?

PRECAUTIONS FOR THE USAGE OF MEDICINES

Medicines can help you feel better. But if medicines are taken incorrectly, they can actually make you feel worse. The following precautions should be kept in mind:

1. Dosage
2. Expiry Date
3. Self Medication
4. Duration
5. Treatment Discontinuation
6. Dosage for Children
7. Darkness
8. Carriage
9. Children's Reach
10. Tampered Medicines

1. **Dosage:**
Always check the instructions on doctor's prescription slip and make sure you take the doses of medicine strictly as your doctor prescribed.

2. **Expiry Date:**
Always check the expiry date printed on the medicine pack. The expired medicines may prove poisonous.

3. **Self Medication:**
Never take medicines prescribed for someone else, even if you think you have the same medical problem.

4 **Duration:**
Some medicines - such as antibiotics - must be taken for a specific number of days. Make sure you take the medicine for the fated time. Otherwise the problem may come back again.

5. **Treatment Discontinuation:**
Always check with your doctor before you top t king a medicine or consider a new treatment.

6. **Dosage for Children:**
Some medicines are not suitable for children, and there are special children's dosages for many medicines.

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7. **Darkness:**
Do not take medicine in the dark.

8. **Carriage:**
If your prescription medicines are crucial for your heath and life, carry medicines and dosage instructions with you, when ever you are out of home.

9. **Children's Reach:**
Always keep healthcare products out of the reach of children.

10. **Tampered Medicines:**
Do not use the medicine if there are signs of tampering. Inform the pharmacist and the manufacture of the medicine, about it.

Q. No. 4 Write a note on addictive drugs.

ADDICTIVE DRUGS

Definition:
The drugs that make person dependent on them or addicted are called addictive drugs.

Effect:
By using addictive drug, the person's body becomes familiar to it and the user can not function well with out it.

CATEGORIES OF ADDICTIVE DRUGS

The following are major categories of addictive drugs:

1. Sedatives
2. Narcotics
3. Hallucinogens

1. Sedatives:
These drugs induce sedation by reducing irritability or excitement.

Mode of Action:
These drugs interact with central nervous system to depress its activities.

Effects: Sedative drugs induce:

- Dizziness
- Lethargy
- Slow brain function
- Depression

Long Term Use:
Long term use of sedative drugs induces suicidal thoughts.

2. Narcotics:
Narcotics are strong pain killers.

Prescription:
These drugs are often prescribed in conjunction with other less potent pain killers like paracetamol or aspirin.

Usage:
These are used to relieve pain for patients with chronic diseases like cancer. These are also used to relieve acute pain after operations.

Drug Abuse:
But some people may abuse narcotics for ecstatic effects.

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Examples:

Morphine:
Morphine is derived from opium (poppy). It acts directly on central nervous system to relieve pain. Morphine has a high potential for addiction.

Codeine:
It is also derived from opium.

Heroin:
It is the most commonly abused narcotic. It is semi-synthetic drug from morphine. It effects on central nervous system and causes drowsiness.

Usage in Western Countries:

In many western countries, heroin is prescribed as a strong analgesic under the name diamorphine. Its use includes treatment for acute pain, such as:

- Severe physical trauma
- Myocardial infarction
- Post-surgical pain

3. **Hallucinogens:**

Hallucinogens are the drugs that cause changes in:

- Perception
- Thought
- Emotion
- Consciousness

Mode of Action:

Physiologically, hallucinogens affect on the sympathetic nervous system causing:

- Dilation of pupils
- Constriction of some arteries
- Rise in blood pressure

Examples:

Mescaline:
Mescaline is derived from cactus.

Psilocin:
Psilocin is derived from a mushroom.

Marijuana (Hashish):
Marijuana is a hallucinogen, which is smoked.

Sources:
It is obtained from the flowers, stems and leaves of the marijuana plant.

- *Cannabis sativa*
- *Cannabis indica*

Less Dosage:
Small doses of marijuana result in a feeling of well being that lasts for two to three hours.

High Dosage:
High doses increase heart rate.

Adverse Effects:
It also effects the production of sperms in men and also weakens the short-term memory.

Usage:
Marijuana is one of the most commonly used drugs in the world, followed by caffeine, nicotine and alcoholic beverages in popularity.

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Q. No. 5 Explain problems with drug addiction.

PROBLEMS WITH DRUG ADDICTION

There is a long list of drug related associated problems, some of them are as follow:

Withdrawal of Social Contact:

Drug abusers go through withdrawal of social contact or communication. The addicts are very weak in their social behaviour. They face social stigma i.e. the society dislikes them because of their unpredictable behaviours.

Problems for Government:

The jails and prisons of our country are full of such people who have committed no other crime than the illegal possession of narcotics.

Increase in Crime Rate:

Many studies by the experts of social sciences prove that there exists a close relationship between drug addiction and crime. The compulsion for narcotic drug makes every drug addict a criminal.

Law Violator:

The drug addicts are law violators. Mere possession of a narcotic drug is violation of the law. Thus, every drug addict is subject to arrest by the police.

Psychic Patients:

Drug addicts may commit violent crimes since so many become psychic patients.

Other Crimes:

Most narcotic addicts get involved in various types of crimes, like:

- Robbery
- Shop lifting
- Burglary
- Embezzlement

Q. No. 6 What are antibiotics? Explain major groups of antibiotics.

ANTIBIOTICS

Definition:

An antibiotic is a drug that kills or retards the growth (reproduction) of bacteria. They are the chemicals produced by or derived from microorganisms like bacteria and fungi.

Types of antibiotics:

There are two main types of antibiotics:

Bactericidal Antibiotics

The antibiotics that kill the bacteria are called bactericidal antibiotics.

Bacteriostatic Antibiotics

The antibiotics that work by stopping the bacterial growth are called bacteriostatic antibiotics.

GROUPS OF ANTIBIOTICS

There are three major group of antibiotics:

1. Cephalosporins
2. Tetracyclines
3. Sulpha Drugs – Sulfonamides

1. Cephalosporins:

Mode of action:

Cephalosporins interfere with the synthesis of bacterial cell wall.

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Category:

These are bactericidal antibiotics.

Indications:

Cephalosporins are used to treat

- Pneumonia
- Sore throat
- *Tonsillitis*
- Bronchitis

2. Tetracyclines:

Mode of Action:

Tetracyclines inhibit bacterial protein synthesis.

Category:

These are broad-spectrum bacteriostatic antibiotics.

Indications:

Tetracyclines are used in the treatment of infections of:

- Respiratory tract
- Urinary tract
- Intestine

Contraindications:

Tetracyclines are not used in children under the age of 8, and especially during periods of tooth development.

3. Sulpha Drugs – Sulfonamides:

Composition:

Sulpha drugs are synthetic antibiotics that contain sulfonamide group.

Category:

Sulfonamides are broad spectrum bacteriostatic antibiotics.

Mode of Action:

They inhibit the folic acid synthesis of bacteria.

Indications:

They are used to treat:

- Pneumonia
- Urinary tract infections

Q. No. 7 Write a note on antibiotic resistance.

ANTIBIOTIC RESISTANCE

Antibiotics are extremely important in medicine, but unfortunately bacteria are capable of developing resistance to them. Such bacteria are not affected by commonly used antibiotics.

Definition:

The ability of bacteria not to be affected by the particular antibiotic is called antibiotic resistance.

Reason:

When bacteria are exposed to the same antibiotic over and over, they can change and are no longer affected by the drug.

WAYS OF DEVELOPING RESISTANCE

Bacteria have number of ways of developing resistance.

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Stoppage by Internal Mechanism:

Sometimes, their internal mechanism stops the working of antibiotic.

Transfer of Genes:

Bacteria can also transfer the genes responsible for antibiotic resistance between them. So such resistance bacteria make it possible for other bacteria to acquire resistance.

Unethical Usage:

Another reason for increasing antibiotic resistance in bacteria is their use in diseases in which they have no efficacy e.g. antibiotics are not effective against infections caused by viruses.

A Growing Problem:

Resistance to antibiotics poses a serious and growing problem, because some infectious diseases are becoming more difficult to treat. Some of the resistant bacteria can be treated with more powerful antibiotics, but there are some infeetions that do not eliminate even with new antibiotics.

Q. No. 8 Write a note on vaccines. Explain mode of action of vaccines.

VACCINES

Definition:

A material containing weakened or killed pathogens and is used to produce immunity to a disease by stimulating the production of antibodies is called a vaccine.

Work of Edward Jenner:

In 1796, a British physician, Edward Jenner, infected a young boy with cowpox, by injecting pus cells. After the boy had recovered from cowpox, Jenner injected the pus cells from a smallpox patient into him. The boy did not get smallpox.

Result:

So it became clear that intentional infection with cowpox protected people from smallpox.

Vaccination:

This method was named "vaccination" and the substance used to vaccinate was called a "vaccine".

MODE OF ACTION OF VACCINES

Antigens:

Pathogens contain special proteins called "antigens".

Antibodies:

When pathogens enter the body (blood) of host, these proteins stimulate the immune response in host i.e. synthesis of "antibodies". Antibodies bind to pathogens and destroy them.

Production of Memory Cells:

In addition, "memory cells" are produced, which remain in blood and provide protection against future infections with the same pathogen.

Stimulation of White Blood Cells:

When a vaccine i.e. weakened or dead pathogen is introduced into bloodstream, the white blood cells are' stimulated.

Recognition by B-lymphocytes:

B-lymphocytes recognize the weakened or dead pathogens as enemies and start producing antibodies against them.

Protection against Pathogens:

These antibodies remain in blood and provide protection against pathogens. If real pathogens enter blood, the already present antibodies kill the

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REVIEW QUESTION:

MULTIPLE CHOICE QUESTIONS

1. Antibiotics are used for the:

(a) Treatment of viral infections (b) Treatment of bacterial infections
(c) Immunization against infections (d) Both a and b

2. The substances used for the treatment, cure, prevention or diagnoses of diseases are called:

(a) Medicinal drugs (b) Narcotics
(c) Hallucinogens (d) Sedatives

3. Aspirin is categorized as;

(a) A drug from animals (b) A synthetic drug
(c) A drug from plants (d) A drug from minerals

4. The drugs used to reduce pain are known as;

(a) Analgesics (b) Antiseptics
(c) Antibiotics (d) Sedatives

5. Which of the following drugs obtained from plants?

(a) Aspirin (b) Opium
(c) Cephalosporin (d) Insulin

6. Which of these addictive drugs are also used as painkillers?

(a) Narcotics (b) Sedatives
(c) Hallucinogens (d) All can be used

7. Sulfonamides affect bacteria in the following way;

(a) Break the cell wall
(b) Inhibit protein synthesis
(c) Stop the synthesis of new cell wall
(d) Stop the synthesis of folic acid

8. What is true about vaccines?

(a) Protect against the future viral and bacterial infections
(b) Treat the existing bacterial infections only
(c) Treat existing infections and also protect against future infections
(d) Protect against viral infections only

ANSWER KEY

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	b	2	a	3	b	4	a
5	b	6	a	7	d	8	a

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SHORT QUESTIONS

1.Define pharmacology and distinguish it from pharmacy.

PHARMACOLOGY

The study of drug composition, properties, medical applications sources of drugs is called pharmacology.

Difference from Pharmacy:

Pharmacology is not synonymous with pharmacy, which is the name used for a profession, though in common usage the two terms are confused.

2. Differentiate between medicinal drug and addictive drug.

Consult Long Question No. 1 and 4

3. Differentiate between analgesic and antibiotic.

Consult Long Question No. 2

4. What is marijuana? To which category of addictive drugs it belongs?

Consult Long Question No. 4

5. Differentiate between narcotics and hallucinogens.

Consult Long Question No. 4

UNDERSTANDING THE CONCEPT

1. What are the sources of drugs? Give examples.

Consult Long Question No. 1

2. Write a note on sedatives, narcotics and hallucinogens.

Consult Long Question No. 2 and 4

3. Describe the main groups of antibiotics.

Consult Long Question No. 6

4. Write a note on resistance against antibiotics.

Consult Long Question No. 7

5. Describe the mode of action of vaccines.

Consult Long Question No. 8

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THE TERMS TO KNOW

Addictive Drug:

The drug which makes the person dependent on it or addicted

Analgesic:

The medicines that reduce pain

Antibiotics:

The medicines that inhibit or kill bacteria

Aspirin:

Acetaminophen; A pain like medicine

Bactericidal:

The antibiotics that work by killing bacteria

Bacteriostatic:

The antibiotics that work by stopping bacteria multiplying

Cardiotonic:

Medicines for giving strength to heart muscles

Cephalosporins:

A group of antibiotics; interfere with synthesis of bacterial cell wall

Hallucinogen:

Drug that causes changes in perception, thought, emotion and consciousness

Heroin:

A commonly abused narcotic; derived from morphine; affects the central nervous system and causes drowsiness, disorientation, hypotension etc.

Marijuana:

A hallucinogen and addictive drug; obtained from the flowers, stems and leaves of the marijuana plant

Medicinal Drug:

Any chemical substance intended for use in the medical diagnosis, cure, treatment or prevention of disease

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Morphine:

A commonly used narcotic; derived from the juice of opium; acts directly on the CNS to relieve pain; has a high potential for addiction

Narcotics:

Strong painkiller drugs; also used as addictive drugs; commonly abused narcotics include heroin, morphine, methadone etc.

Pharmacology:

The study of drug composition, properties and medical applications

Sedatives:

Types of drugs that interact with the central vervous system to depress its activities; make a person calm or drowsy

Sulfonamides:

Sulpha drugs; synthetic antibiotics that contain the sulfonamidé group; bacteriostatic in action

Tetracyclines:

Broad spectrum bacteriostatic antibiotics; inhibit bacterial protein synthesis

Vaccines:

The material used to produce immunity to a disease by stimulating the production of antibodies.

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